**Dimitrios S. Nikolopoulos**
John W. Hancock Professor of Engineering
Professor, Department of Computer Science
IEEE Fellow; ACM Distinguished Member

Virginia Tech, Blacksburg, VA
dsn@vt.edu
dsniko.github.io
Google Scholar · ORCID 0000-0003-0217-8307

## At a Glance

| | |
|---|---|
| **Publications** | 292 refereed: 155 conference, 69 journal, 43 workshop, 9 chapters, 16 posters |
| **Citations** | 8,086 citations and an h-index of 44 (Google Scholar, October 2026) |
| **Funding** | $82.9M in grants as PI and $58.2M as Co-PI |
| **Advising** | 27 Ph.D. graduates, 9 current Ph.D. students, and 17 postdoctoral fellows |

## Employment History

| | |
|---|---|
| **John W. Hancock Professor of Engineering**, Virginia Tech | Aug. 2019–present |
| **Professor**, Department of Computer Science, Virginia Tech | Aug. 2019–present |
| **Professor (by courtesy)**, Department of Electrical and Computer Engineering, Virginia Tech | Aug. 2020–present |
| **Associate Director**, Stacks@CS Center for Computer Systems Research, Virginia Tech | Aug. 2022–present |
| **Honorary Professor**, School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast | Aug. 2022–present |
| **Director**, Queen's University Belfast Research Institute in Electronics, Communications and Information Technology (ECIT) | Jan. 2018–Aug. 2019 |
| **Head of School**, School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast | Jan. 2016–Jan. 2018 |
| **Royal Society Wolfson Research Fellow** | Sep. 2015–present |
| **Professor & Chair in High Performance and Distributed Computing**, School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast | Jan. 2012–Aug. 2019 |
| **Director**, Center for Data Science and Scalable Computing, Queen's University Belfast | Feb. 2016–Aug. 2019 |
| **Director**, High Performance and Distributed Computing Research Cluster, Queen's University Belfast | Jan. 2012–Feb. 2016 |
| **Adjunct Professor**, Department of Computer Science, Virginia Tech | Oct. 2016–Aug. 2019 |
| **Adjunct Professor**, Department of Computer Science, Old Dominion University | Oct. 2013–present |
| **Associate Professor with Tenure**, Department of Computer Science, University of Crete | Sep. 2009–Jan. 2012 |
| **Affiliate Professor**, Institute of Computer Science, Foundation for Research and Technology (FORTH) | Sep. 2009–Jan. 2012 |
| **Associate Professor with Tenure**, Department of Computer Science, Virginia Tech | Aug. 2008–Sep. 2009 |
| **Associate Professor, Tenure-Track**, Department of Computer Science, Virginia Tech | Aug. 2006–Aug. 2008 |
| **Assistant Professor, Tenure-Track**, Department of Computer Science, College of William & Mary | Aug. 2002–Aug. 2006 |

**Visiting Assistant Professor**, Department of Electrical and Computer Engineering, University of Illinois at Urbana–Champaign — Jan. 2001–Aug. 2002

## Earned Degrees

| | | |
|---|---|---|
| **Ph.D.** | Computer Engineering and Informatics, University of Patras | 2000 |
| **M.Eng.** | Computer Engineering and Informatics, University of Patras | 1997 |
| **Diploma**[1] | Computer Engineering and Informatics, University of Patras | 1996 |

[1] A European Diploma in Engineering is a five-year university degree, academically positioned between a Bachelor's and Master's in the U.S. system.

## Primary Research Areas

**High-Performance Computing and Computational Science; Computer Systems**

## Current Technical Research Interests

- **Efficient and resilient AI systems across the computing continuum:** federated learning; edge/microcloud infrastructure; and runtime/orchestration for robustness under heterogeneity and failures.
- **Performance engineering for modern accelerators:** GPU/CPU overlap; memory and KV-cache redesign; long-context inference/training efficiency; and cost/energy-aware optimization.
- **Systems for data-intensive science and cyberinfrastructure:** streaming-plus-learning pipelines; scalable experimentation/digital twins; and deployable research platforms with measurable impact.

## Honors and Awards

[44] **Outstanding Service Award**, Department of Computer Science, Virginia Tech, 2026
[43] **Sigma Xi Full Member**, The Scientific Research Honor Society, 2025
[42] **IEEE Fellow**, "For contributions to dynamic execution environments and multiprocessor memory management", 2024
[41] **IEEE Computer Society Distinguished Visitor**, 2024
[40] **Fellow, International Artificial Intelligence Industry Alliance**, 2024
[39] **Fellow, Asia-Pacific Artificial Intelligence Association**, 2024
[38] **Distinguished Contributor, IEEE Computer Society**, 2022
[37] **Best Paper Award**, Design Automation and Test in Europe Conference (DATE), 2020
[36] **IEEE Award for Editorial Excellence**, IEEE Transactions on Parallel and Distributed Systems, 2020
[35] **HiPEAC Paper Award**, twice in 2020 (for ASPLOS 2020 and MICRO 2020 papers)
[34] **Best Paper Award Nomination**, IEEE/ACM International Symposium on Microarchitecture (MICRO), 2020
[33] **John W. Hancock Chair in Engineering**, Virginia Tech, 2019–present
[32] **Elsevier Distinguished Editorial Service Award**, 2019
[31] **Distinguished Member of the ACM**, 2018
[30] **IEEE Outstanding Service Award**, 2018
[29] **Fellow of the IET**, 2017
[28] **Investors in People Silver Award**, 2017
[27] **Royal Society Wolfson Research Merit Award**, 2015 (extended to lifetime)
[26] **SFI-DEL Investigator Award**, 2015

[25] **Fellow of the British Computer Society**, 2014
[24] **IEEE Outstanding Service Award**, 2014
[23] **Best Paper Award**, ACM International Workshop on Code Optimization for Multi and Many Cores (COSMIC), 2013
[22] **ACM Senior Member**, 2011
[21] **IEEE Senior Member**, 2010
[20] **IEEE Outstanding Service Award**, 2010
[19] **Best Paper Award Nomination**, International Conference on Parallel Processing (ICPP), 2010
[18] **Marie Curie Fellow**, 2009
[17] **HiPEAC Fellow**, 2008
[16] **IBM Faculty Award**, 2007
[15] **Best Paper Award**, ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPOPP), 2007
[14] **Best Paper Nomination**, ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC), 2006
[13] **DOE Early Career Principal Investigator Award**, 2005
[12] **Best Paper Award**, International Workshop on OpenMP (IWOMP), 2005
[11] **NSF CAREER Award**, 2004
[10] **Best Paper Award**, International Symposium on High-Performance Computing (ISHPC), 2003
[9] **Best Paper Award**, IEEE/ACM International Symposium on Cluster Computing and the Grid (CCGRID), 2002
[8] **Best Paper Award**, IEEE/ACM International Parallel and Distributed Processing Symposium (IPDPS), 2002
[7] **Best Paper Award Nomination**, ACM International Conference on Supercomputing (ICS), 2001
[6] **Best Paper Award Nomination**, IEEE/ACM Supercomputing: High-Performance Computing, Networking, Storage, and Analysis Conference (SC), 2001
[5] **Best Paper Award**, IEEE/ACM Supercomputing: High-Performance Computing, Networking, Storage and Analysis Conference (SC), 2000
[4] **Best Paper Award Nomination**, ACM International Conference on Supercomputing (ICS), 1999
[3] **Outstanding Academic Performance Award**, Technical Chamber of Greece, 1996
[2] **Outstanding Academic Performance Award**, Technical Chamber of Greece, 1993
[1] **Outstanding Academic Performance Award**, National Scholarships Foundation of Greece, 1992

## Publications

An asterisk (*) marks a co-author who was my student at the time of the work.

## Refereed Publications

## Papers in Archival Conference Proceedings

[292] K. H. I. Arif*, J. Yoon, D. S. Nikolopoulos, H. Vandierendonck, D. John, and B. Ji, "Intent-Aware Caching for Efficient LLM Serving," in *Proceedings of the 40th Annual Conference on Neural Information Processing Systems (NeurIPS)*, To appear, 2026.
[291] Y. Li, A. Kumar, P. Vishnu, D. S. Nikolopoulos, B. Ji, H. Vandierendonck, and D. John, "ZTA-Q: An Open-Source RISC-V Platform for Accurate Quantized CNN Inference," in *Proceedings of the 33rd*

*IEEE International Conference on Electronics Circuits and Systems **(ICECS)***, Thessaloniki, Greece, Nov. 2026.

[290] F. Amin*, S. Afroz, and D. S. Nikolopoulos, "CAI-dLLM: Convergence Aware Inference for Diffusion Language Models," in *Proceedings of the 19th International Natural Language Generation Conference **(INLG)***, Utrecht, The Netherlands, Oct. 2026.

[289] Ajay Kumar M, Vishnu Ps, Y. Li, S. Shanker, D. S. Nikolopoulos, B. Ji, H. Vandierendonck, and D. John, "A Framework for Accelerating Transformer Inference on RISC-V for Edge AI," in *Proceedings of the 33rd IEEE International Conference on Electronics Circuits and Systems **(ICECS)***, Thessaloniki, Greece, Nov. 2026.

[288] F. Amin*, S. Afroz, K. Gharami, M. Moghadampanah*, and D. S. Nikolopoulos, "DiffPro: Joint Timestep and Layer-Wise Precision Optimization for Efficient Diffusion Inference," in *Proceedings of the 19th European Conference on Computer Vision **(ECCV)***, Malmö, Sweden, Sep. 2026. DOI: 10.1007/978-3-032-37252-9_7

[287] Y. Shen* and D. S. Nikolopoulos, "Stride-Aware Page Prefetching for GPU Unified Memory via Markov Pattern Prediction," in *Proceedings of the 2026 ACM SIGPLAN International Symposium on Memory Management **(ISMM)***, Boulder, CO, Jun. 2026. DOI: 10.1145/3814942.3816131

[286] J. Fan*, Y. Zhang, X. Li*, and D. S. Nikolopoulos, "dLLM-Serve: Bridging the Memory Gap in Diffusion Language Model Serving," in *Proceedings of the 40th ACM International Conference on Supercomputing **(ICS)***, Belfast, Northern Ireland, UK, 2026. DOI: 10.1145/3797905.3800532

[285] A. Nafisi*, V. Cvetkovic*, C. Shin*, K. Reece*, K. Reddy*, B. Torres, S. Heidari*, M. Ellis, and D. S. Nikolopoulos, "MARCO: Multi-Agent Code Optimization for High-Performance Computing," in *ACM Capital Region Celebration of Women in Computing Conference **(CAPWIC)***, Mar. 2026.

[284] J. Fan*, Y. Zhang, X. Li*, and D. S. Nikolopoulos, "APEX: Asynchronous Parallel CPU-GPU Execution for Online LLM Inference on Constrained GPUs," in *Proceedings of the 40th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, New Orleans, LA, 2026. DOI: 10.1109/IPDPS65963.2026.00099

[283] S. Ghafouri, M. Fayyaz, X. Li*, D. John, B. Ji, D. Nikolopoulos, and H. Vandierendonck, "Polymorph: Energy-Efficient Multi-Label Classification for Video Streams on Embedded Devices," in *IEEE/CVF Winter Conference on Applications of Computer Vision **(WACV)***, Tucson, AZ, 2026. DOI: 10.1109/WACV61042.2026.00622

[282] K. H. I. Arif*, J. Yoon, D. S. Nikolopoulos, H. Vandierendonck, D. John, and B. Ji, "HiRED: Attention-Guided Token Dropping for Efficient Inference of High-Resolution Vision-Language Models in Resource-Constrained Environments," in *Proceedings of the 39th Annual AAAI Conference on Artificial Intelligence **(AAAI'25)***, vol. 39, Philadelphia, PA, Feb. 2025, pp. 1773–1781. DOI: 10.1609/aaai.v39i2.32171

[281] X. Li*, D. Spatharakis, S. Ghafouri, J. Fan*, H. Vandierendonck, D. John, B. Ji, and D. S. Nikolopoulos, "SLED: A Speculative LLM Decoding Framework for Efficient Edge Serving," in *Proceedings of the Tenth ACM/IEEE Symposium on Edge Computing **(SEC)***, Article 26, pages 1–8, Washington, D.C, Dec. 2025. DOI: 10.1145/3769102.3770608

[280] M. Lee, S. Seong, M. Kang, J. Lee, G.-J. Na, I.-G. Chun, D. S. Nikolopoulos, and C.-H. Hong, "ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments," in *Proceedings of Supercomputing: The International Conference for High-Performance Computing, Networking, Storage, and Analysis **(SC)***, Article No.: 42, Nov. 2024, pp. 1–14. DOI: 10.1109/SC41406.2024.00048

[279] Y. Li*, S. Yao, J. Mobin, M. M. Rafique, D. S. Nikolopoulos, K. Sundararajah, H. Li, and A. R. Butt, "Towards an Efficient Python Interpreter for Tiered Memory Systems," in *Proccedings of the 22nd USENIX Conference on File and Storage Systems **(FAST)***, WIP Session, Santa Clara, CA, Feb. 2024.

[278] M. Arif, A. Maurya, M. Rafique, D. S. Nikolopoulos, and A. R. Butt, "Application-Attuned Memory Management for Containerized HPC Workflows," in *Proceedings of the 38th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, San Francisco, CA, May 2024. DOI: 10.1109/IPDPS57955.2024.00019

[277] M. Cameron, M. Ellis, and D. S. Nikolopoulos, "Parallel Islands: A Parallel Computing Educational Video Game," in *Proceedings of the SIGCSE Technical Symposium **SIGCSETS***, Mar. 2024. DOI: 10.1145/3626253.3635591

[276] X. Li, H. Vandierendonck, D. S. Nikolopoulos, B. Ji, B. Cardiff, and D. John, "Decentralized Biomedical Signal Classification using Early Exits," in *Proceedings of the 21st IEEE Interregional NEWCAS Conference **NEWCAS***, Edinburgh, UK, Jun. 2023, pp. 1–2. DOI: 10.1109/NEWCAS57931.2023.10198098

[275] K. Assogba, M. Arif, M. M. Rafique, and D. S. Nikolopoulos, "On Realizing Efficient Deep Learning using Serverless Computing," in *Proceedings of the 22nd IEEE/ACM International Symposium on Cluster, Cloud, and Internet Computing **(CCGRID)***, May 2022. DOI: 10.1109/CCGRID54584.2022.00031

[274] S. Barbhuiya, P. Kilpatrick, and D. S. Nikolopoulos, "Linear Regression Based DDoS Attack Detection," in *Proceedings of the 13th International Conference on Machine Learning and Computing **(ICMLC)***, Jan. 2021, pp. 568–574. DOI: 10.1145/3457682.3457769

[273] E. Chawla, R. Jian, F. Shiely, K. Waki, D. Nikolopoulos, P. Henn, P. Fadahunsi, B. Hayhoe, A. Majeed, K. Balanda, J. Gallagher, S. Oakes, and J. O'Donoghue, "Combining Leadership and Patient Empowerment through Intelligent Data Access and Remote Consultations (CLEAR): A Systematic Review," in *New Horizons Research Conference*, Accepted, Cork, Ireland, Nov. 2020.

[272] L. Mukhanov, D. S. Nikolopoulos, and G. Karakonstantis, "DStress: Automatic Synthesis of DRAM Reliability Stress Viruses using Genetic Algorithms," in *Proceedings of the 53rd IEEE/ACM International Symposium on Microarchitecture **(MICRO)***, **Best Paper Award Nomination**, Oct. 2020, pp. 298–312. DOI: 10.1109/MICRO50266.2020.00035

[271] J. Liu, Z. Xie, D. S. Nikolopoulos, and D. Li, "RIANN: Real-time Incremental Learning with Approximate Nearest Neighbor on Mobile Devices," in *Proceedings of the 2020 USENIX Conference on Operational Machine Learning **(OpML)***, Santa Clara, CA, Jul. 2020.

[270] K. Chen, P. Kilpatrick, D. S. Nikolopoulos, and B. Varghese, "Cross Architectural Power Modeling," in *Proceedings of the 20th IEEE/ACM International Symposium on Cluster, Grid and Internet Computing **(CCGRID)***, Melbourne, Australia, May 2020, pp. 390–399. DOI: 10.1109/CCGRID49817.2020.00-54

[269] A. Abubakar*, S. Barbhuiya, D. S. Nikolopoulos, V. Ngo, and P. Kilpatrick, "Fast Analysis and Prediction in Large Scale Virtual Machines Resource Utilization," in *Proceedings of the 10th International Conference on Cloud Computing and Services Science **(CLOSER)***, Prague, Czech Republic, May 2020, pp. 115–126. DOI: 10.5220/0009408701150126

[268] K. Tovletoglou*, L. Mukhanov, D. S. Nikolopoulos, and G. Karakonstantis, "HaRMony: Heterogeneous-Reliability Memory and QoS-Aware Energy Management on Virtualized Servers," in *Proceedings of the 2020 ACM International Conference on Architectural Support for Programming Languages and Operating Systems **(ASPLOS)***, Lausanne, Switzerland, Mar. 2020, pp. 575–590. DOI: 10.1145/3373376.3378489

[267] I. Tsiokanos*, L. Mukhanov, G. Georgakoudis, D. S. Nikolopoulos, and G. Karakonstantis, "DEFCON: Generating and Detecting Failure-Prone Instruction Sequences via Stochastic Search," in *Proceedings of the 2020 Design Automation and Test in Europe Conference **(DATE)***, **Best Paper Award**, Grenoble, France, Mar. 2020, pp. 1121–1126. DOI: 10.23919/DATE48585.2020.9116363

[266] S. Barbhuiya, P. Kilpatrick, and D. S. Nikolopoulos, "DroidLight: Lightweight Anomaly-Based Intrusion Detection System for Smartphone Devices," in *Proceedings of the 21st International Conference on Distributed Computing and Networking **(ICDCN)***, Kolkata, India, Jan. 2020, 31:1–31:10. DOI: 10.1145/3369740.3369796

[265] L. Mukhanov, K. Tovletoglou*, H. Vandierendonck, D. S. Nikolopoulos, and G. Karakonstantis, "Comprehensive Workload-Aware DRAM Error Prediction using Machine Learning," in *Proceedings of the 2019 IEEE International Symposium on Workload Characterization **(IISWC)***, Orlando, FL, Nov. 2019, pp. 106–118. DOI: 10.1109/IISWC47752.2019.9041963

[264] K. Dichev and D. S. Nikolopoulos, "Implementing Efficient Message Logging Protocols as MPI Application Extensions," in *Proceedings of the 26th International Conference on Recent Advances in Message Passing Interface **EuroMPI***, Article 8, Zurich, Switzerland, Sep. 2019, pp. 1–11. DOI: 10.1145/3343211.3343219

[263] S. Kyle, D. Nolan, M. Price, W. Zhang, T. Robinson, D. S. Nikolopoulos, and S. Barbhuiya*, "Bio-inspired Growth: Introducing Emergence into Computational Design," in *Advances in Manufacturing Technology XXXII*, 2019, pp. 379–385. DOI: 10.3233/atde190067

[262] W. Zhang, M. Price, T. Robinson, D. Nolan, D. S. Nikolopoulos, S. Barbhuiya*, and S. Kyle, "Design Gene Representations for Emergent Innovative Design," in *Advances in Manufacturing Technology XXXII*, **Best Paper Award Runner Up**, Sep. 2019, pp. 386–392. DOI: 10.3233/atde190068

[261] S. Barbhuiya*, D. S. Nikolopoulos, M. Price, T. Robinson, D. Nolan, W. Zhang, and S. Kyle, "SmartMaaS: A Framework for Smart Manufacturing-as-a-Service," in *Advances in Manufacturing Technology XXXII*, **Best Paper Award**, Sep. 2019, pp. 16–21. DOI: 10.3233/atde190005

[260] G. Georgakoudis, I. Laguna, H. Vandierendonck, D. S. Nikolopoulos, and M. Schulz, "SAFIRE: Scalable and Accurate Fault Injection For Parallel Multithreaded Applications," in *Proceedings of the 33rd IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Rio de Janeiro, Brazil, May 2019, pp. 890–899. DOI: 10.1109/IPDPS.2019.00097

[259] J. Sun, H. Vandierendonck, and D. S. Nikolopoulos, "VEBO: A Vertex- and Edge-Balanced Ordering Heuristic to Load Balance Parallel Graph Processing," in *Proceedings of the 2019 ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming **(PPoPP)***, Washington, DC, USA, Feb. 2019, pp. 391–392. DOI: 10.1145/3293883.3295703

[258] R. Istrate*, F. Sheidegger, G. Mariani, D. S. Nikolopoulos, C. Bekas, and C. Malossi, "TAPAS: Train-Less Accuracy Predictor for Architecture Search," in *Proceedings of the 33rd AAAI International Conference on Artificial Intelligence **(AAAI)***, Honolulu, Hawaii, Jan. 2019, pp. 3927–3934. DOI: 10.1609/aaai.v33i01.33013927

[257] D. Fenacci, H. Vandierendonck, and D. S. Nikolopoulos, "Code and Data Transformations to Address Garbage Collector Performance in Big Data Processing," in *Proceedings of the 25th IEEE International Conference on High-Performance Computing, Data, and Analytics **(HiPC)***, Bangaluru, India, Dec. 2018. DOI: 10.1109/HIPC.2018.00040

[256] B. Wang, H. Vandierendonck, G. Karakonstantis, and D. S. Nikolopoulos, "Userspace Hypervisor Data Characterization in Virtualized Environment," in *Proceedings of the 24th IEEE International Conference on Parallel and Distributed Systems **(ICPADS)***, Singapore, Dec. 2018, pp. 638–645. DOI: 10.1109/PADSW.2018.8644612

[255] K. Dichev, K. W. Cameron, and D. S. Nikolopoulos, "Energy-Efficient Localized Rollback after Failures via Data Flow Analysis," in *Proceedings of the 25th International Conference on Recent Advances in Message Passing Interface **(EuroMPI)***, Barcelona, Spain, Sep. 2018. DOI: 10.1145/3236367.3236379

[254] E. Barlaskar*, K. Dichev, I. Spence, P. Kilpatrick, and D. S. Nikolopoulos, "Supporting Cloud IaaS Users in Detecting Performance-based Violation for Streaming Applications," in *Proceedings of the 15th IEEE International Conference on Autonomic Computing (**ICAC**)*, Trento, Italy, Sep. 2018. DOI: 10.1109/ICAC.2018.00027

[253] I. Tsiokanos*, L. Mukhanov, D. S. Nikolopoulos, and G. Karakonstantis, "Variation-Aware Pipelined Cores through Path Shaping and Dynamic Cycle Adjustment: Case Study on a Floating-Point Unit," in *Proceedings of the 2018 IEEE/ACM International Symposium on Low Power Electronics and Design **(ISLPED)***, Washington, DC, USA, Jul. 2018, 52:1–52:6. DOI: 10.1145/3218603.3218617

[252] I. Tsiokanos*, L. Mukhanov, D. S. Nikolopoulos, and G. Karakonstantis, "Minimization of Timing Failures in Pipelined Designs via Path Shaping and Operand Truncation," in *Proceedings of the 24th IEEE International Symposium on On-Line Testing and Robust System Design **(IOLTS)***, Costa Brava, Spain, Jul. 2018. DOI: 10.1109/IOLTS.2018.8474084

[251] L. Mukhanov, K. Tovletoglou*, D. S. Nikolopoulos, and G. Karakonstantis, "DRAM Characterization under Relaxed Refresh Period Considering System Level Effects within a Commodity Server," in *Proceedings of the 24th IEEE International Symposium on On-Line Testing and Robust System Design **(IOLTS)***, Costa Brava, Spain, Jul. 2018. DOI: 10.1109/IOLTS.2018.8474184

[250] L. Mukhanov, K. Tovletoglou*, D. S. Nikolopoulos, and G. Karakonstantis, "Characterization of HPC Workloads on an ARMv8 based Server under Relaxed DRAM Refreshed and Thermal Stress," in *Proceedings of the 18th International Conference on Embedded Computer Systems: Architectures, Modeling, and Simulation **SAMOS***, Jul. 2018, pp. 230–235. DOI: 10.1145/3229631.3236091

[249] C. Kachris, D. Soudris, S. Mavridis, M. Pavlidakis, C. Symeonidou, C. Kozanitis, A. Bilas, D. Fenacci, S. V. Bogaraju, H. Vandierendonck, and D. S. Nikolopoulos, "The VINEYARD Integrated Framework for Hardware Accelerators in the Cloud," in *Proceedings of the 18th International Conference on Embedded Computer Systems: Architectures, Modeling, and Simulation, Pythagorion, Greece, July 15-19, 2018.*, 2018, pp. 236–243. DOI: 10.1145/3229631.3236093

[248] G. Karakonstantis et al., "An Energy-Efficient and Error-Resilient Server Ecosystem Exceeding Conservative Scaling Limits," in *Proceedings of the 2018 Design Automation and Test in Europe Conference **(DATE)***, Dresden, Germany, Mar. 2018, pp. 1099–1104. DOI: 10.23919/DATE.2018.8342175

[247] C. Malossi, M. Schaffner, A. Molnos, L. Gamaitoni, G. Tagliavini, A. Emerson, A. Tomás, D. S. Nikolopoulos, E. Flamand, and N. Wehn, "The Transprecision Computing Paradigm: Concept, Design and Applications," in *Proceedings of the 2018 Design Automation and Test in Europe Conference **(DATE)***, Dresden, Germany, Mar. 2018, pp. 1105–1110. DOI: 10.23919/DATE.2018.8342176

[246] E. Barlaskar*, P. Kilpatrick, I. T. Spence, and D. S. Nikolopoulos, "Using Docker Swarm with a User-Centric Decision-Making Framework for Cloud Application Migration," in *Proceedings of the 2017 International Conference on Cloud Computing and Services Sciences **(CLOSER)** Lecture Notes in Computer Science, Communications in Computer and Information Science*, vol. 864, 2018, pp. 186–195. DOI: 10.1007/978-3-319-94959-8_5

[245] P. Thoman, K. Hasanov, K. Dichev, R. Iakymchuk, X. Aguilar, P. Gschwandtner, E. Laure, H. Jordan, P. Lemarinier, K. Katrinis, D. S. Nikolopoulos, and T. Fahringer, "A Taxonomy of Task-Based Technologies for High-Performance Computing," in *Proceedings of the 12th International Conference on Parallel Processing and Applied Mathematics **(PPAM)***, Lublin, Poland, Sep. 2017. DOI: 10.1007/978-3-319-78054-2_25

[244] C. D. Antonopoulos, S. Lallis, N. Bellas, G. Karakonstantis, D. S. Nikolopoulos, D. Gizopoulos, and P. Lawthers, "Energy Efficiency in ARMv8-based Microservers by Hardware Margins Identification," in *Proceedings of the 2017 **ARM Research Summit***, Cambridge, UK, Sep. 2017.

[243] C. D. Antonopoulos, S. Lallis, N. Bellas, G. Karakonstantis, D. S. Nikolopoulos, D. Gizopoulos, and P. Lawthers, "Reliability-Aware System Software Support on ARM Microservers," in *Proceedings of the **2017 ARM Research Summit***, Cambridge, UK, Sep. 2017.

[242] G. Georgakoudis*, I. Laguna, D. S. Nikolopoulos, and M. Schulz, "REFINE: Realistic Fault Injection via Compiler-Based Instrumentation for Accuracy, Portability, and Speed," in *Proceedings of Supercomputing: International Conference on High-Performance Computing, Networking, Storage and Analysis (**SC**)*, Denver, CO, USA, Nov. 2017, 29:1–29:14. DOI: 10.1145/3126908.3126972

[241] J. Sun*, H. Vandierendonck, and D. S. Nikolopoulos, "Accelerating Graph Analytics by Utilizing the Memory Locality of Graph Partitioning," in *Proceedings of the 46th International Conference on Parallel Processing **(ICPP)***, Bristol, UK, Aug. 2017, pp. 181–190. DOI: 10.1109/ICPP.2017.27

[240] K. Tovletoglou, D. S. Nikolopoulos, and G. Karakonstantis, "Access-Aware DRAM Failure-Rate Estimation under Relaxed Refresh Operations," in *Proceedings of the 2017 International Conference on Embedded Computer Systems: Architectures, Modeling and Simulation **(SAMOS)***, Samos, Greece, Jul. 2017. DOI: 10.1109/SAMOS.2017.8344643

[239] K. Tovletoglou*, D. S. Nikolopoulos, and G. Karakonstantis, "Relaxing DRAM Refresh Rate through Access Pattern Scheduling: A Case Study on Stencil-based Algorithms," in *Proceedings of the 23rd IEEE International Symposium on On-Line Testing and Robust System Design **(IOLTS)***, Thessaloniki, Greece, Jul. 2017, pp. 45–50. DOI: 10.1109/IOLTS.2017.8046197

[238] J. Sun*, H. Vandierendonck, and D. S. Nikolopoulos, "GraphGrind: Addressing Load Imbalance of Graph Partitioning," in *Proceedings of the ACM International Conference on Supercomputing **(ICS)***, Chicago, IL, USA, Jun. 2017, pp. 16.1–16.10. DOI: 10.1145/3079079.3079097

[237] E. Barlaskar*, P. Kilpatrick, I. Spence, and D. S. Nikolopoulos, "MyMinder: A User-Centric Decision Making Framework for Inter-Cloud Migration," in *Proceedings of the 7th International Conference on Cloud Computing and Services Science **(CLOSER)***, Porto, Portugal, Apr. 2017, pp. 560–567. DOI: 10.5220/0006355905880595

[236] H. Vandierendonck, K. Murphy, M. Arif, and D. S. Nikolopoulos, "HPTA: High-Performance Text Analytics," in *Proceedings of the 2016 IEEE International Conference on Big Data (**IEEE BigData 2016**)*, Washington, DC, Dec. 2016, pp. 416–423. DOI: 10.1109/BIGDATA.2016.7840632

[235] B. Varghese, N. Wang*, S. Barbhuiya*, P. Kilpatrick, and D. S. Nikolopoulos, "Challenges and Opportunities in Edge Computing," in *Proceedings of the 2016 IEEE International Conference on Smart Cloud **(IEEE SmartCloud)***, Nov. 2016, pp. 20–26. DOI: 10.1109/SmartCloud.2016.18

[234] G. Georgakoudis*, C. Gillan, A. Hassan*, U. Minhas, G. Tzenakis, I. Spence, H. Vandierendonck, R. Woods, D. S. Nikolopoulos, M. Shyamsundar, P. Barber, M. Russell, A. Bilas, S. Kaloutsakis, H. Giefers, P. Staar, C. Bekas, N. Horlock, R. Faloon, and C. Pattison, "NanoStreams: Codesigned Microservers

for Edge Analytics in Real Time," in *Proceedings of the 16th International Conference on Embedded Computer Systems: Architectures, Modeling and Simulation **(SAMOS-XVI)***, Samos, Greece, Jul. 2016, pp. 180–187. DOI: 10.1109/SAMOS.2016.7818346

[233] Y. Wu, D. S. Nikolopoulos, and R. Woods, "Runtime Support for Adaptive Power Capping on Heterogeneous SoCs," in *Proceedings of the 16th International Conference on Embedded Computer Systems: Architectures, Modeling, and Simulation **(SAMOS-XVI)***, Samos, Greece, Jul. 2016, pp. 71–78. DOI: 10.1109/SAMOS.2016.7818333

[232] C. Trehan, G. Karakonstantis, D. S. Nikolopoulos, and H. Vandierendonck, "Energy Optimization of Memory Intensive Parallel Workloads," in *Proceedings of the 28th ACM Symposium on Parallelism in Algorithms and Architectures **(SPAA)***, Asilomar State Beach, CA, Jun. 2016, pp. 251–252. DOI: 10.1145/2935764.2935811

[231] C. Kachris, D. Soudris, G. Gaydadjiev, H.-N. Nguyen, D. S. Nikolopoulos, A. Bilas, N. Morgan, C. Strydis, C. Tsalidis, J. Balafas, R. Jiménez-Peris, and A. Almeida, "The VINEYARD Project: Versatile, Integrated, Accelerator-Based, Heterogeneous Data Centres," in *Proceedings of the Fifth International Conference on Modern Circuits and Systems Technologies **(MOCAST)***, Thessaloniki, Greece, May 2016, pp. 1–4. DOI: 10.1109/MOCAST.2016.7495121

[230] C. Kachris, D. Soudris, G. Gaydadjiev, H.-N. Nguyen, D. S. Nikolopoulos, A. Bilas, N. Morgan, C. Strydis, C. Tsalidis, J. Balafas, R. Jiménez-Peris, and A. Almeida, "The VINEYARD Approach: Versatile, Integrated, Accelerator-Based, Heterogeneous Data Centres," in *Proceedings of the 12th International Symposium on Applied Reconfigurable Computing **(ARC)***, ser. Lecture Notes in Computer Science, vol. 9625, Mangaratiba, Brazil, Mar. 2016, pp. 3–13. DOI: 10.1007/978-3-319-30481-6_1

[229] M. Marcu, O. Boncalo, M. Ghenea, A. Amaricai, J. Weinstock, R. Leupers, Z. Wang, G. Georgakoudis*, D. S. Nikolopoulos, L. B. Cosmin Cernazanu-Glavanand, and M. Ionascu, "Low-Cost Hardware Infrastructure for Runtime Thread Level Energy Accounting," in *Proceedings of the 2016 International Conference on Architecture of Computing Systems **(ARCS)***, ser. Lecture Notes in Computer Science, vol. 9637, Mar. 2016, pp. 277–289. DOI: 10.1007/978-3-319-30695-7_21

[228] I. Mavroidis, I. Papaefstathiou, L. Lavagno, D. S. Nikolopoulos, D. Koch, J. Goodacre, I. Sourdis, V. Papaefstathiou, M. Coppola, and M. Palormino, "ECOSCALE: Reconfigurable Computing and Runtime System for Future Exascale Systems," in *Proceedings of the 2016 International Conference on Design, Automation, and Test in Europe **(DATE)***, Dresden, Germany, Mar. 2016, pp. 696–671. DOI: 10.3850/9783981537079_1021

[227] P. Petoumenos, L. Mukhanov, Z. Wang, H. Leather, and D. S. Nikolopoulos, "Power Capping: What Works, What Does Not," in *Proceedings of the 21st IEEE International Conference on Parallel and Distributed Systems **(ICPADS)***, Melbourne, Australia, Dec. 2015, pp. 525–534. DOI: 10.1109/ICPADS.2015.72

[226] L. Mukhanov, D. S. Nikolopoulos, and B. R. de Supinski, "ALEA: Fine-Grain Energy Profiling with Basic Block Sampling," in *Proceedings of the 24th International Conference on Parallel Architectures and Compilation Techniques **(PACT)***, San Francisco, CA, Oct. 2015, pp. 87–98. DOI: 10.1109/PACT.2015.16

[225] A. Hassan*, H. Vandierendonck, and D. S. Nikolopoulos, "Energy-Efficient Hybrid DRAM/NVM Main Memory," in *Proceedings of the 24th International Conference on Parallel Architectures and Compilation Techniques **(PACT)***, ACM Student Research Competition (SRC), San Francisco, CA, Oct. 2015, pp. 492–493. DOI: 10.1109/PACT.2015.58

[224] C.-Y. Su*, D. Roberts, E. A. León, K. W. Cameron, B. R. de Supinski, G. Loh, and D. S. Nikolopoulos, "HpMC: An Energy-Aware Management System for Multi-Level Memory Architectures," in *Proceedings of the First International Symposium on Memory Systems **(MEMSYS)***, Washington, DC, Oct. 2015, pp. 167–178. DOI: 10.1145/2818950.2818974

[223] S. Svorobej, J. Byrne, P. Liston, P. Byrne, C. Stier, H. Groenda, Z. Papazachos, and D. S. Nikolopoulos, "Towards Automated Data-Driven Cloud Computing Simulation Model Creation," in *Proceedings of the Eighth International Conference on Simulation Tools and Techniques **(SIMUTOOLS)***, Athens, Greece, Aug. 2015, pp. 248–255. DOI: 10.4108/eai.24-8-2015.2261129

[222] S. Barbhuiya*, D. S. Nikolopoulos, P. Kilpatrick, and Z. Papazachos, "A Lightweight Tool for Anomaly Detection in Cloud Data Centres," in *Proceedings of the Fifth International Conference on Cloud Com-*

*puting and Services Science* ***(CLOSER)***, **Best Paper Award Nominee**, Lisbon, Portugal, May 2015, pp. 343–351. DOI: 10.5220/0005453403430351

[221] V. Vassiliadis, C. Chalios*, K. Parasyris, C. Antonopoulos, S. Lalis, N. Bellas, H. Vandierendonck, and D. S. Nikolopoulos, "A Significance-Driven Programming Framework for Energy-Constrained Approximate Computing," in *Proceedings of the ACM International Conference on Computing Frontiers* ***(CF)***, Article 9, Ischia, Italy, May 2015. DOI: 10.1145/2742854.2742857

[220] A. Hassan*, H. Vandierendonck, and D. S. Nikolopoulos, "Software-Managed Energy-Efficient Hybrid DRAM/NVM Main Memory," in *Proceedings of the ACM International Conference on Computing Frontiers* ***(CF)***, Article 23, Ischia, Italy, May 2015. DOI: 10.1145/2742854.2742886

[219] O. G. Lorenzo, T. F. Pena, J. C. Cabaleiro, J. C. Pichel, F. F. Rivera, and D. S. Nikolopoulos, "Power and Energy Implications of the Number of Threads Used on the Intel Xeon Phi," in *Proceedings of the Second Congress on Multicore and GPU Programming* ***(PPMG)***, ISBN: 978-84-606-6036-1, Caceres, Spain, Mar. 2015, pp. 1–8.

[218] V. Vassiliadis, K. Parasyris, C. Chalios*, C. D. Antonopoulos, S. Lalis, N. Bellas, H. Vandierendonck, and D. S. Nikolopoulos, "A Programming Model and Runtime System for Significance-Aware Energy-Efficient Computing," in *Proceedings of the 20th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming* ***(PPoPP)***, San Francisco, CA, USA, Feb. 2015, pp. 275–276. DOI: 10.1145/2688500.2688546 Extended version *CoRR* **abs/1412.5150**, presented in *First HiPEAC Workshop on Approximate Computing* ***(WAPCO)***, Amsterdam, The Netherlands, January 2015.

[217] P.-O. Östberg et al., "The CACTOS Vision of Context-Aware Cloud Topology Optimization and Simulation," in *Proceedings of the Sixth IEEE International Conference on Cloud Computing Technology and Science* ***(CloudCom)***, Singapore, Dec. 2014, pp. 26–31. DOI: 10.1109/CLOUDCOM.2014.62

[216] Y. Wu, J. Nunez-Yanez, R. Woods, and D. S. Nikolopoulos, "Power Modeling and Capping for Heterogeneous ARM/FPGA SoCs," in *Proceedings of the 2014 International Conference on Field-Programmable Technology* ***(FPT)***, Shanghai, China, Dec. 2014, pp. 231–234. DOI: 10.1109/FPT.2014.7082782

[215] S. Imamura, K. Inoue, H. Sasaki, and D. S. Nikolopoulos, "Power-Capped DVFS and Thread Allocation with ANN Models on Modern NUMA Systems," in *Proceedings of the 32nd IEEE International Conference on Computer Design* ***(ICCD)***, Seoul, Korea, Oct. 2014, pp. 324–331. DOI: 10.1109/ICCD.2014.6974701

[214] C. Gillan, D. S. Nikolopoulos, I. Spence, A. Bilas, and C. Bekas, "Advancing the Hardware and Software Stack for Real-Time Analytics on Fast Data Streams," in *Proceedings of the IEEE 2014* ***eChallenges e-2014*** *Conference*, Belfast, UK, Oct. 2014, pp. 1–8.

[213] G. Georgakoudis*, D. S. Nikolopoulos, H. Vandierendonck, and S. Lalis, "Fast Dynamic Binary Rewriting for Flexible Thread Migration on Shared-ISA Heterogeneous MPSoCs," in *Proceedings of the International Conference on Embedded Computer Systems: Architectures, Modeling and Simulation* ***(ICSAMOS)***, Jul. 2014, pp. 156–163. DOI: 10.1109/SAMOS.2014.6893207

[212] J.-s. Yeom*, A. Batele, K. R. Bisset, E. Bohm, A. Gupta, L. V. Kale, M. Marathe, D. S. Nikolopoulos, M. Schulz, and L. Wesolowski, "Overcoming the Scalability Challenges of Epidemic Simulations on Blue Waters," in *Proceedings of the 28th IEEE International Parallel and Distributed Processing Symposium* ***(IPDPS)***, Phoenix, AZ, USA, May 2014, pp. 755–764. DOI: 10.1109/IPDPS.2014.83

[211] H. Vandierendonck, K. Chronaki*, and D. S. Nikolopoulos, "Deterministic Scale-Free Pipeline Parallelism with Hyperqueues," in *Proceedings of Supercomputing: International Conference for High Performance Computing, Networking, Storage and Analysis* ***(SC)***, Article No. 32, Denver, CO, USA: ACM, Nov. 2013. DOI: 10.1145/2503210.2503233

[210] F. Zakkak*, D. Chasapis, P. Pratikakis, A. Bilas, and D. S. Nikolopoulos, "Inference and Declaration of Independence in Task-Parallel Programs," in *Proceedings of the 10th International Conference on Advanced Parallel Processing Technology* ***(APPT)***, ser. Lecture Notes in Computer Science, vol. 8299, Stockholm, Sweden, Sep. 2013, pp. 1–16. DOI: 10.1007/978-3-642-45293-2_1

[209] G. Tzenakis, A. Papatriantafyllou*, H. Vandierendonck, P. Pratikakis, and D. S. Nikolopoulos, "BDDT: Block-Level Dynamic Dependence Analysis for Deterministic Task-Based Parallelism," in *Proceedings of the 10th International Conference on Advanced Parallel Processing Technology* ***(APPT)***, ser. Lecture Notes in Computer Science, vol. 8299, Stockholm, Sweden, Sep. 2013, pp. 17–31. DOI: 10.1007/978-3-642-45293-2_2

[208] C. Symeonidou*, P. Pratikakis, D. S. Nikolopoulos, and A. Bilas, "DRASync: Distributed Region-Based Memory Allocation and Synchronization," in *Proceedings of the 20th International Conference on Recent Advances in Message Passing Interface **(EuroMPI)***, Madrid, Spain, Sep. 2013, pp. 49–54.

[207] V. Papaefstathiou*, M. Katevenis, D. S. Nikolopoulos, and D. Pnevmatikatos, "Prefetching and Cache Management using Task Lifetimes," in *Proceedings of the 27th ACM International Conference on Supercomputing **(ICS)***, Eugene, OR, USA, Jun. 2013, pp. 325–334. DOI: 10.1145/2464996.2465443

[206] C.-Y. Su*, D. Li*, D. S. Nikolopoulos, K. W. Cameron, B. R. de Supinski, and E. A. Leon, "Model-Based, Memory-Centric Performance and Power Optimization on NUMA Multiprocessors," in *Proceedings of the 2012 IEEE International Symposium on Workload Characterization **(IISWC)***, San Diego, CA, Nov. 2012, pp. 164–173. DOI: 10.1109/IISWC.2012.6402921

[205] I. Manousakis* and D. S. Nikolopoulos, "BTL: A Framework for Measuring and Modeling Energy in Memory Hierarchies," in *Proceedings of the 24th International Symposium on Computer Architectures and High Performance Computing **(SBAC-PAD)***, New York City, NY, Oct. 2012, pp. 139–146. DOI: 10.1109/SBAC-PAD.2012.38

[204] F. Zakkak*, D. Chasapis, P. Pratikakis, D. S. Nikolopoulos, and A. Bilas, "Inference and Declaration of Independence: Impact on Deterministic Task Parallelism," in *Proceedings of the 21st International Conference on Parallel Architectures and Compilation Techniques **(PACT)***, Minneapolis, MN, USA, Sep. 2012, pp. 453–454.

[203] A. Khasymski*, M. M. Rafique*, A. R. Butt, S. S. Vazhkudai, and D. S. Nikolopoulos, "On the Use of GPUs in Realizing Cost-Effective Distributed RAID," in *Proceedings of the 20th IEEE International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems **(MASCOTS)***, Washington, DC, USA, Aug. 2012, pp. 469–478. DOI: 10.1109/MASCOTS.2012.59

[202] S. Lyberis*, P. Pratikakis, D. S. Nikolopoulos, M. Schulz, T. Gamblin, and B. R. de Supinski, "The Myrmics Memory Allocator: Hierarchical Message-Passing Allocation for Global Address Spaces," in *Proceedings of the 2012 ACM SIGPLAN International Symposium on Memory Management **(ISMM)***, Beijing, China, Jun. 2012, pp. 15–24. DOI: 10.1145/2258996.2259001

[201] G. Georgakoudis*, S. Lalis, and D. S. Nikolopoulos, "Dynamic Binary Rewriting and Migration for Shared-ISA Asymmetric, Multicore Processors," in *Proceedings of the 21st International ACM Symposium on High Performance Parallel and Distributed Computing **(HPDC)***, Delft, The Netherlands, Jun. 2012, pp. 127–128. DOI: 10.1145/2287076.2287096

[200] S. Lyberis*, G. Kalokerinos, M. Lygerakis, V. Papaefstathiou*, D. Tsaliagkos, M. Katevenis, D. Pnevmatikatos, and D. S. Nikolopoulos, "Formic: Cost-Efficient and Scalable Prototyping of Manycore Architectures," in *Proceedings of the 20th Annual International IEEE Symposium on Field-Programmable Custom Computing Machines **(FCCM)***, Toronto, Ontario, Canada, Apr. 2012, pp. 61–64. DOI: 10.1109/FCCM.2012.20

[199] G. Tzenakis, A. Papatriantafyllou*, J. Kesapides, P. Pratikakis, H. Vandierendonck, and D. S. Nikolopoulos, "Block-level Dynamic Dependence Analysis for Deterministic Task-Based Parallelism," in *Proceedings of the 17th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming **(PPOPP)***, New Orleans, LA, USA, Feb. 2012, pp. 301–302.

[198] H. Vandierendonck, G. Tzenakis, and D. S. Nikolopoulos, "A Unified Scheduler for Recursive and Task-Based Parallelism," in *Proceedings of the 20th International Conference on Parallel Architectures and Compilation Techniques, **(PACT)***, Galveston, TX, USA, Oct. 2011, pp. 1–11. DOI: 10.1109/PACT.2011.7

[197] A. Papagiannis* and D. S. Nikolopoulos, "Scalable Runtime Support for Data-Intensive Applications on the Single-Chip Cloud Computer," in *Proceedings of the 3rd Intel Many-core Applications Research Community Symposium **(MARC)***, Ettlingen, Germany, Jul. 2011, pp. 25–30.

[196] M. Alvanos*, G. Tzenakis, A. Bilas, and D. S. Nikolopoulos, "Design and Evaluation of a Task-based Parallel H.264 Video Encoder for Heterogeneous Processors," in *Proceedings of SAMOS XI: International Conference on Embedded Computer Systems: Architectures, Modeling and Simulation **(IC-SAMOS)***, Samos, Greece, Jul. 2011, pp. 217–224.

[195] D. Li*, D. S. Nikolopoulos, K. W. Cameron, B. R. D. Supinski, and M. Schulz, "Scalable Memory Registration for High-Performance Networks Using Helper Threads," in *Proceedings of the 8th ACM International Conference on Computing Frontiers **(CF)***, Article No. 38, Ischia, Italy: ACM, May 2011. DOI: 10.1145/2016604.2016652

[194] P. Tendulkar*, V. Papaefstathiou*, G. Nikiforos, S. Kavadias, D. S. Nikolopoulos, and M. Katevenis, "Fine-Grain OpenMP Runtime Support with Explicit Communication Hardware Primitives," in *Proceedings of the 2011 International Conference on Design, Automation & Test in Europe **(DATE)***, Grenoble, France, Mar. 2011, pp. 891–894. DOI: 10.1109/DATE.2011.5763299

[193] J. Yeom* and D. S. Nikolopoulos, "Strider: Runtime Support for Optimizing Strided Data Accesses on Multi-cores with Explicitly Managed Memories," in *Proceedings of ACM/IEEE Supercomputing'2010: International Conference on High Performance Computing, Networking, Storage, and Analysis **(SC)***, New Orleans, LA, USA: IEEE, Nov. 2010, pp. 1–11. DOI: 10.1109/SC.2010.52

[192] A. Papagiannis* and D. S. Nikolopoulos, "Rearchitecting MapReduce for Heterogeneous Multicore Processors with Explicitly Managed Memories," in *Proceedings of the 39th International Conference on Parallel Processing **(ICPP)***, San Diego, CA, USA, Sep. 2010, pp. 121–130. DOI: 10.1109/ICPP.2010.21

[191] K. Singh, M. Curtis-Maury, S. A. McKee, F. Blagojevic, D. S. Nikolopoulos, B. R. D. Supinski, and M. Schulz, "Comparing Scalability Prediction Strategies on an SMP of CMPs," in *Proceedings of the 16th International European Conference on Parallel and Distributed Computing **(EUROPAR)***, ser. Lecture Notes in Computer Science, vol. 6271, Ischia, Italy, Aug. 2010, pp. 143–155. DOI: 10.1007/978-3-642-15277-1_14

[190] S. Schneider*, H. Andrade, B. Gedik, K.-L. Wu, and D. S. Nikolopoulos, "Evaluation of Streaming Aggregation on Parallel Hardware Architectures," in *Proceedings of the Fourth ACM International Conference on Distributed Event-Based Systems **(DEBS)***, Cambridge, United Kingdom, Jul. 2010, pp. 248–257. DOI: 10.1145/1827418.1827467

[189] S. Kavadias, Manolis G. H. Katevenis, M. Zampetakis*, and D. S. Nikolopoulos, "On-chip Communication and Synchronization Mechanisms with Cache-Integrated Network Interfaces," in *Proceedings of the Seventh ACM International Conference on Computing Frontiers **(CF)***, Bertinoro, Italy, May 2010, pp. 217–226. DOI: 10.1145/1787275.1787328

[188] M. M. Rafique*, A. R. Butt, and D. S. Nikolopoulos, "Designing Accelerator-Based Distributed Systems for High Performance," in *Proceedings of the 10th IEEE/ACM International Symposium on Cluster, Cloud, and Grid Computing **(CCGRID)***, Melbourne, Australia, May 2010, pp. 165–174. DOI: 10.1109/CCGRID.2010.109

[187] D. Li*, D. S. Nikolopoulos, K. W. Cameron, B. R. D. Supinski, and M. Schulz, "Power-aware MPI Task Aggregation Prediction for High-End Computing Systems," in *Proceedings of the 24th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Atlanta, GA, USA: IEEE, Apr. 2010, pp. 1–12. DOI: 10.1109/IPDPS.2010.5470464

[186] D. Li*, B. R. D. Supinski, M. Schulz, K. W. Cameron, and D. S. Nikolopoulos, "Hybrid MPI/OpenMP Power-Aware Computing," in *Proceedings of the 24th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Atlanta, GA, USA: IEEE, Apr. 2010, pp. 1–12. DOI: 10.1109/IPDPS.2010.5470463

[185] G. Tzenakis*, K. Kapelonis, M. Alvanos*, K. Koukos, D. S. Nikolopoulos, and A. Bilas, "Tagged Procedure Calls (TPC): Efficient Runtime Support for Task-Based Parallelism on the Cell Processor," in *Proceedings of the Fifth International Conference on High-Performance Embedded Architectures and Compilers **(HIPEAC)***, ser. Lecture Notes in Computer Science, vol. 5952, Pisa, Italy, Jan. 2010, pp. 307–321. DOI: 10.1007/978-3-642-11515-8_23

[184] F. Blagojevic*, C. Iancu, K. A. Yelick, D. S. Nikolopoulos, B. Rose*, and M. Curtis-Maury*, "Scheduling Dynamic Parallelism on Accelerators," in *Proceedings of the Sixth ACM International Conference on Computing Frontiers **(CF)***, Ischia, Italy, May 2009, pp. 161–170. DOI: 10.1145/1531743.1531769

[183] M. M. Rafique*, B. Rose*, A. R. Butt, and D. S. Nikolopoulos, "CellMR: A Framework for Supporting MapReduce on Asymmetric Cell-based Clusters," in *Proceedings of the 23rd IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Rome, Italy: IEEE, May 2009, pp. 1–12. DOI: 10.1109/IPDPS.2009.5161062

[182] S. Schneider*, J. Yeom*, B. Rose*, J. C. Linford*, A. Sandu, and D. S. Nikolopoulos, "A Comparison of Programming Models for Multiprocessors with Explicitly Managed Memory Hierarchies," in *Proceedings of the 14th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming **(PPOPP)***, Raleigh, NC, USA, Feb. 2009, pp. 131–140. DOI: 10.1145/1504176.1504197

[181] M. Curtis-Maury*, A. Shah*, F. Blagojevic*, D. S. Nikolopoulos, B. R. de Supinski, and M. Schulz, "Prediction Models for Multi-dimensional Power-Performance Optimization on Many Cores," in *Pro-*

*ceedings of the 17th International Conference on Parallel Architectures and Compilation Techniques **(PACT)***, Toronto, Ontario, Canada, Oct. 2008, pp. 250–259. DOI: 10.1145/1454115.1454151

[180] F. Blagojevic*, M. Curtis-Maury*, J.-S. Yeom*, S. Schneider*, and D. S. Nikolopoulos, "Scheduling Asymmetric Parallelism on a PlayStation3 Cluster," in *Proceedings of the 8th IEEE International Symposium on Cluster Computing and the Grid **(CCGRID)***, Lyon, France, May 2008, pp. 146–153. DOI: 10.1109/CCGRID.2008.64

[179] M. M. Rafique*, A. R. Butt, and D. S. Nikolopoulos, "DMA-based Prefetching for I/O-Intensive Workloads on the Cell Architecture," in *Proceedings of the Fifth ACM International Conference on Computing Frontiers **(CF)***, Ischia, Italy, May 2008, pp. 23–32. DOI: 10.1145/1366230.1366236

[178] A. Aji, F. Blagojevic*, W. Feng, and D. S. Nikolopoulos, "Cell-Swat: Modeling and Scheduling Wavefront Computations on the Cell BE," in *Proceedings of the Fifth ACM International Conference on Computing Frontiers **(CF)***, Ischia, Italy, May 2008, pp. 13–22. DOI: 10.1145/1366230.1366235

[177] F. Blagojevic*, X. Feng, K. W. Cameron, and D. S. Nikolopoulos, "Modeling Multi-grain Parallelism on Heterogeneous Multicore Processors: A Case Study of the Cell BE," in *Proceedings of the Third International Conference on High-Performance Embedded Architectures and Compilers **(HIPEAC)***, ser. Lecture Notes in Computer Science, vol. 4917, Göteborg, Sweden, Jan. 2008, pp. 38–52. DOI: 10.1007/978-3-540-77560-7_4

[176] A. Chernikov, C. D. Antonopoulos, N. Chrisochoides, S. Schneider*, and D. S. Nikolopoulos, "Experience with Memory Allocators for Parallel Mesh Generation on Multi-core Architectures," in *Proceedings of the 10th International Conference on Numerical Grid Generation **(ISGG)***, Heraklion, Greece, Sep. 2007, pp. 159–168.

[175] F. Blagojevic*, A. Stamatakis, C. Antonopoulos, and D. S. Nikolopoulos, "RAxML-CELL: Parallel Phylogenetic Tree Construction on the Cell Broadband Engine," in *Proceedings of the 21st IEEE/ACM International Parallel and Distributed Processing Symposium **(IPDPS)***, Long Beach, CA, USA: IEEE, Mar. 2007, pp. 1–10. DOI: 10.1109/IPDPS.2007.370267

[174] F. Blagojevic*, D. S. Nikolopoulos, A. Stamatakis, and C. Antonopoulos, "Dynamic Multigrain Parallelization on the Cell Broadband Engine," in *Proceedings of the 12th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming **(PPOPP)***, **Best Paper Award**, San Jose, CA, USA, Mar. 2007, pp. 90–100. DOI: 10.1145/1229428.1229445

[173] M. Curtis-Maury*, C. D. Antonopoulos, and D. S. Nikolopoulos, "A Comparison of Online and Offline Strategies for Program Adaptation," in *Proceedings of the 45th Annual ACM Southeast Conference **(ACMSE)***, Winston-Salem, NC, USA, Mar. 2007, pp. 162–167. DOI: 10.1145/1233341.1233371

[172] M. Curtis-Maury*, C. D. Antonopoulos, and D. S. Nikolopoulos, "PACMAN: A Performance Counters Manager for Intel Hyperthreaded Processors," in *Proceedings of the 3rd International Conference on the Quantitative Evaluation of Systems **(QEST)***, Riverside, CA, USA, Sep. 2006, pp. 141–144. DOI: 10.1109/QEST.2006.41

[171] M. Curtis-Maury*, J. Dzierwa*, C. D. Antonopoulos, and D. S. Nikolopoulos, "Online Power-Performance Adaptation of Multithreaded Programs using Event-Based Prediction," in *Proceedings of the 20th ACM International Conference on Supercomputing **(ICS)***, Queensland, Australia, Jun. 2006, pp. 157–166. DOI: 10.1145/1183401.1183426

[170] C. Yue, R. T. Mills, A. Stathopoulos, and D. S. Nikolopoulos, "Runtime Support for Memory Adaptation in Scientific Workloads via Local Disk and Remote Memory," in *Proceedings of the 15th IEEE International Symposium on High Performance Distributed Computing **(HPDC)***, **Best Paper Award Nominee** (one of five papers), Paris, France, Jun. 2006, pp. 183–194. DOI: 10.1109/HPDC.2006.1652149

[169] S. Schneider*, C. D. Antonopoulos, and D. S. Nikolopoulos, "Scalable Locality-Conscious Multithreaded Memory Allocation," in *Proceedings of the 2006 ACM SIGPLAN International Symposium on Memory Management **(ISMM)***, Ottawa, Ontario, Canada, Jun. 2006, pp. 84–94. DOI: 10.1145/1133956.1133968

[168] X. Ding, D. S. Nikolopoulos, S. Jiang, and X. Zhang, "MESA: Reducing Cache Conflicts by Integrating Static and Run-Time Methods," in *Proceedings of the 2006 IEEE International Symposium on Performance Analysis of Systems and Software **(ISPASS)***, Austin, TX, USA, Mar. 2006, pp. 189–198. DOI: 10.1109/ISPASS.2006.1620803

[167] S. Schneider*, C. D. Antonopoulos, and D. S. Nikolopoulos, "Factory: An Object-Oriented Parallel Programming Substrate for Deep Multiprocessors," in *Proceedings of the 7th IEEE International Conference on High Performance Computing and Communications **(HPCC)***, Lecture Notes in Computer Science, vol. 3726, Sorrento, Italy, Sep. 2005, pp. 223–232. DOI: 10.1007/11557654_28

[166] M. Curtis-Maury*, T. Wang, C. D. Antonopoulos, and D. S. Nikolopoulos, "Integrating Multiple Forms of Multithreaded Execution on SMT Processors: A Quantitative Study with Scientific Workloads," in *Proceedings of the Second International Conference on the Quantitative Evaluation of Systems **(QEST)***, Torino, Italy, Sep. 2005, pp. 199–209. DOI: 10.1109/QEST.2005.16

[165] T. Wang, C. D. Antonopoulos, and D. S. Nikolopoulos, "smt-SPRINTS: Software Precomputation with Intelligent Streaming for Resource-Constrained SMTs," in *Proceedings of 11th 2005 International European Conference on Parallel and Distributed Computing **(EUROPAR)***, Lecture Notes in Computer Science, vol. 3648, Lisbon, Portugal, Aug. 2005, pp. 710–719. DOI: 10.1007/11549468_78

[164] C. D. Antonopoulos, X. Ding, A. Chernikov, F. Blagojevic*, D. S. Nikolopoulos, and N. Chrisochoides, "Multigrain Parallel Delaunay Mesh Generation: Challenges and Opportunities for Multithreaded Architecture," in *Proceedings of the 19th ACM International Conference on Supercomputing **(ICS)***, Cambridge, MA, USA, Jun. 2005, pp. 367–376. DOI: 10.1145/1088149.1088198

[163] R. L. McGregor*, C. D. Antonopoulos, and D. S. Nikolopoulos, "Scheduling Algorithms for Effective Thread Pairing on Hybrid Multiprocessors," in *Proceedings of the 19th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Denver, CO, USA: IEEE, Apr. 2005, 28a. DOI: 10.1109/IPDPS.2005.390

[162] C. D. Antonopoulos, D. S. Nikolopoulos, and T. S. Papatheodorou, "Realistic Workload Scheduling Policies for Taming the Memory Bandwidth Bottleneck of SMPs," in *Proceedings of the 11th International Conference on High Performance Computing **(HIPC)***, Lecture Notes in Computer Science, vol. 3296, Bangalore, India, Dec. 2004, pp. 286–296. DOI: 10.1007/978-3-540-30474-6_33

[161] R. T. Mills*, A. Stathopoulos, and D. S. Nikolopoulos, "Adapting to Memory Pressure from within Scientific Applications on Multiprogrammed COWs," in *Proceedings of the 18th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Santa Fe, NM, USA: IEEE, Apr. 2004. DOI: 10.1109/IPDPS.2004.1303002

[160] D. S. Nikolopoulos, "Code and Data Transformations for Improving Shared Cache Performance on SMT Processors," in *Proceedings of the 5th International Symposium on High Performance Computing **(ISHPC)***, Lecture Notes in Computer Science, **Best Paper Award**, vol. 2858, Tokyo-Odaiba, Japan, Oct. 2003, pp. 54–69. DOI: 10.1007/978-3-540-39707-6_5

[159] C. D. Antonopoulos, D. S. Nikolopoulos, and T. S. Papatheodorou, "Scheduling Algorithms with Bus Bandwidth Considerations for SMPs," in *Proceedings of the 32nd International Conference on Parallel Processing **(ICPP)***, Kaohsiung, Taiwan, Oct. 2003, pp. 547–554. DOI: 10.1109/ICPP.2003.1240622

[158] D. S. Nikolopoulos, "Malleable Memory Mapping: User-Level Control of Memory Bounds for Effective Program Adaptation," in *Proceedings of the 17th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Nice, France, Apr. 2003. DOI: 10.1109/IPDPS.2003.1213074

[157] D. S. Nikolopoulos and C. D. Polychronopoulos, "Adaptive Scheduling under Memory Pressure on Multiprogrammed Clusters," in *Proceedings of the Second IEEE/ACM International Symposium on Cluster Computer and the Grid **(CCGRID)***, **Best Paper Award**, Berlin, Germany, May 2002, pp. 22–29. DOI: 10.1109/CCGRID.2002.1017108

[156] D. S. Nikolopoulos, "Quantifying and Resolving Remote Memory Access Contention on Hardware DSM Multiprocessors," in *Proceedings of the 16th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, **Best Paper Award**, Fort Lauderdale, FL, USA, Apr. 2002. DOI: 10.1109/IPDPS.2002.1015503

[155] D. S. Nikolopoulos and C. D. Polychronopoulos, "Adaptive Scheduling under Memory Pressure on Multiprogrammed SMPs," in *Proceedings of the 16th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Fort Lauderdale, FL, USA, Apr. 2002. DOI: 10.1109/IPDPS.2002.1015481

[154] D. S. Nikolopoulos, E. Ayguadé, and C. D. Polychronopoulos, "Scaling Irregular Parallel Codes with Minimal Programming Effort," in *Proceedings of the ACM/IEEE Supercomputing'2001: High Performance Computing and Networking Conference **(SC)***, **Best Paper Award Nominee**, Denver, CO, USA: IEEE, Nov. 2001, p. 5. DOI: 0.1109/SC.2001.10013

[153] I. E. Venetis, D. S. Nikolopoulos, and T. S. Papatheodorou, "A Transparent Operating System Infrastructure for Embedding Adaptability to Thread-Based Programming Models," in *Proceedings of the 7th International European Conference on Parallel and Distributed Computing **(EUROPAR)***, Lecture Notes in Computer Science, vol. 2150, Manchester, United Kingdom, Aug. 2001, pp. 504–513. DOI: 10.1007/3-540-44681-8_75

[152] C. D. Antonopoulos, D. S. Nikolopoulos, and T. S. Papatheodorou, "Informing Algorithms for Efficient Scheduling of Synchronizing Threads on Multiprogrammed SMPs," in *Proceedings of the 30th International Conference on Parallel Processing **(ICPP)***, Valencia, Spain, Sep. 2001, pp. 123–130. DOI: 10.1109/ICPP.2001.952054

[151] D. S. Nikolopoulos, E. Ayguadé, J. Labarta, T. S. Papatheodorou, and C. D. Polychronopoulos, "The Trade-Off Between Implicit and Explicit Data Distribution in Shared-Memory Programming Paradigms," in *Proceedings of the 15th ACM International Conference on Supercomputing **(ICS)***, Sorrento, Italy, Jun. 2001, pp. 23–37. DOI: 10.1145/377792.377801

[150] D. S. Nikolopoulos, T. S. Papatheodorou, C. D. Polychronopoulos, J. Labarta, and E. Ayguadé, "Is Data Distribution Necessary in OpenMP?" In *Proceedings of ACM/IEEE Supercomputing'2000: High Performance Computing and Networking Conference **(SC)***, Article No. 47, ISBN: ISBN:0-7803-9802-5, **Best Technical Paper Award**, Dallas, TX, USA, Nov. 2000. DOI: 10.1109/SC.2000.10025

[149] D. S. Nikolopoulos, T. S. Papatheodorou, C. D. Polychronopoulos, J. Labarta, and E. Ayguadé, "Leveraging Transparent Data Distribution in OpenMP via User-Level Dynamic Page Migration," in *Proceedings of the 3rd International Symposium on High Performance Computing **(ISHPC)***, Lecture Notes in Computer Science, vol. 1940, Oct. 2000, pp. 415–427. DOI: 10.1007/3-540-39999-2_40

[148] D. S. Nikolopoulos, T. S. Papatheodorou, C. D. Polychronopoulos, J. Labarta, and E. Ayguadé, "User-Level Dynamic Page Migration for Multiprogrammed Shared-Memory Multiprocessors," in *Proceedings of the 29th International Conference on Parallel Processing **(ICPP)***, Toronto, Ontario, Canada, Aug. 2000, pp. 95–103. DOI: 10.1109/ICPP.2000.876083

[147] C. D. Antonopoulos, I. E. Venetis, D. S. Nikolopoulos, and T. S. Papatheodorou, "Efficient Dynamic Parallelism with OpenMP on Linux-Based SMPs," in *Proceedings of the 6th International Conference on Parallel and Distributed Processing Techniques and Applications **(PDPTA)***, vol. V, Las Vegas, NV, USA, Jul. 2000, pp. 2507–2514.

[146] D. S. Nikolopoulos, T. S. Papatheodorou, C. D. Polychronopoulos, J. Labarta, and E. Ayguadé, "A Case for User-Level Page Migration," in *Proceedings of the 14th ACM International Conference on Supercomputing **(ICS)***, Santa Fe, NM, USA, May 2000, pp. 119–130. DOI: 10.1145/335231.335243

[145] D. S. Nikolopoulos and T. S. Papatheodorou, "Fast Synchronization on Scalable Cache-Coherent Multiprocessors using Hybrid Primitives," in *Proceedings of the 14th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Cancun, Mexico, May 2000, pp. 711–719. DOI: 10.1109/IPDPS.2000.846056

[144] D. S. Nikolopoulos, C. D. Antonopoulos, I. E. Venetis, P. E. Hadjidoukas, E. D. Polychronopoulos, and T. S. Papatheodorou, "Achieving Multiprogramming Scalability of Parallel Programs on Intel SMP Platforms: Nanothreading in the Linux Kernel," in *Parallel Computing Fundamentals and Applications: Proceedings of the International Conference ParCo'99 **(PARCO)***, Delft, The Netherlands, Aug. 1999, pp. 623–630. DOI: 10.1142/9781848160170_0074

[143] E. D. Polychronopoulos, D. S. Nikolopoulos, T. S. Papatheodorou, X. Martorell, N. Navarro, and J. Labarta, "An Efficient Kernel-Level Scheduling Methodology for Multiprogrammed Shared Memory Multiprocessors," in *Proceedings of the 12th International Conference on Parallel and Distributed Computing Systems **(PDCS)***, Fort Lauderdale, FL, USA, Aug. 1999, pp. 148–155.

[142] D. S. Nikolopoulos and T. S. Papatheodorou, "System Software Support for Reducing Memory Latency on Distributed Shared-Memory Multiprocessors," in *Proceedings of the 7th Pan-Hellenic Conference on Informatics **(PCI)***, vol. 4, Ioannina, Greece, Aug. 1999, pp. 61–68.

[141] D. S. Nikolopoulos, E. D. Polychronopoulos, and T. S. Papatheodorou, "Fine-Grain and Multiprogramming-Conscious Nanothreading with the Solaris Operating System," in *Proceedings of the 5th International Conference on Parallel and Distributed Processing Techniques and Applications **(PDPTA)***, vol. IV, Las Vegas, NV, USA, Jul. 1999, pp. 1797–1803.

[140] D. S. Nikolopoulos and T. S. Papatheodorou, "A Quantitative Evaluation of Synchronization Algorithms and Disciplines on ccNUMA Systems: The Case of the SGI Origin2000," in *Proceedings of the*

*13th ACM International Conference on Supercomputing **(ICS)***, Rhodes, Greece, Jun. 1999, pp. 319–328. DOI: 10.1145/305138.305209

[139] D. S. Nikolopoulos, E. D. Polychronopoulos, and T. S. Papatheodorou, "Enhancing the Performance of Autoscheduling with Locality-Based Partitioning on Distributed Shared Memory Multiprocessors," in *Proceedings of 4th International European Conference on Parallel and Distributed Computing **(EUROPAR)***, Lecture Notes in Computer Science, vol. 1470, Southampton, United Kingdom, Aug. 1998, pp. 491–501. DOI: 10.1007/bfb0057892

[138] E. D. Polychronopoulos, X. Martorell, D. S. Nikolopoulos, T. S. Papatheodorou, J. Labarta, and N. Navarro, "Kernel-Level Scheduling for the Nano-Threads Programming Model," in *Proceedings of the 12th ACM International Conference on Supercomputing **(ICS)***, Melbourne, Australia, Jul. 1998, pp. 337–344. DOI: 10.1145/277830.277911

## Journal Articles

[137] X. Li*, S. Ghafouri, H. Vandierendonck, D. John, and D. S. Nikolopoulos, "SWEET: Serving Workload-balanced End-to-End Efficient and Tailored Edge Inference via Quantization and Partitioning," ***Frontiers in Complex Systems***, Apr. 2026, Section on Multi- and Cross-Disciplinary Complexity. DOI: 10.3389/fcpxs.2026.1801157

[136] X. Li*, J. Fan*, Q. Wang, D. Spatharakis, S. Ghafouri, H. Vandierendonck, D. John, B. Ji, A. R. Butt, and D. S. Nikolopoulos, "WISP: Waste- and Interference-Suppressed Distributed Speculative LLM Serving at the Edge via Dynamic Drafting and SLO-Aware Batching," ***Proceedings of the ACM on Measurement and Analysis of Computing Systems (ACM POMACS-SIGMETRICS)***, 2026. DOI: 10.1145/3805653

[135] A. Kumar, C. O'Mahoney, P. K. Werle, S. Shanker, D. S. Nikolopoulos, B. Ji, H. Vandierendonck, and D. John, "MARVEL: An End-to-End Framework for Generating Model-Class Aware Custom RISC-V Extensions for Lightweight AI," ***IEEE Open Journal of Circuits and Systems***, vol. 6, pp. 445–456, 2025. DOI: 10.1109/OJCAS.2025.3589132

[134] M. J. Abdel-Rhaman, E. Mazied*, H. Fahid, T. Kory, A. McKenzie, S. Midkiff, K. Cardoso, and D. S. Nikolopoulos, "On Robust Optimal Joint Deployment and Assignment of RAN Intelligent Controllers in O-RANs," ***IEEE Open Journal of the Communications Society***, vol. 5, pp. 2358–2376, Jan. 2024. DOI: 10.1109/OJCOMS.2024.3383607

[133] E. A. Mazied*, D. S. Nikolopoulos, Y. Hanafy, and S. F. Midkiff, "Auto-Scaling Edge Cloud for Network Slicing," ***Frontiers in High Performance Computing***, vol. 1, 2023. DOI: 10.3389/fhpcp.2023.1167162

[132] C.-H. Hong, M. Lee, H. Ahn, and D. S. Nikolopoulos, "gShare: A Centralized GPU Memory Management Framework to Enable GPU Memory Sharing for Containers," ***Future Generation Computer Systems***, vol. 130, pp. 181–192, May 2022. DOI: 10.1016/j.future.2021.12.016

[131] K. Dichev, D. di Sensi, I. Spence, K. W. Cameron, and D. S. Nikolopoulos, "Power Log'n'Roll: Power-Efficient Localized Rollback for MPI Applications using Logging Protocols," ***IEEE Transactions on Parallel and Distributed Systems***, vol. 33, no. 6, pp. 1276–1288, Jun. 2022. DOI: 10.1109/TPDS.2021.3107745

[130] U. Minhas, R. Woods, D. S. Nikolopoulos, and G. Karakonstantis, "Efficient, Dynamic Multi-task Execution on FPGA-based Computing Systems," ***IEEE Transactions on Parallel and Distributed Systems***, vol. 33, no. 3, pp. 710–722, Mar. 2022. DOI: 10.1109/TPDS.2021.3101153

[129] J. Lee, H. Vandierendonck, and D. S. Nikolopoulos, "Mixed Precision Kernel Recursive Least Squares," ***IEEE Transactions on Neural Networks and Learning Systems***, vol. 33, no. 3, pp. 1284–1298, Mar. 2022. DOI: 10.1109/TNNLS.2020.3041677

[128] L. Mukhanov, K. Tovletoglou*, H. Vandierendonck, D. S. Nikolopoulos, and G. Karakonstantis, "Revealing DRAM Operating Guardbands through Workload-Aware Error Predictive Modeling," ***IEEE Transactions on Computers***, vol. 70, no. 11, pp. 1976–1987, Nov. 2021. DOI: 10.1109/TC.2020.3033627

[127] N. Wang*, B. Varghese, M. Matthaiou, and D. S. Nikolopoulos, "ENORM: A Framework For Edge NOde Resource Management," ***IEEE Transactions on Services Computing***, vol. 13, no. 6, pp. 1086–1099, 2020. DOI: 10.1109/TSC.2017.2753775

[126] J. Sun, H. Vandierendonck, and D. S. Nikolopoulos, "Fast Load Balance Parallel Graph Analytics with an Automatic Data Structure Selection Algorithm," ***Future Generation Computer Systems***, vol. 112, pp. 612–623, Nov. 2020. DOI: 10.1016/j.future.2020.06.005

[125] J. Lee, G. T. Peterson, H. Vandierendonck, and D. S. Nikolopoulos, "AIR: Iterative Refinement Acceleration using Arbitrary Dynamic Precision," ***Parallel Computing***, vol. 97, Sep. 2020, Article: 102663. DOI: 10.1016/j.parco.2020.102663

[124] N. Wang*, B. Varghese, M. Matthaiou, and D. S. Nikolopoulos, "DYVERSE: DYnamic VERtical Scaling in Multi-tenant Edge Environments," ***Future Generation Computer Systems***, vol. 108, pp. 598–612, Jul. 2020. DOI: 10.1016/j.future.2020.02.043

[123] H. Vandierendonck and D. S. Nikolopoulos, "Hyperqueues: Design and Implementation of Deterministic Concurrent Queues," ***ACM Transactions on Parallel Computing***, vol. 6, no. 4, pp. 1–35, Nov. 2019, Article 23. DOI: 10.1145/3365660

[122] A. Hassan, H. Vandierendonck, and D. S. Nikolopoulos, "Fast and Energy-Efficient OLAP Data Management on Hybrid Main Memory Systems," ***IEEE Transactions on Computers***, vol. 63, no. 11, pp. 1597–1611, Nov. 2019. DOI: 10.1109/TC.2019.2919287

[121] I. Tsiokanos*, L. Mukhanov, D. S. Nikolopoulos, and G. Karakonstantis, "Significance-Driven Data Truncation for Preventing Timing Failures," ***IEEE Transactions on Device and Materials Reliability***, vol. 19, no. 1, pp. 25–36, Mar. 2019. DOI: 10.1109/TDMR.2019.2898949

[120] K. Tovletoglou*, L. Mukhanov, D. S. Nikolopoulos, and G. Karakonstantis, "Shimmer: Implementing a Heterogeneous-Reliability DRAM Framework on a Commodity Server," ***IEEE Computer Architecture Letters***, vol. 18, no. 1, pp. 26–29, Jun. 2019. DOI: 10.1109/LCA.2019.2893189

[119] U. Minhas, M. Russell, S. Kaloutsakis, P. Barber, R. Woods, G. Georgakoudis*, C. Gillan, and D. S. Nikolopoulos, "NanoStreams: A Microserver Architecture for Real-Time Analytics on Fast Data Streams," ***IEEE Transactions on MultiScale Computing Systems***, vol. 4, no. 3, pp. 396–409, 2018. DOI: 10.1109/TMSCS.2017.2764087

[118] J.-k. Lee, H. Vandierendonck, M. Arif, G. Peterson, and D. S. Nikolopoulos, "Energy-Efficient Iterative Refinement using Dynamic Precision," ***IEEE Journal on Emerging and Selected Topics in Circuits and Systems***, vol. 8, no. 4, pp. 722–735, Dec. 2018. DOI: 10.1109/JETCAS.2018.2850665

[117] C. Gillan, A. Novakovic, A. H. Marshall, M. Shyamsundar, and D. S. Nikolopoulos, "Expediting Assessments of Database Performance for Streams of Respiratory Parameters," ***Computers in Biology and Medicine***, vol. 100, no. 1, pp. 186–195, Sep. 2018. DOI: 10.1016/j.compbiomed.2018.05.028

[116] C. Reano, F. Silla, D. S. Nikolopoulos, and B. Varghese, "Intra-Node Memory Safe GPU Co-Scheduling," ***IEEE Transactions on Parallel and Distributed Systems***, vol. 29, no. 5, pp. 1089–1102, 2018. DOI: 10.1109/TPDS.2017.2784428

[115] P. Thoman, K. Dichev, T. Heller, R. Iakymchuk, X. Aguilar, K. Hasanov, P. Gschwandtner, P. Lemarinier, S. Makridis, H. Jordan, T. Fahringer, K. Katrinis, E. Laure, and D. S. Nikolopoulos, "A Taxonomy of Task-Based Parallel Programming Technologies for High-Performance Computing," ***The Journal of Supercomputing***, vol. 74, no. 4, pp. 1422–1434, Apr. 2018. DOI: 10.1007/s11227-018-2238-4

[114] C. Chalios*, G. Georgakoudis*, K. Tovletoglou*, G. Karakonstantis, H. Vandierendonck, and D. S. Nikolopoulos, "DARE: Data Access Aware Refresh via Spatio-Temporal Application Resilience on Commodity Servers," ***International Journal of High Performance Computing Applications***, vol. 32, no. 1, pp. 74–88, 2018. DOI: 10.1177/1094342017718612

[113] G. Karakonstantis, D. S. Nikolopoulos, D. Gizopoulos, P. Trancoso, Y. Sazeides, C. D. Antonopoulos, S. Venugopal, and S. Das, "Error-Resilient Server Ecosystems for Edge and Cloud Datacenters," ***IEEE Computer***, vol. 50, no. 12, pp. 78–82, Dec. 2017. DOI: 10.1109/MC.2017.4451208

[112] C. Black, O. P. Chevalier, S. A. Haughey, J. Balog, S. Stead, S. D. Pringler, M. V. Riina, F. Martucci, P. L. Acutis, M. Morris, D. S. Nikolopoulos, Z. Takats, and C. T. Elliott, "A Real-Time Metabolomic Profiling Approach to Detecting Fish Fraud using Rapid Evaporative Ionization Mass Spectrometry," ***Metabolomics***, vol. 13, no. 153, Nov. 2017. DOI: 10.1007/s11306-017-1291-y

[111] G. Georgakoudis*, H. Vandierendonck, P. Thoman, B. R. de Supinski, T. Fahringer, and D. S. Nikolopoulos, "SCALO: Scalability-Aware Parallelism Orchestration for Multi-Threaded Workloads," ***ACM Transactions on Architecture and Code Optimization***, vol. 14, no. 4, 54:1–54:25, 2017. DOI: 10.1145/3158643

[110] C.-H. Hong, I. T. A. Spence, and D. S. Nikolopoulos, "FairGV: Fair and Fast GPU Virtualization," ***IEEE Transactions on Parallel and Distributed Systems***, vol. 28, no. 12, pp. 3472–3485, Jun. 2017. DOI: 10.1109/TPDS.2017.2717908

[109] C.-H. Hong, I. Spence, and D. S. Nikolopoulos, "GPU Virtualization and Scheduling Methods: A Comprehensive Survey," ***ACM Computing Surveys***, vol. 50, no. 3, 35:1–35:37, Jan. 2017, Article No. 35. DOI: 10.1145/3068281

[108] L. Mukhanov, P. Petoumenos, Z. Wang, N. Parasyris, D. S. Nikolopoulos, B. R. de Supinski, and H. Leather, "ALEA: A Fine-Grain Energy Profiling Tool," ***ACM Transactions on Architecture and Code Optimization***, vol. 14, no. 1, pp. 1–25, Nov. 2017. DOI: 10.1145/3050436

[107] R. Montella, G. Giunta, G. Laccetti, M. Lapegna, C. Palmieri, C. Ferraro, V. Pelliccia, C.-H. Hong, I. Spence, and D. S. Nikolopoulos, "On the Virtualization of CUDA-based GPU Remoting on ARM and X86 Machines in the GVirtuS Framework," ***International Journal of Parallel Programming***, vol. 45, no. 5, pp. 1142–1163, 2017. DOI: 10.1007/s10766-016-0462-1

[106] V. Vassiliadis, C. Chalios*, K. Parasyris, C. D. Antonopoulos, S. Lalis, N. Bellas, H. Vandierendonck, and D. S. Nikolopoulos, "Exploiting Significance of Computations for Energy-Constrained Approximate Computing," ***International Journal of Parallel Programming***, vol. 44, no. 5, pp. 1078–1098, 2016. DOI: 10.1007/s10766-016-0409-6

[105] E. O'Neill*, J. McGlone, P. Kilpatrick, and D. S. Nikolopoulos, "Managed Acceleration for In-Memory Database Analytic Workloads," ***International Journal of Parallel, Emergent, and Distributed Systems***, vol. 32, no. 4, pp. 406–427, May 2017. DOI: 10.1080/17445760.2016.1170832

[104] G. Georgakoudis*, C. J. Gillan, A. Sayed, I. Spence, R. Faloon, and D. S. Nikolopoulos, "Methods and Metrics for Fair Server Assessment under Real-Time Financial Workloads," ***Concurrency and Computation: Practice and Experience***, vol. 28, no. 3, pp. 916–928, Mar. 2016. DOI: 10.1002/cpe.3704

[103] C. Chalios*, S. Catalán, E. S. Quintana-Orti, and D. S. Nikolopoulos, "Evaluating Asymmetric Multicore Systems-on-Chip and the Cost of Fault Tolerance using Iso-Metrics," ***IET Computers & Digital Techniques***, vol. 10, no. 2, pp. 85–92, Feb. 2016. DOI: 10.1049/iet-cdt.2015.0056

[102] H. Vandierendonck, A. Hassan*, and D. S. Nikolopoulos, "On The Energy-Efficiency of Byte-Addressable Non-Volatile Memory," ***IEEE Computer Architecture Letters***, vol. 14, no. 2, pp. 144–147, Jul. 2015. DOI: 10.1109/LCA.2014.2355195

[101] G. Georgakoudis*, C. J. Gillan, A. Sayed, I. Spence, R. Faloon, and D. S. Nikolopoulos, "Iso-Quality of Service: Fairly Ranking Servers for Real-Time Data Analytics," ***Parallel Processing Letters***, vol. 25, no. 3, 2015. DOI: 10.1142/s0129626415410042

[100] A. Khasymski* and D. S. Nikolopoulos, "Scalable Black-Box Prediction Models for Multi-Dimensional Adaptation on NUMA Multi-Cores," ***International Journal of Parallel, Emergent, and Distributed Systems***, vol. 30, no. 3, pp. 193–210, Apr. 2015. DOI: 10.1080/17445760.2014.895346

[99] D. S. Nikolopoulos, H. Vandierendonck, N. Bellas, C. D. Antonopoulos, S. Lalis, G. Karakonstantis, A. Burg, and U. Naumann, "Energy Efficiency through Significance-Based Computing," ***IEEE Computer***, vol. 47, no. 7, pp. 82–85, Jul. 2014. DOI: 10.1109/MC.2014.182

[98] I. Manousakis*, F. S. Zakkak, P. Pratikakis, and D. S. Nikolopoulos, "TProf: An Energy Profiler for Task-Parallel Programs," ***Sustainable Computing: Informatics and Systems***, vol. 5, pp. 1–13, Mar. 2015. DOI: 10.1016/j.suscom.2014.07.004

[97] P. Gschwandtner, C. Chalios*, D. S. Nikolopoulos, H. Vandierendonck, and T. Fahringer, "On the Potential of Significance-Driven Execution for Energy-Aware HPC," ***Computer Science – Research and Development***, vol. 30, no. 2, pp. 197–206, Feb. 2015. DOI: 10.1007/s00450-014-0265-9

[96] C. Symeonidou*, P. Pratikakis, D. S. Nikolopoulos, and A. Bilas, "Distributed Region-Based Memory Allocation and Synchronization," ***International Journal of High Performance Computing Applications***, vol. 28, no. 4, pp. 406–414, Nov. 2014. DOI: 10.1177/1094342014552863

[95] A. Papagiannis* and D. S. Nikolopoulos, "Hybrid Address Spaces: A Methodology for Implementing Scalable High-Level Programming Models on Non-Coherent Many-core Architectures," ***Journal of Systems and Software***, vol. 97, pp. 47–64, Nov. 2014. DOI: 10.1016/j.jss.2014.06.058

[94] S. Lyberis*, G. Kalokerinos, M. Lygerakis, I. Mavroidis, V. Papaefstathiou*, M. Katevenis, D. Pnevmatikatos, and D. S. Nikolopoulos, "FPGA Prototyping of Emerging Manycore Architectures for Par-

allel Programming Research using Formic Boards," ***Journal of Systems Architecture***, vol. 60, no. 6, pp. 481–493, Jun. 2014. DOI: 10.1016/j.sysarc.2014.03.002

[93] H. Vandierendonck, G. Tzenakis, and D. S. Nikolopoulos, "Analysis of Dependence Tracking Algorithms for Task Dataflow Execution," ***ACM Transactions on Architecture and Code Optimization***, vol. 10, no. 4, pp. 1–24, Dec. 2013, Article No. 61. DOI: 10.1145/2541228.2555316

[92] D. Li*, B. R. D. Supinski, M. Schulz, D. S. Nikolopoulos, and K. W. Cameron, "Strategies for Energy Efficient Resource Management of Hybrid Programming Models," ***IEEE Transactions on Parallel and Distributed Systems***, vol. 24, no. 1, pp. 144–157, Jan. 2013. DOI: 10.1109/TPDS.2012.95

[91] I. Manousakis* and D. S. Nikolopoulos, "EPC: A Power Instrumentation Controller for Embedded Applications," ***ACM SIGBED Review***, vol. 9, no. 2, pp. 28–32, Jun. 2012. DOI: 10.1145/2318836.2318841

[90] C.-Y. Su*, D. Li*, D. S. Nikolopoulos, M. Grove, K. W. Cameron, and B. R. D. Supinski, "Critical Path-Based Thread Placement for NUMA Systems," ***ACM SIGMETRICS Performance Evaluation Review***, vol. 40, no. 2, pp. 106–112, Sep. 2012. DOI: 10.1145/2381056.2381079

[89] S. Kavadias, M. Katevenis, and D. S. Nikolopoulos, "Cache-Integrated Network Interfaces: Flexible On-chip Communication and Synchronization for Large-scale CMPs," ***International Journal of Parallel Programming***, vol. 40, no. 6, pp. 583–604, Dec. 2012. DOI: 10.1007/s10766-011-0173-6

[88] M. M. Rafique*, A. R. Butt, and D. S. Nikolopoulos, "A Capabilities-Aware Framework for Using Computational Accelerators in Data-Intensive Computing," ***Journal of Parallel and Distributed Computing***, vol. 71, no. 2, pp. 185–197, Feb. 2011. DOI: 10.1016/j.jpdc.2010.09.004

[87] R. Ferrer, P. Bellens, J. Yeom*, S. Schneider*, K. Koukos, M. Alvanos*, V. Beltran, M. González, X. Martorell, R. M. Badia, D. S. Nikolopoulos, A. Bilas, and E. Ayguadé, "Parallel Programming Models for Heterogeneous Multi-Core Architectures," ***IEEE Micro***, vol. 30, no. 5, pp. 42–53, Oct. 2010.

[86] M. Katevenis, V. Papaefstathiou*, S. Kavadias, D. Pnevmatikatos, F. Silla, and D. S. Nikolopoulos, "Explicit Communication and Synchronization in SARC," ***IEEE Micro***, vol. 30, no. 5, pp. 30–41, Oct. 2010. DOI: 10.1109/MM.2010.77

[85] S. Schneider*, J. Yeom*, and D. S. Nikolopoulos, "Programming Multiprocessors with Explicitly Managed Memory Hierarchies," ***IEEE Computer***, vol. 42, no. 12, pp. 28–34, Dec. 2009. DOI: 10.1109/MC.2009.407

[84] C. D. Antonopoulos, F. Blagojevic*, A. Chernikov, N. Chrisochoides, and D. S. Nikolopoulos, "A Multi-grain Delaunay Mesh Generation Method for Multicore SMT-based Architectures," ***Journal of Parallel and Distributed Computing***, vol. 69, no. 7, pp. 589–600, Jul. 2009. DOI: 10.1016/j.jpdc.2009.03.009

[83] C. D. Antonopoulos, F. Blagojevic*, A. Chernikov, D. S. Nikolopoulos, and N. Chrisochoides, "Algorithm, Software, and Hardware Optimizations for Delaunay Mesh Generation on Simultaneous Multithreaded Architectures," ***Journal of Parallel and Distributed Computing***, vol. 69, no. 7, pp. 601–612, Jul. 2009. DOI: 10.1016/j.jpdc.2009.03.005

[82] M. M. Rafique*, B. Rose*, A. R. Butt, and D. S. Nikolopoulos, "Supporting MapReduce on Asymmetric Multi-core Clusters," ***ACM SIGOPS Operating Systems Review***, vol. 43, no. 2, pp. 25–34, 2009. DOI: 10.1145/1531793.1531800

[81] M. Curtis-Maury*, F. Blagojevic*, C. D. Antonopoulos, and D. S. Nikolopoulos, "Prediction-Based Power-Performance Adaptation of Multithreaded Scientific Codes," ***IEEE Transactions on Parallel and Distributed Systems***, vol. 19, no. 10, pp. 1396–1410, Oct. 2008. DOI: 10.1109/TPDS.2007.70804

[80] F. Blagojevic*, D. S. Nikolopoulos, A. Stamatakis, C. D. Antonopoulos, and M. Curtis-Maury*, "Runtime Scheduling of Dynamic Parallelism on Accelerator-Based Multi-core Systems," ***Parallel Computing***, vol. 33, no. 10–11, pp. 700–719, Nov. 2007. DOI: 10.1016/j.parco.2007.09.004

[79] A. Stamatakis, F. Blagojevic*, D. S. Nikolopoulos, and C. D. Antonopoulos, "Exploring new Search Algorithms and Hardware for Phylogenetics: RAxML meets the IBM Cell," ***Journal of VLSI Signal Processing***, vol. 48, no. 3, pp. 271–286, Aug. 2007. DOI: 10.1007/s11265-007-0067-4

[78] R. T. Mills, C. Yue, A. Stathopoulos, and D. S. Nikolopoulos, "Runtime and Programming Support for Memory Adaptation in Scientific Applications via Local Disk and Remote Memory," ***Journal of Grid Computing***, vol. 5, no. 2, pp. 213–234, Jun. 2007. DOI: 10.1007/s10723-007-9075-7

[77] D. S. Nikolopoulos, "Dynamic Tiling for Effective Use of Shared Caches on Multithreaded Processors," ***International Journal of High Performance Computing and Networking***, vol. 2, no. 1, pp. 22–35, 2004. DOI: 10.1504/ijhpcn.2004.009265

[76] D. S. Nikolopoulos, "Quantifying Contention and Balancing Memory Load on Hardware DSM Multiprocessors," ***Journal of Parallel and Distributed Computing***, vol. 63, no. 9, pp. 866–886, Sep. 2003. DOI: 10.1016/s0743-7315(03)00105-9

[75] D. S. Nikolopoulos and C. D. Polychronopoulos, "Adaptive Scheduling under Memory Constraints on Non-Dedicated Computational Farms," ***Future Generation Computer Systems***, vol. 19, no. 4, pp. 505–519, May 2003. DOI: 10.1016/s0167-739x(03)00031-1

[74] D. S. Nikolopoulos, E. Artiaga, E. Ayguadé, and J. Labarta, "Scaling Non-Regular Shared-Memory Codes by Reusing Custom Loop Schedules," ***Scientific Programming***, vol. 11, no. 2, pp. 143–158, Apr. 2003. DOI: 10.1155/2003/379739

[73] D. S. Nikolopoulos, E. Ayguadé, and C. D. Polychronopoulos, "Runtime vs. Manual Data Distribution for Architecture-Agnostic Shared-Memory Programming Models," ***International Journal of Parallel Programming***, vol. 30, no. 4, pp. 225–254, Aug. 2002. DOI: 10.1023/a:1019899812171

[72] D. S. Nikolopoulos, T. S. Papatheodorou, C. D. Polychronopoulos, J. Labarta, and E. Ayguadé, "Scheduler-Activated Dynamic Page Migration for Multiprogrammed DSM Multiprocessors," ***Journal of Parallel and Distributed Computing***, vol. 62, no. 6, pp. 1069–1103, Jun. 2002. DOI: 10.1006/jpdc.2001.1817

[71] D. S. Nikolopoulos, E. Artiaga, E. Ayguadé, and J. Labarta, "Exploiting Memory Affinity in OpenMP through Schedule Reuse," ***ACM Computer Architecture News***, vol. 29, no. 5, pp. 49–55, Dec. 2001. DOI: 10.1145/563647.563657

[70] D. S. Nikolopoulos and T. S. Papatheodorou, "The Architectural and Operating System Implications on the Performance of Synchronization on ccNUMA Multiprocessors," ***International Journal of Parallel Programming***, vol. 29, no. 3, pp. 249–282, Jun. 2001. DOI: 10.1023/a:1011168003859

[69] D. S. Nikolopoulos, T. S. Papatheodorou, C. D. Polychronopoulos, J. Labarta, and E. Ayguadé, "A Transparent Runtime Data Distribution Engine for OpenMP," ***Scientific Programming***, vol. 8, no. 3, pp. 143–162, Dec. 2000. DOI: 10.1155/2000/417570

## Papers in Workshops with Archival Proceedings

[68] F. Amin*, S. Afroz, M. Moghadampanah*, and D. S. Nikolopoulos, "Serving Masked Diffusion LLMs: Characterization and Design Principles from Real Hardware," in *Proceedings of the Workshop on AI on HPC: Performance Engineering, Challenges and Opportunities **(AI on HPC)**, in conjunction with SC26*, Chicago, IL, Nov. 2026.

[67] S. Yao, K. T. Lowe, and D. S. Nikolopoulos, "DART: ML-Accelerated Surrogate for Real-Time CFD Digital Twins," in *Proceedings of the 8th Annual Workshop on Extreme-Scale Experiment-in-the-Loop Computing **(XLOOP)**, in conjunction with SC26*, Chicago, IL, Nov. 2026.

[66] X. Li*, S. Ghafouri, J. Fan*, H. Vandierendonck, and D. S. Nikolopoulos, "ConfigSpec: Profiling-Based Configuration Selection for Distributed Edge–Cloud Speculative LLM Serving," in *Proceedings of the 4th International Workshop on Testing Distributed Internet of Things Systems (**TDIS**)*, Edinburgh, UK, 2026. DOI: 10.1145/3802513.3803483

[65] M. Moghadampanah*, A. R. Shahmirzadi, F. Amin*, and D. S. Nikolopoulos, "Modality Inflation: Energy Characterization and Optimization Opportunities for MLLM Inference," in *Proceedings of the 21st International Workshop on Automatic Performance Tuning (**iWAPT**)*, New Orleans, LA, 2026. DOI: 10.1109/IPDPSW71298.2026.00035

[64] Y. Li*, S. Yao, J. Mobin, T. Zhan, M. M. Rafique, D. S. Nikolopoulos, K. Sundararajah, and A. R. Butt, "Memory Tiering in the Python Virtual Machine," in *Proceedings of the 17th SPLASH Workshop on Virtual Machines and Language Implementations **(VMIL)***, Singapore, Oct. 2025, pp. 33–44. DOI: 10.1145/3759548.3763372

[63] M. Jackson*, B. Ji, and D. S. Nikolopoulos, "FrameFeedback: A Closed-Loop Control System for Dynamic Offloading Real-Time Edge Inference," in *2024 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, 2024, pp. 584–591. DOI: 10.1109/IPDPSW63119.2024.00116

[62] E. A. Mazied*, D. S. Nikolopoulos, and S. Midkiff, "Fine-Grain Slicing of Edge Cloud Servers for Radio Workloads," in *IEEE Workshop on Hot Topics in System Infrastructure **(HotInfra'23)**, in conjunction with ACM FCRC 2023*, Orlando, FL, 2023.

[61] R. A. Istrate*, C. I. Malossi, C. Bekas, and D. S. Nikolopoulos, "Incremental Training of Deep Convolutional Neural Networks," in *Proceedings of the International Workshop on Automatic Selection, Configuration, and Composition of Machine Learning Algorithms **(AutoML)***, Held in conjunction with **ECML/PKDD**, 2017, pp. 41–48.

[60] K. Chen, B. Varghese, P. Kilpatrick, and D. S. Nikolopoulos, "Power Modeling for Heterogeneous Cloud-Edge Datacentres," in *Proceedings of the Minisymposium on Edge Computing held in conjunction with the Parallel Computing 2017 Conference **(PARCO)***, ser. Advances in Parallel Computing, 2017, pp. 804–813. DOI: 10.3233/978-1-61499-843-3-804

[59] B. Varghese, N. Wang*, J. Li, and D. S. Nikolopoulos, "Edge-as-a-Service: Towards Distributed Cloud Architectures," in *Proceedings of the Minisymposium on Edge Computing held in conjunction with the Parallel Computing 2017 Conference **(PARCO)***, 2017, pp. 784–793. DOI: 10.3233/978-1-61499-843-3-784

[58] K. Tovletoglou* et al., "An Energy-Efficient and Error-Resilient Server Ecosystem Exceeding Conservative Scaling Limits," in *Proceedings of the First Workshop on Energy-Efficient Servers for Cloud and Edge Computing **(ENESCE)***, In conjunction with the HiPEAC'17 Conference, Stockholm, Sweden, Jan. 2017.

[57] Y. Wu, C. Gillan, A. Novakovic, K. Tovletoglou, G. Tzenakis, H. Vandierendonck, G. Karakonstantis, D. S. Nikolopoulos, S. Barbhuiya*, and U. Minhas, "Heterogeneous Servers based on Programmable Cores and Dataflow Engines," in *Proceedings of the First Workshop on Energy-Efficient Servers for Cloud and Edge Computing **(ENESCE)***, In conjunction with the HiPEAC'17 Conference, Stockholm, Sweden, Jan. 2017.

[56] M. Arif, H. Vandierendonck, D. S. Nikolopoulos, and B. R. de Supinski, "A Scalable and Composable Map-Reduce System," in *Proceedings of the Third Workshop on Advances in Software and Hardware for Big Data to Knowledge Discovery (**ASH**)*, 2016 IEEE International Conference on Big Data **(Big Data)**, Washington, D.C., Dec. 2016, pp. 2233–2242. DOI: 10.1109/BIGDATA.2016.7840854

[55] D. Playfair, A. Trehan, and D. S. Nikolopoulos, "Big Data Availability: Selective Partial Checkpointing for In-Memory Database Queries," in *Proceedings of the Fourth Workshop on Scalable Cloud Data Management **(SCDM)***, 2016 IEEE International Conference on Big Data **(Big Data)**, Washington, D.C., Dec. 2016, pp. 2785–2794. DOI: 10.1109/BIGDATA.2016.7840926

[54] K. Dichev and D. S. Nikolopoulos, "TwinPCG: Dual Thread Redundancy with Forward Recovery for Preconditioned Conjugate Gradient Methods," in *Second International Workshop on Fault Tolerant Systems, **IEEE FTS***, Held in conjunction with the IEEE International Conference on Cluster Computing, **CLUSTER**, Taipei, Taiwan, Sep. 2016, pp. 506–514. DOI: 10.1109/CLUSTER.2016.99

[53] P. Harvey, K. Bakanov, I. Spence, and D. S. Nikolopoulos, "A Scalable Runtime for FPGA-Based Heterogeneous Exascale Hardware," in *Proceedings of the Sixth International Workshop on Runtime and Operating Systems for Supercomputers **(ROSS)***, Article No. 7, Kyoto, Japan, Jun. 2016. DOI: 10.1145/2931088.2931090

[52] H. Vandierendonck, K. Murphy, M. Arif, J. Sun*, and D. S. Nikolopoulos, "Operator and Workflow Optimization for High-Performance Analytics," in *Proceedings of the First International Workshop on Multi-Engine Data Analytics **(MEDAL)***, ser. EDBT/ICDT Workshops, Bordeaux, France, Mar. 2016.

[51] C. Trehan, H. Vandierendonck, G. Karakonstantis, and D. S. Nikolopoulos, "Energy Optimization of Parallel Workloads on Unreliable Hardware," in *Proceedings of the Second Workshop on Approximate Computing **(WAPCO)***, In conjunction with the HiPEAC 2016 Conference., Prague, Czech Republic, Jan. 2016.

[50] J. I. Aliaga, S. Catalán, C. Chalios*, D. S. Nikolopoulos, and E. S. Quintana-Orti, "Performance and Fault Tolerance of Preconditioned Iterative Solvers on Low-Power ARM Architectures," in *Workshop on Energy and Resilience in Parallel Programming **(ERPP)***, Held in conjunction with the **ParCo2015** Conference, Edinburgh, United Kingdom, Sep. 2015. DOI: 10.3233/978-1-61499-621-7-711

[49] F. Alessi, P. Thoman, G. Georgakoudis*, T. Fahringer, and D. S. Nikolopoulos, "Application-Level Energy Awareness for OpenMP," in *Proceedings of the 11th International Workshop on OpenMP*

***(IWOMP)***, ser. Lecture Notes in Computer Science, vol. 9342, Aachen, Germany, Oct. 2015, pp. 219–232. DOI: 10.1007/978-3-319-24595-9_16

[48] A. Hassan*, H. Vandierendonck, and D. S. Nikolopoulos, "Energy-Efficient In-Memory Data Stores on Hybrid Memory Hierarchies," in *Proceedings of the 11th International Workshop on Data Management on New Hardware **(DAMON)**, in conjunction with ACM SIGMOD/PODS 2015*, Article No. 1, Melbourne, Australia, Jun. 2015. DOI: 10.1145/2771937.2771940

[47] C. Chalios*, E. S. Quintana-Orti, and D. S. Nikolopoulos, "Evaluating Asymmetric Multi-core Systems-on-Chip using Iso-Metrics," *CoRR*, Jan. 2015, Presented at the *First HiPEAC Workshop on Energy Efficiency with Heterogeneous Computing **(EEHCO)***. DOI: 10.13140/RG.2.1.3042.5120 [Online]. Available: http://arxiv.org/abs/1503.08104

[46] C. J. Gillan, D. S. Nikolopoulos, G. Georgakoudis*, R. Faloon, G. Tzenakis, and I. Spence, "On the Viability of Microservers for Financial Analytics," in *Proceedings of the Seventh ACM SIGHPC Workshop on High Performance Computational Finance **(WHPCF)***, New Orleans, LA, USA, Nov. 2014, pp. 29–36. DOI: 10.1109/WHPCF.2014.11

[45] G. Georgakoudis*, D. S. Nikolopoulos, and S. Lalis, "Fast Dynamic Binary Rewriting to Support Thread Migration in Shared-ISA Asymmetric Multicores," in *Proceedings of the First International Workshop on Code Optimization for Multi and Many Cores **(COSMIC)***, Article No. 4, **Best Paper Award**, Shenzhen, China: ACM, Feb. 2013. DOI: 10.1145/2446920.2446924

[44] P. Pratikakis, H. Vandierendonck, and D. S. Nikolopoulos, "A Programming Model for Deterministic Task-based Parallelism," in *Proceedings of the 2011 ACM SIGPLAN Workshop on Memory Systems Performance and Correctness **(MSPC)***, San Jose, CA, USA, Jun. 2011, pp. 7–12. DOI: 10.1145/1988915.1988918

[43] H. Vandierendonck, P. Pratikakis, and D. S. Nikolopoulos, "Parallel Programming of General-Purpose Programs using Task-Based Programming Models," in *Proceedings of the 3rd USENIX Workshop on Hot Topics on Parallelism **(HotPar)***, Berkeley, CA, USA, May 2011, pp. 1–6.

[42] J. Yeom* and D. S. Nikolopoulos, "A Runtime Framework for Optimizing Multi-dimensional Array Accesses on Multi-core Processors," Presented in *First International Workshop on Programming Models for Emerging Architectures **(PMEA)***, held in conjunction with the *18th International Conference on Parallel Architectures and Compilation Techniques **(PACT)***, Raleigh, NC, USA, Sep. 2009. DOI: 10.13140/2.1.4304.2086

[41] D. S. Nikolopoulos, G. Back, J. Tripathi*, and M. Curtis-Maury*, "VT-ASOS: Holistic System Software Customization for Many Cores," in *Proceedings of the Workshop on the NSF Next Generation Software Program **(NSFNGS)***, Held in conjunction with the *22nd IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Miami, FL, USA: IEEE, Apr. 2008, pp. 1–5. DOI: 10.1109/IPDPS.2008.4536390

[40] M. Curtis-Maury*, K. Singh, S. A. McKee, F. Blagojevic*, D. S. Nikolopoulos, B. R. de Supinski, and M. Schulz, "Identifying Energy-Efficient Concurrency Levels using Machine Learning," in *Proceedings of the First International Workshop on Green Computing **(GREENCOM)***, Held in conjunction with the *2007 IEEE International Conference on Cluster Computing **(CLUSTER)***, Austin, TX, USA, Sep. 2007, pp. 488–495. DOI: 10.1109/CLUSTR.2007.4629274

[39] D. S. Nikolopoulos and K. W. Cameron, "Synthesizing Parallel Programming Models for Asymmetric Multi-Core Systems," Presented in *11th Workshop on High Performance Embedded Computing **(HPEC)***, Lexington, MA, USA, Sep. 2007. DOI: 10.13140/2.1.3779.9200

[38] G. Back and D. S. Nikolopoulos, "Application-Specific Customization on Many-Core Platforms: The VT-ASOS Framework," Presented in *Second Workshop on Software and Tools for Multi-Core Systems **(STMCS)***, held in conjunction with the *2007 International Symposium on Code Generation and Optimization (CGO)*, San Jose, CA, USA, Mar. 2007.

[37] M. Curtis-Maury*, J. Dzierwa*, C. D. Antonopoulos, and D. S. Nikolopoulos, "Online Strategies for High-Performance Power-Aware Thread Execution on Emerging Multiprocessors," in *Proceedings of the Second Workshop on High-Performance Power-Aware Computing **(HPPAC)***, Held in conjunction with the *20th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Rhodes, Greece, Apr. 2006. DOI: 10.1109/IPDPS.2006.1639598

[36] M. Curtis-Maury*, D. S. Nikolopoulos, and C. D. Antonopoulos, "Dynamic Program Stirring on Multiple Cores: How Hardware Performance Monitors Can Help Regulate Performance, Power, and

Temperature Simultaneously," Presented in *Second Workshop on Functionality of Hardware Performance Monitors **(FHPM)***, held in conjunction with the *39th IEEE/ACM International Symposium on Microarchitecture **(MICRO)***, Orlando, FL, USA, Dec. 2006.

[35] C. D. Antonopoulos and D. S. Nikolopoulos, "Using Hardware Counters for Continuous Online Optimization: Lessons and Challenges," Presented in *First Workshop on Hardware Performance Monitor Design and Functionality **(FHPM)***, held in conjunction with the *11th International Symposium on High Performance Computer Architecture **(HPCA)***, San Francisco, CA, USA, Feb. 2005. DOI: 10.13140/2.1.4173.1362

[34] M. Curtis-Maury*, X. Ding, C. D. Antonopoulos, and D. S. Nikolopoulos, "An Evaluation of OpenMP on Current and Emerging Multithreaded Processors," in *Proceedings of the First International Workshop on OpenMP **(IWOMP)***, Lecture Notes in Computer Science, **Best Paper Award**, vol. 4315, Eugene, OR, USA, Jun. 2005, pp. 133–142. DOI: 10.1007/978-3-540-68555-5_11

[33] B. Lawson, C. Yue, E. Smirni, and D. S. Nikolopoulos, "Power-Aware Resource Allocation via Online Simulation with Multiple-Queue Backfilling," in *Proceedings of the 7th Workshop on Performability Modeling of Computer and Communication Systems **(PMCCS)***, Held in conjunction with the *Second International Conference on the Quantitative Evaluation of Systems **(QEST)***, Torino, Italy, Sep. 2005. DOI: 10.13140/2.1.2026.8324

[32] T. Wang, F. Blagojevic*, and D. S. Nikolopoulos, "Runtime Support for Integrating Precomputation and Thread-Level Parallelism on Simultaneous Multithreaded Processors," in *Proceedings of the 7th ACM SIGPLAN Workshop on Languages, Compilers and Runtime Support for Scalable Systems **(LCR)***, ser. ACM International Conference Proceedings Series, vol. 81, Houston, TX, USA: ACM, Oct. 2004, pp. 1–12. DOI: 10.1145/1066650.1066667

[31] W. Ko, M. Yankelevsky, D. S. Nikolopoulos, and C. D. Polychronopoulos, "Effective Cross-Platform Multilevel Parallelization via Dynamic Adaptive Execution," in *Proceedings of the 7th International Workshop on High-Level Programming Models and Supportive Environments **(HIPS)***, Held in conjunction with the *16th IEEE International Parallel and Distributed Processing Symposium **(IPDPS)***, Fort Lauderdale, FL, USA, Apr. 2002. DOI: 10.1109/IPDPS.2002.1016495

[30] D. S. Nikolopoulos and E. Ayguadé, "A Study of Transparent Implicit Data Distribution Mechanisms for OpenMP using the SPEC Benchmarks," in *Proceedings of the Second International Workshop on OpenMP Applications and Tools **(WOMPAT)***, Lecture Notes in Computer Science, vol. 2104, West Lafayette, IN, USA, Jul. 2001, pp. 115–129. DOI: 10.1007/3-540-44587-0_11

[29] D. Craig, F. Breg, S. Carroll, D. S. Nikolopoulos, and C. D. Polychronopoulos, "Improving Java Server Performance with Interruptlets," in *Proceedings of the First International Conference on Computational Science **(ICCS)***, Lecture Notes in Computer Science, vol. 2073, San Francisco, CA, USA, May 2001, pp. 223–232. DOI: 10.1007/3-540-45545-0_31

[28] D. S. Nikolopoulos, T. S. Papatheodorou, C. D. Polychronopoulos, J. Labarta, and E. Ayguadé, "UPMlib: A Runtime System for Tuning the Memory Performance of OpenMP Programs on Distributed Shared Memory Multiprocessors," in *Proceedings of the 5th ACM SIGPLAN Workshop on Languages, Compilers, and Runtime Systems for Scalable Computers **(LCR)***, Lecture Notes in Computer Science, vol. 1915, Rochester, NY, USA, May 2000, pp. 85–99. DOI: 10.1007/3-540-40889-4_7

[27] X. Martorell, J. Corbalan, D. S. Nikolopoulos, N. Navarro, E. D. Polychronopoulos, and T. S. Papatheodorou, "A Tool to Schedule Parallel Applications on Multiprocessors: The NANOS CPU Manager," in *Proceedings of the 6th International Workshop on Job Scheduling Strategies for Parallel Processing **(JSSPP)***, Lecture Notes in Computer Science, vol. 1911, Cancun, Mexico, May 2000, pp. 87–112. DOI: 10.1007/3-540-39997-6_7

[26] D. S. Nikolopoulos, E. D. Polychronopoulos, and T. S. Papatheodorou, "Efficient Runtime Thread Management for the Nano-Threads Programming Model," in *Proceedings of the Second International Workshop on Runtime Systems for Parallel Programming **(RTSPP)***, Lecture Notes in Computer Science, vol. 1388, Orlando, FL, USA, Apr. 1998, pp. 183–194. DOI: 10.1007/3-540-64359-1_688

## Book Chapters

[25] D. Moyer* and D. S. Nikolopoulos, "Punching Holes in the Cloud: Direct Communication between Serverless Functions," in *Lecture Notes on Data Engineering and Communication Technologies*, R. Kr-

ishnamurthi, A. Kumar, S. S. Gill, and R. Buyya, Eds., vol. 162, Springer Nature, 2023, pp. 15–41. DOI: 10.1007/978-3-031-26633-1_2

[24] B. Varghese, N. Wang*, D. S. Nikolopoulos, and R. Buyya, "Feasibility of Fog Computing," in ***Handbook of Integration of Cloud Computing, Cyber-Physical Systems and Internet of Things***, R. Ranjan, K. Mitra, P. P. Jayaraman, L. Wang, and A. Zomaya, Eds., Springer, Apr. 2019. DOI: 10.1007/978-3-030-43795-4_5

[23] S. Barbhuiya*, Z. Papazachos, P. Kilpatrick, and D. S. Nikolopoulos, "LS-ADT: Lightweight and Scalable Anomaly Detection for Cloud Datacentres," in ***Communications in Computer and Information Science: Cloud Computing and Services Science***, D. F. Markus Helfert Víctor Méndez Muñoz, Ed., vol. 581, ISBN: 978-3-319-29581-7, Springer International Publishing, Switzerland, 2016, pp. 135–152. DOI: 10.1007/978-3-319-29582-4_8

[22] A. Khasymski*, M. M. Rafique, A. R. Butt, S. S. Vazhkudai, and D. S. Nikolopoulos, "Realizing Accelerated Cost-Effective Distributed RAID," in ***Handbook on Data Centers***, S. U. Khan and A. Y. Zomaya, Eds., ISBN: 978-1-4939-2091-4, Springer, 2015, pp. 729–753. DOI: 10.1007/978-1-4939-2092-1_25

[21] M. Curtis-Maury and D. S. Nikolopoulos, "Energy-efficient Multithreading through Runtime Adaptation," in ***The Green Computing Book: Tackling Energy Efficiency at Large Scale***, W. Feng, Ed., ISBN: 978-1439819876, Chapman & Hall/CRC Computational Science, Jul. 2014, pp. 115–148.

[20] M. M. Rafique, A. R. Butt, and D. S. Nikolopoulos, "Programming and Managing Resources on Accelerator-Enabled Clusters," in ***Programming Multi-core and Many-core Computing Systems***, ser. Wiley Series on Parallel and Distributed Computing, S. Pllana and F. Xhafa, Eds., ISBN: 978-0-470-93690-0, Wiley-Blackwell, Mar. 2014. DOI: 10.1002/9781119332015.ch20

[19] D. Li*, D. S. Nikolopoulos, and K. W. Cameron, "Modeling and Algorithms for Scalable and Energy Efficient Execution on Multicore Systems," in ***Scalable Computing: Theory and Practice***, S. U. Khan, L. Wang, and A. Y. Zomaya, Eds., ISBN: 978-1-118-16265-1, Wiley–IEEE Computer Society Press, Mar. 2013, pp. 157–184.

[18] H. Vandierendonck, D. S. Nikolopoulos, and P. Pratikakis, "Parallel Programming," in ***Encyclopedia of Software Engineering***, Taylor & Francis, Feb. 2013, ch. 62, pp. 1–14.

[17] C. D. Antonopoulos, D. S. Nikolopoulos, and T. S. Papatheodorou, "Scheduling Algorithms with Bus Bandwidth Considerations for SMPs," in ***High Performance Computing: Paradigm and Infrastructure***, L. T. Yang and M. Guo, Eds., ISBN: 978-0-471-65471-1, John Wiley & Sons, Dec. 2005, ch. 16, pp. 313–332. DOI: 10.1002/0471732710.ch16

## Posters

[16] M. Moghadampanah*, X. Li*, C.-H. Hong, and D. S. Nikolopoulos, "CoEncode: Reclaiming Idle Encoder GPUs in Multi-Tenant Multimodal LLM Serving," in *Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis* ***(SC)***, Poster presentation, Chicago, IL, Nov. 2026.

[15] M. Moghadampanah*, X. Li*, F. Amin*, and D. S. Nikolopoulos, "E-dLLM: Energy-Efficient Diffusion LLM Inference via Phase-Aware DVFS," in *Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis* ***(SC)***, Poster presentation, Chicago, IL, Nov. 2026.

[14] A. R. Shahmirzadi, S. Shabihi, F. Huang, and D. S. Nikolopoulos, "HydraCache: Long-Context Prefill Parallelization via Distributed Cache Blending," in *Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis* ***(SC)***, Poster presentation, St. Louis, MO, Nov. 2025.

[13] F. Amin*, K. Gharami, and D. S. N. **(SC)**, "DiffPro: Joint Timestep and Layer-Wise Precision Optimization for Efficient Diffusion Inference," in *Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis*, Poster presentation, St. Louis, MO, Nov. 2025.

[12] F. Amin*, K. Gharami, and D. S. Nikolopoulos, "Divide, Conquer, and Denoise: Hybrid Parallel Diffusion with Memory-Aware Coarse-to-Fine Inference," in *Proceedings of the International Conference on*

*High Performance Computing, Networking, Storage and Analysis* ***(SC)***, Poster presentation, St. Louis, MO, Nov. 2025.

[11] M. Moghadampanah*, A. R. Shahmirzadi, and D. S. Nikolopoulos, "Energy-Efficient Multimodal LLM Inference: Stage-Level Characterization and Input-Aware Controls," in *Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis* ***(SC)***, Poster presentation, St. Louis, MO, Nov. 2025.

[10] K. Tovletoglou*, G. Karakonstantis, and D. S. Nikolopoulos, "Implementation of a Heterogeneous Reliability Framework," in *Proceedings of the 27th International Conference on Parallel Architectures and Compilation Technique* ***(PACT)***, Poster Session, Limassol, Cyprus, Nov. 2018.

[9] J.-K. Lee, D. S. Nikolopoulos, and H. Vandierendonck, "Energy-Efficient Transprecision Techniques for Iterative Refinement," in *Proceedings of Supercomputing: International Conference on High-Performance Computing, Networking, Storage and Analysis* ***(SC)***, Poster Session, Denver, CO, USA, Nov. 2017.

[8] K. Dichev and D. S. Nikolopoulos, "TwinCG: Dual Thread Redundancy with Forward Recovery for Preconditioned Conjugate Gradient Methods," in *IEEE International Conference on Cluster Computing* ***CLUSTER***, Taipei, Taiwan, Sep. 2016, pp. 162–163. DOI: 10.1109/CLUSTER.2016.86

[7] S. Barbhuiya*, Y. Wu, K. Murphy, H. Vandierendonck, G. Karakonstantis, and D. S. Nikolopoulos, "Accelerating Data Center Applications with Reconfigurable DataFlow Engines," in *Proceedings of the Second International Workshop on Heterogeneous High Performance Reconfigurable Computing* ***(H2RC'16)***, Held in conjunction with the SC'16 International Conference on High Performance Computing, Networking, Storage and Analysis, Salt Lake City, UT, Nov. 2016.

[6] V. Vassiliadis, K. Parasyris, C. D. Antonopoulos, N. Bellas, S. Lalis, U. Naumann, J. Riehme, J. Deussen, and D. S. Nikolopoulos, "SCoRPiO: Significance Based Computing for Reliability and Power Optimization," in *Proceedings of the 2016 International Symposium on Code Generation and Optimization* ***(CGO)***, Barcelona, Spain, Mar. 2016.

[5] F. S. Zakkak*, D. Chassapis*, P. Pratikakis, D. S. Nikolopoulos, and A. Bilas, *C Source Level Transformations & Optimizations for Task-Based Parallelism*, Student Poster Session, 2011 International Symposium on Code Generation and Optimization ***(CGO)***, Chamonix, France, Apr. 2011.

[4] D. Li*, K. W. Cameron, D. S. Nikolopoulos, M. Schulz, and B. R. D. Supinski, *Model-Based Hybrid MPI/OpenMP Power-Aware Computing*, Poster Session, ACM/IEEE Supercomputing'2009: High-performance Computing, Networking, Storage and Analysis ***(SC)***, Portland, OR, USA, Nov. 2009.

[3] M. M. Rafique*, A. R. Butt, and D. S. Nikolopoulos, "Supporting I/O-intensive Workloads on the Cell Architecture," Presented in *6th USENIX Conference on File and Storage Systems* ***(FAST)***, San Jose, CA, USA, Feb. 2008. DOI: 10.13140/2.1.1682.7689

[2] F. Blagojevic*, C. Iancu, K. A. Yelick, D. S. Nikolopoulos, B. Rose*, and M. Curtis-Maury*, *Scheduling Dynamic Parallelism on the Cell BE*, Proceedings of the 15th Meeting of the IBM HPC Systems Scientific Computing User Group ***(SCICOMP)***, Barcelona, Spain, May 2009.

[1] M. Yankelevsky, W. Ko, D. S. Nikolopoulos, and C. D. Polychronopoulos, *Using Machine Descriptors to Select Parallelization Models and Strategies on Hierarchical Systems*, Poster Session, ACM/IEEE Supercomputing'2001: High Performance Networking and Computing Conference ***(SC)***, Denver, CO, USA, Nov. 2001.

## Non-Refereed Publications

## Edited Proceedings Volumes

[80] S. Kuppannagari, M. Koyuturk, A. Goldman, and D. S. Nikolopoulos, Eds., *HPDC '26: Proceedings of the 35th International Symposium on High-Performance Parallel and Distributed Computing*, Cleveland, USA: Association for Computing Machinery, 2026, ISBN: 9798400726408.

[79] K. A. Gallivan, E. Gallopoulos, D. S. Nikolopoulos, and R. Beivide, "ICS'23: Proceedings of the 37th ACM International Conference on Supercomputing," in *Proceedings of the 37th ACM International Conference on Supercomputing (ICS'23)*, ISBN 979-8-4007-0056-9, Association for Computing Machinery, Association for Computing Machinery, Jun. 2023.

[78] L. Rauchwerger, K. W. Cameron, D. S. Nikolopoulos, and D. N. Pnevmatikatos, "ICS'22: 2022 ACM International Conference on Supercomputing, Proceedings," in *Proceedings of the 36th ACM International Conference on Supercomputing (ICS'22)*, ISBN 978-1-4503-9281-5, Association for Computing Machinery, Association for Computing Machinery, Jun. 2022.

[77] C. Reaño and D. S. Nikolopoulos, "Deployment and Use of Accelerators, International Workshop held in conjunction with the International Conference on Parallel Processing," in *ACM International Conference Proceedings Series, VI*, Aug. 2022.

[76] C. Reaño and D. S. Nikolopoulos, "Deployment and Use of Accelerators, International Workshop held in conjunction with the International Conference on Parallel Processing," in *ACM International Conference Proceedings Series, VI*, Aug. 2021.

[75] D. S. Nikolopoulos and B. R. D. Supinski, "Proceedings: 2018 IEEE International Conference on Cluster Computing **(CLUSTER)**," in *Proceedings of the 2018 IEEE International Conference on Cluster Computing (CLUSTER)*, Oct. 2018.

[74] D. S. Nikolopoulos and B. R. D. Supinski, "Proceedings: 2018 IEEE International Symposium on Performance Analysis of Systems and Software **(ISPASS)**," in *Proceedings of the 2018 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, May 2018.

[73] C. D. Antonopoulos and D. S. Nikolopoulos, "MiniSymposium on Edge Computing," in *Parallel Computing is Everywhere, Proceedings of the International Conference on Parallel Computing, ParCo 2017, 12-15 September 2017, Bologna, Italy*, 2017, p. 783. DOI: 10.3233/978-1-61499-843-3-783

[72] K. W. Cameron, T. Gamblin, and D. S. Nikolopoulos, Eds., ***VarSys Introduction.*** *First IEEE International Workshop on Variability in Parallel and Distributed Systems.* Held in conjunction with the 2016 IEEE International Parallel and Distributed Processing Symposium, Chicago, IL, 2016, p. 1068.

[71] D. S. Nikolopoulos and C. D. Antonopoulos, Eds., *Mini-Symposium on Energy and Resilience in Parallel Programming, Parallel Computing on the Road to Exascale, Proceedings of the International Conference on Parallel Computing,* ***ParCo 2015***, Edinburgh, Scotland, Sep. 2015, pp. 709–709. DOI: 10.3233/978-1-61499-621-7-709

[70] J. L. Núñez-Yáñez, J. M. Moreno, and D. S. Nikolopoulos, "Guest Editorial: Special Issue on Energy Efficient Computing with Adaptive and Heterogeneous Architectures," ***IET Computers & Digital Techniques***, vol. 9, no. 1, pp. 1–2, 2015. DOI: 10.1049/iet-cdt.2014.0215

[69] K. W. Cameron, A. Hoisie, D. K. Lowenthal, D. S. Nikolopoulos, S. Yalamanchili, L. Carrington, and J. B. Manzano, "Energy Efficient Supercomputing Workshop (E2SC'15) Proceedings," in *Proceedings of the 3rd International Workshop on Energy Efficient Supercomputing* ***(E2SC'15)***, ISBN 978-1-4503-3994-0, Austin, Texas, USA: IEEE, Nov. 2015.

[68] K. W. Cameron, A. Hoisie, D. K. Lowenthal, D. S. Nikolopoulos, S. Yalamanchili, and A. Marquez, "Energy Efficient Supercomputing Workshop (E2SC'14) Proceedings," in *Proceedings of the 2nd International Workshop on Energy Efficient Supercomputing* ***(E2SC'14)***, ISBN 978-1-4799-7036-0, New Orleans, Louisiana, USA: IEEE, Nov. 2014.

[67] B. R. de Supinski, B. Krammer, K. Fürlinger, J. Labarta, and D. S. Nikolopoulos, "Topic 1: Support Tools and Environments - (Introduction)," in ***Euro-Par 2013 Parallel Processing*** *- 19th International Conference, Aachen, Germany, August 26-30, 2013. Proceedings*, F. Wolf, B. Mohr, and D. an Mey, Eds., ser. Lecture Notes in Computer Science, vol. 8097, 2013, p. 3. DOI: 10.1007/978-3-642-40047-6_2

[66] A. Ramírez, D. S. Nikolopoulos, D. R. Kaeli, and S. Matsuoka, "Topic 16: GPU and Accelerators Computing," in *Euro-Par 2012 Parallel Processing - 18th International Conference,* ***Euro-Par 2012****, Rhodes Island, Greece, August 27-31, 2012. Proceedings*, ser. Lecture Notes in Computer Science, vol. 7484, Aug. 2012, pp. 857–858. DOI: 10.1007/978-3-642-32820-6_84

[65] Y. Cotronis, A. Danalis, D. S. Nikolopoulos, and J. Dongarra, Eds., ***Recent Advances in the Message Passing Interface - 18th European MPI Users' Group Meeting, EuroMPI 2011.*** *Proceedings*, Lecture Notes in Computer Science, vol. 6960, Santorini, Greece: Springer, Sep. 18–21, 2011.

## Invited Keynote Talks

[64] D. S. Nikolopoulos, "AVA: Accelerator Virtualization Anywhere," in *Proceedings of the 13th Workshop on Virtualization in High-Performance Cloud Computing* ***(VHPC)***, Jun. 2018.

[63] D. S. Nikolopoulos, "Energy Efficient Computing using Computational Significance Abstractions," in *UK-China Workshop on Shaping Low Carbon Energy Future*, Keynote talk, Belfast, UK, Aug. 2016.

[62] D. S. Nikolopoulos, "Using Computational Significance and Resilience in System Software Stacks," in *First Workshop on Energy-Aware High Performance Computing*, Keynote talk. Held in conjunction with the 2016 International Supercomputing Conference **(ISC)**, Frankfurt, Germany, Jun. 2016.

[61] D. S. Nikolopoulos, "Why Energy-Efficient High Performance Computing is Harder than Energy-Efficient Embedded Computing," in *Workshop on Power & Energy-Aware High Performance Computing on Emerging Technology*, Keynote talk. Held in conjunction with the 2015 International Supercomputing Conference **(ISC)**, Frankfurt, Germany, Jul. 2015.

[60] D. S. Nikolopoulos, "Programming the Energy Efficiency of HPC Systems," in *Proceedings of the 4th International Conference on Energy-Aware High Performance Computing*, **Keynote Talk**, Dresden, Germany, Sep. 2013.

[59] D. S. Nikolopoulos, "Connecting the Dots between Parallel Programming and Energy," in *Proceedings of the 21st Euromicro International Conference on Parallel, Distributed and Network-Based Processing*, **Keynote Talk**, Belfast, Northern Ireland, UK, Mar. 2013.

[58] D. S. Nikolopoulos, "To Program or not to Program the Memory Hierarchy?" In *Fourth Workshop on Programmability Issues for Heterogeneous Multicores **(MULTIPROG)***, **Keynote Address**, Heraklion, Greece, Jan. 2011.

[57] D. S. Nikolopoulos, "Facing the Challenges of Multicore Processor Technologies using Autonomic System Software," in *Proceedings of the 7th Workshop on Parallel and Distributed Scientific and Engineering Computing with Applications **(PDSEC)***, Held in conjunction with the 20th IEEE International Parallel and Distributed Processing Symposium (IPDPS), 1pp., **Keynote Address**, Rhodes, Greece, Apr. 2006. DOI: 10.1109/IPDPS.2006.1639604

## Invited Papers

[56] D. S. Nikolopoulos, "New Approaches to Memory Reliability Management for Big Data Workloads," in *Abstracts of the 2018 SIAM Conference on Parallel Processing for Scientific Computing **(SIAM PP)***, Tokyo, Japan, Mar. 2018.

[55] D. S. Nikolopoulos, "NanoStreams: A Hardware and Software Stack for Real-Time Analytics on Fast Data Streams," *HiPEAC Info*, no. 38, pp. 15–16, May 2014.

[54] D. S. Nikolopoulos, "Reconciling Explicit with Implicit Parallelism," in *Abstracts of the 2012 SIAM Conference on Parallel Processing for Scientific Computing **(SIAM PP)***, Savannah, Georgia, USA, Feb. 2012.

[53] D. S. Nikolopoulos, "Region-Based Memory Management for Task Dataflow Models," in *Joint ENCORE & PEPPHER Workshop on Programmability and Portability for Emerging Architectures **(EPoPPEA)***, Held in conjunction with the 7th International Conference on High Performance and Embedded Architectures and Compilers (HIPEAC), Paris, France, Jan. 2012.

[52] M. Duranton et al., "Computing Systems: Research Challenges Ahead: The HiPEAC Vision 2011/2012," HiPEAC, Tech. Rep., 2011.

[51] J. Kesapides, D. S. Nikolopoulos, and A. Bilas, "ADAM: Automatic Dependence Analysis & Monitoring," in *Proceedings of the Sixth International Summer School on Advanced Computer Architecture and Compilation for Embedded Systems **(ACACES)***, 4pp, Barcelona, Spain, Jul. 2010.

[50] D. S. Nikolopoulos, "Green Building Blocks: Software Stacks for Energy-efficient Clusters and Data Centres," *ERCIM News*, no. 79, pp. 29–30, Oct. 2009.

[49] D. S. Nikolopoulos and M. Katevenis, "Processors: The Challenge of Cooperation," ***Economist***, no. 71, Dec. 2009, In Greek.

[48] M. Alvanos*, G. Tzenakis*, D. S. Nikolopoulos, and A. Bilas, "Parallelization and Performance of an H.264 Video Encoder on the Cell B.E.," in *Proceedings of the Fifth International Summer School on Advanced Computer Architecture and Compilation for Embedded Systems **(ACACES)***, 4pp, Barcelona, Spain, Jul. 2009.

[47] D. S. Nikolopoulos, "Set-top Supercomputing: Scalable Software for Scientific Simulations on Game Consoles," *ERCIM News*, no. 74, pp. 44–45, Jul. 2008.

[46] D. S. Nikolopoulos, "Unified Scheduling of Polymorphic Parallelism on the Cell Processor," in *Abstracts of the 2008 SIAM Conference on Parallel Processing for Scientific Computing, Mini-workshop on the Cell Processor **(SIAM PP)***, Atlanta, GA, USA, Mar. 2008.

[45] D. S. Nikolopoulos, "System Software Challenges and Opportunities on Asymmetric Multi-core Processors," in *Proceedings of the 2007 **Fall Creek Falls Conference** – Panel on Key Challenges Presented by Next Generation Hardware Systems*, Nashville, TN, USA, Sep. 2007.

[44] F. Blagojevic* and D. S. Nikolopoulos, *Exploring Programming Models and Optimizations for the Cell Broadband Engine using RAxML*, 2006 Virginia Tech **High-End Computing Challenge**, 14 pp. **Awarded Best Entry for Performance**, Sep. 2006.

[43] C. D. Antonopoulos, N. Chrisochoides, and D. S. Nikolopoulos, "2-D Parallel Constrained Delaunay Mesh Generation: A Multigrain Approach on Deep Multiprocessors," in *Abstracts of the Workshop in Programming Models for HPCS Ultra-Scale Applications **(PMUA)***, Held in conjunction with the 19th ACM International Conference on Supercomputing (ICS). Invited presentation, Cambridge, MA, USA, Jun. 2005.

[42] C. D. Antonopoulos, N. Chrisochoides, and D. S. Nikolopoulos, "Exploiting Simultaneous Multithreading for Parallel Mesh Generation: A Multigrain Approach on Deep Multiprocessors," in *13th International Meshing Roundtable **(IMR)**, Poster Session*, Williamsburg, VA, USA, Sep. 2004.

[41] D. S. Nikolopoulos and A. Stathopoulos, "Application Awareness in Adaptation Middleware: Balancing Transparency with Performance and Adaptivity," in *Abstracts of the 2004 SIAM Conference on Parallel Processing for Scientific Computing **(SIAM PP)**, Mini-workshop on Adaptivity in Parallel and Distributed Computing through Interoperating Systems and Applications*, 1 pp, San Francisco, CA, USA, Feb. 2004.

[40] D. S. Nikolopoulos, "Programming Environments for Multigrain Parallelization," in *Abstracts of the 2003 **EURESCO** Conference on Advanced Environments and Tools for High-Performance Computing*, Invited presentation, Albufeira, Portugal, Jun. 2003.

## Technical Reports

[39] Y. Li, A. Kumar, P. Vishnu, D. S. Nikolopoulos, B. Ji, H. Vandierendonck, and D. John, *Zta-q: An open-source risc-v platform for accurate quantized cnn inference*, 2026. arXiv: 2610.01867 [cs.AR]. [Online]. Available: https://arxiv.org/abs/2610.01867

[38] S. Yao, D. S. Nikolopoulos, and A. R. Butt, *XMPIaaS: Towards Cloud Native MPI via Cooperative Process Migration*, 2026. arXiv: 2609.16531 [cs.DC]. [Online]. Available: https://arxiv.org/abs/2609.16531

[37] F. Amin*, S. Afroz, M. Moghadampanah*, and D. S. Nikolopoulos, *Serving Masked Diffusion LLMs: Characterization and Design Principles from Real Hardware*, 2026. arXiv: 2608.23807 [cs.AI]. [Online]. Available: https://arxiv.org/abs/2608.23807

[36] K. Crumpacker* and D. S. Nikolopoulos, *LLM-Guided Runtime Parameter Optimization for Energy-Efficient Model Inference*, 2026. arXiv: 2604.27032 [cs.SE]. [Online]. Available: https://arxiv.org/abs/2604.27032

[35] S. Heidari* and D. S. Nikolopoulos, *FACT: Compositional Kernel Synthesis with a Three-Stage Agentic Workflow*, 2026. arXiv: 2604.26666 [cs.DC]. [Online]. Available: https://arxiv.org/abs/2604.26666

[34] X. Li*, J. Fan*, Q. Wang, D. Spatharakis, S. Ghafouri, H. Vandierendonck, D. John, B. Ji, A. R. Butt, and D. S. Nikolopoulos, *WISP: Waste- and Interference-Suppressed Distributed Speculative LLM Serving at the Edge via Dynamic Drafting and SLO-Aware Batching*, 2026. arXiv: 2601.11652 [cs.DC]. [Online]. Available: https://arxiv.org/abs/2601.11652

[33] J. Fan*, Y. Zhang, X. Li*, and D. S. Nikolopoulos, *Taming the Memory Footprint Crisis: System Design for Production Diffusion LLM Serving*, 2025. arXiv: 2512.17077 [cs.DC]. [Online]. Available: https://arxiv.org/abs/2512.17077

[32] A. K. M, C. O'Mahoney, P. K. Werle, S. Shanker, D. S. Nikolopoulos, B. Ji, H. Vandierendonck, and D. John, *MARVEL: An End-to-End Framework for Generating Model-Class Aware Custom RISC-V Extensions for Lightweight AI*, 2025. arXiv: 2508.01800 [cs.AR]. [Online]. Available: https://arxiv.org/abs/2508.01800

[31] A. Rahman*, V. Cvetkovic*, K. Reece*, A. Walters*, Y. Hassan, A. Tummeti*, B. Torres*, D. Cooney*, M. Ellis, and D. S. Nikolopoulos, *MARCO: Multi-Agent Code Optimization with Real-Time Knowledge Integration for High-Performance Computing*, 2025. arXiv: 2505.03906 [cs.DC]. [Online]. Available: https://arxiv.org/abs/2505.03906

[30] S. Ghafouri, M. Fayyaz, X. Li*, D. John, B. Ji, D. Nikolopoulos, and H. Vandierendonck, *Polymorph: Energy-Efficient Multi-Label Classification for Video Streams on Embedded Devices*, 2025. arXiv: 2507.14959 [cs.CV]. [Online]. Available: https://arxiv.org/abs/2507.14959

[29] X. Li*, S. Ghafouri, B. Ji, H. Vandierendonck, D. John, and D. S. Nikolopoulos, *QPART: Adaptive Model Quantization and Dynamic Workload Balancing for Accuracy-aware Edge Inference*, 2025. arXiv: 2506.23934 [cs.DC]. [Online]. Available: https://arxiv.org/abs/2506.23934

[28] X. Li*, D. Spatharakis, S. Ghafouri, J. Fan*, H. Vandierendonck, D. John, B. Ji, and D. Nikolopoulos, *SLED: A Speculative LLM Decoding Framework for Efficient Edge Serving*, 2025. arXiv: 2506.09397 [cs.DC]. [Online]. Available: https://arxiv.org/abs/2506.09397

[27] J. Fan*, Y. Zhang, X. Li*, and D. S. Nikolopoulos, *Parallel CPU-GPU Execution for LLM Inference on Constrained GPUs*, 2025. arXiv: 2506.03296 [cs.DC]. [Online]. Available: https://arxiv.org/abs/2506.03296

[26] F. Amin*, S. Afroz, K. Gharami, M. Moghadampanah*, and D. S. Nikolopoulos, *DiffPro: Joint Timestep and Layer-Wise Precision Optimization for Efficient Diffusion Inference*, 2025. arXiv: 2511.11446 [cs.LG]. [Online]. Available: https://arxiv.org/abs/2511.11446

[25] S. Chen, J. Huang, S. Yang, J. Liu, H. Li, D. Nikolopoulos, J. Ryu, J. Baek, K. Shin, and D. Li, *Tuning Fast Memory Size based on Modeling of Page Migration for Tiered Memory*, 2024. arXiv: 2410.00328 [cs.PF]. [Online]. Available: https://arxiv.org/abs/2410.00328

[24] K. H. I. Arif*, J. Yoon, D. S. Nikolopoulos, H. Vandierendonck, D. John, and B. Ji, *HiRED: Attention-Guided Token Dropping for Efficient Inference of High-Resolution Vision-Language Models*, 2024. arXiv: 2408.10945 [cs.CV]. [Online]. Available: https://arxiv.org/abs/2408.10945

[23] M. Lee, S. Seong, M. Kang, J. Lee, G.-J. Na, I.-G. Chun, D. Nikolopoulos, and C.-H. Hong, *ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments*, 2024. arXiv: 2409.14447 [cs.DC]. [Online]. Available: https://arxiv.org/abs/2409.14447

[22] Y. Li*, K. Assogba, A. Tripathy, M. Arif, M. M. Rafique, A. R. Butt, and D. Nikolopoulos, *Towards Persistent Memory based Stateful Serverless Computing for Big Data Applications*, 2023. arXiv: 2309.01662 [cs.DC]. [Online]. Available: https://arxiv.org/abs/2309.01662

[21] N. Wang*, M. Matthaiou, D. S. Nikolopoulos, and B. Varghese, *DYVERSE: DYnamic VERtical Scaling in Multi-tenant Edge Environments*, 2020. arXiv: 1810.04608 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1810.04608

[20] K. Dichev and D. S. Nikolopoulos, *Implementing Efficient Message Logging Protocols as MPI Application Extensions*, 2019. arXiv: 1905.03184 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1905.03184

[19] S. Barbhuiya*, Z. Papazachos, P. Kilpatrick, and D. S. Nikolopoulos, *RADS: Real-time Anomaly Detection System for Cloud Data Centres*, 2018. arXiv: 1811.04481 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1811.04481

[18] J. Sun*, H. Vandierendonck, and D. S. Nikolopoulos, *VEBO: A Vertex- and Edge-Balanced Ordering Heuristic to Load Balance Parallel Graph Processing*, 2018. arXiv: 1806.06576 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1806.06576

[17] K. Dichev, K. Cameron, and D. Nikolopoulos, *Energy-efficient localised rollback after failures via data flow analysis*, 2018. arXiv: 1806.01611 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1806.01611

[16] R. Istrate*, A. C. I. Malossi, C. Bekas, and D. Nikolopoulos, *Incremental Training of Deep Convolutional Neural Networks*, 2018. arXiv: 1803.10232 [cs.LG]. [Online]. Available: https://arxiv.org/abs/1803.10232

[15] K. Chen, B. Varghese, P. Kilpatrick, and D. S. Nikolopoulos, *Power Modelling for Heterogeneous Cloud-Edge Data Centers*, 2017. arXiv: 1710.10325 [cs.PF]. [Online]. Available: https://arxiv.org/abs/1710.10325

[14] K. Dichev, H. Jordan, K. Tovletoglou*, T. Heller, D. S. Nikolopoulos, G. Karakonstantis, and C. Gillan, *Dependency-Aware Rollback and Checkpoint-Restart for Distributed Task-Based Runtimes*, 2017. arXiv: 1705.10208 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1705.10208

[13] B. Varghese, N. Wang*, D. S. Nikolopoulos, and R. Buyya, *Feasibility of Fog Computing*, 2017. arXiv: 1701.05451 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1701.05451

[12] A. Labrineas, P. Pratikakis, D. S. Nikolopoulos, and A. Bilas, *BDDT-SCC: A Task-parallel Runtime for Non Cache-Coherent Multicores*, 2016. arXiv: 1606.04288 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1606.04288

[11] K. Dichev and D. S. Nikolopoulos, *TwinCG: Dual Thread Redundancy with Forward Recovery for Conjugate Gradient Methods*, 2016. arXiv: 1605.04580 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1605.04580

[10] G. Georgakoudis*, C. J. Gillan, A. Sayed, I. Spence, R. Faloon, and D. S. Nikolopoulos, *Methods and Metrics for Fair Server Assessment under Real-Time Financial Workloads*, 2014. arXiv: 1501.00048 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1501.00048

[9] C. Trehan, H. Vandierendonck, G. Karakonstantis, and D. S. Nikolopoulos, *Energy Optimization of Memory Intensive Parallel workloads*, 2016. arXiv: 1605.04111 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1605.04111

[8] L. Mukhanov, D. S. Nikolopoulos, and B. R. de Supinski, *ALEA: Fine-grain Energy Profiling with Basic Block Sampling*, 2016. arXiv: 1504.00825 [cs.PF]. [Online]. Available: https://arxiv.org/abs/1504.00825

[7] C. Chalios, D. S. Nikolopoulos, and E. S. Quintana-Orti, *Evaluating Asymmetric Multicore Systems-on-Chip using Iso-Metrics*, 2015. arXiv: 1503.08104 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1503.08104

[6] G. Georgakoudis*, C. J. Gillan, A. Sayed, I. Spence, R. Faloon, and D. S. Nikolopoulos, *Iso-Quality of Service: Fairly Ranking Servers for Real-Time Data Analytics*, 2015. arXiv: 1501.03481 [cs.DC]. [Online]. Available: https://arxiv.org/abs/1501.03481

[5] V. Vassiliadis, K. Parasyris, C. Chalios*, C. D. Antonopoulos, S. Lalis, N. Bellas, H. Vandierendonck, and D. S. Nikolopoulos, *A Programming Model and Runtime System for Significance-Aware Energy-Efficient Computing*, 2014. arXiv: 1412.5150 [cs.PL]. [Online]. Available: https://arxiv.org/abs/1412.5150

[4] S. Lyberis, P. Pratikakis, I. Mavroidis, and D. S. Nikolopoulos, "Myrmics: Scalable, Dependency-Aware Task Scheduling on Heterogeneous Manycores," arXiv, Tech. Rep., Jun. 2016, arXiv:1606.04282. [Online]. Available: https://arxiv.org/abs/1606.04282

[3] E. Ayguadé, C. Calidonna, J. Corbalan, M. Giordano, M. Gonazalez, H.-C. Hoppe, J. Labarta, M. Furnari, X. Martorell, N. Navarro, D. S. Nikolopoulos, J. Oliver, T. S. Papatheodorou, and E. D. Polychronopoulos, "NANOS: Effective Integration of Fine-grain Parallelism Exploitation and Multiprogramming," Universitat Politècnica de Catalunya / Barcelona Supercomputing Center, Tech. Rep., 1999, Technical Report.

## Theses

[2] D. S. Nikolopoulos, "System Software Support for Reducing Memory Latency on CC-NUMA Architectures," PhD Dissertation, Department of Computer Engineering and Informatics, University of Patras, Dec. 2000.

[1] D. S. Nikolopoulos and I. Tsolakis, "Load Balancing in Volunteer Computing Clusters," MSc Thesis, Department of Computer Engineering and Informatics, University of Patras, Jul. 1997.

### Publication Metrics

**Citation metrics** (as of October 2026)

| Source | Citations | h-index | Other |
|---|---|---|---|
| Google Scholar | 8,086 | 44 | i-10 index: 149 |
| Scopus | 4,280 | 31 | Field-Weighted Citation Index: 1.46 |

Top 2% of Computer Scientists Worldwide according to Scopus Standardized Citation Indicators.

## Knowledge and Technology Transfer & Commercialization

**Patents**

Research on memory management for stream processing and in-memory database systems, supported by NSF grants 0715051, 0720673, 0904844, led to foundational results later extended into U.S. Patents (1) 9,112,666 by former student Scott Schneider, and (2) 10,365,997, (3) 10,387,127, (4) 10,474,557, (5) 10,698,732 by former student Ahmad Hassan. In accordance with NSF and other federal funding agency policies on open dissemination, I declined to participate in inventorship but contributed to the original publication. Patent details below:

[1] "Elastic auto-parallelization for stream processing applications based on measured throughput and congestion," U.S. Patent 9,112,666 (granted Aug. 18, 2015).
[2] "Optimizing DRAM memory based on read-to-write ratio of memory access latency," U.S. Patent 10,365,997 (granted Jul. 30, 2019).
[3] "Detecting sequential access data and random access data for placement on hybrid main memory for in-memory databases," U.S. Patent 10,387,127 (granted Aug. 20, 2019).
[4] "Source code profiling for line-level latency and energy consumption estimation," U.S. Patent 10,474,557 (granted Nov. 12, 2019).
[5] "Page ranking in operating system virtual pages in hybrid memory systems," U.S. Patent 10,698,732 (granted Jun. 30, 2020).

**Technology Transfer Activities**

- Train-less deep neural network architecture search (**IBM Watson Studio**, 2019)
- Query planning and data placement for in-memory transactional–analytical databases (**SAP**, 2019)
- High-level program parallelization software for FPGAs (**Crevinn**, 2019)
- Automatic scaling of streaming operators (**IBM**, 2010)
- Adoption of methods for memory management and dynamic software scaling (**Samsung, Red Hat, Microsoft, Oracle, IBM, Alpine Electronics, Thales, Reservoir Labs**, 2008–present)

## Research Grants

**Funding summary** (as of October 10, 2026)

Total value of grants as PI (25 awards): **$82.9M**

Total value of grants as Co-PI (28 awards): **$58.2M**

Currency conversion at year-of-award annual average rates. Major sponsors: NSF, DOE, EU Horizon 2020 & FP7, EPSRC, Royal Society, Marie Curie, Amazon, IBM, Intel, SAP, Sony, Cisco. Totals exclude in-kind allocations unless noted.

*Active*

[54] **[08/2026–07/2027] Eliminating Redundant Computation in Multi-Agent LLM Systems via Hybrid KV-Cache Communication**.
Amazon. PI. $75,000 total (PI share: $37,500).

[53] **[05/2026–05/2027] SCALE-R: Semantic Co-rendering & Adaptive Link-aware Encoding for Real-time Graphics**.
Sony. PI. $150,000 total (PI share: $75,000).

[52] **[10/2025–10/2026] Cloud–Edge–Neuromorphic Computing Integration for Multimodal Sensing and AI-Driven Modeling of Complex Biological Systems**.
Virginia Tech Institute for Critical Technology & Applied Science / Institute for Advanced Computing. Co-PI. $50,000 total (PI share: $12,500).

*Completed*

[51] **[08/2025–08/2026] Advancing AI-Driven Autonomous Experimentation for Turbulent Flows**.
Virginia Tech College of Engineering Major Grant Initiative Program. PI. $50,000.

[50] **[05/2024–11/2024] GPU-Accelerated High-Performance Computing to Supercharge AI Models for Pandemic Prevention**.
National Science Foundation (National AI Resource). Co-PI. In-kind HPC resource allocation (no direct $ amount).

[49] **[10/2023–09/2026] SWEET: Hardware and Software for Sustainable Wearable Edge Intelligence**.
National Science Foundation (#2315851). PI. $600,000.

[48] **[08/2023–07/2024] WASML: Enabling Scalable Serverless Computer Vision**.
Sony Faculty Innovation Award (#VTF-AG3ZURVF). PI. $100,000.

[47] **[02/2022–01/2023] VirtEdgeAI: Seamless Virtualization of AI Accelerators in Edge Computing Substrates**.
Cisco (#VTF-446663). PI. $100,000.

[46] **[06/2021–05/2024] Collaborative Research: CNS Core: Medium: HardLambda: A new FaaS Abstraction for Cross-Stack Resource Management in Disaggregated Datacenters**.
National Science Foundation (#2106634). Co-PI. $600,000 (PI share: $210,000).

[45] **[12/2017–12/2018] Belfast Region City Deal, Queen's Global Innovation Institute**.
UK Treasury (77%); Belfast City Council (23%). PI. £56,000,000 (PI share: £15,000,000).

[44] **[01/2018–06/2021] Biohaviour: Building the Blind Watchmaker**.
EPSRC (#EP/R003564/1). Co-PI. £778,226 (PI share: £194,556).

[43] **[08/2017–07/2018] Variability in Computing Systems**.
Royal Academy of Engineering, Distinguished Visiting Fellowships. Co-PI. £4,100.

[42] **[01/2017–12/2020] Scalable, Virtualized Data Center Acceleration**.
Intel. Co-PI. £3,945 (PI share: £795).

[41] **[01/2017–12/2020] OPRECOMP: Open Transprecision Computing**.
EU, Horizon 2020 (#H2020-732631). Co-PI. €5,999,510 (£5,091,933) (PI share: €705,625 (£599,781)).

[40] **[02/2016–01/2019] UNISERVER: A Universal Micro-server Ecosystem by Exceeding the Energy and Performance Scaling Boundaries**.
EU, Horizon 2020 (#H2020-688540). Co-PI. €4,815,810 (£4,333,717) (PI share: €322,648 (£222,516)).

[39] **[02/2016–01/2019] VINEYARD: Versatile Integrated Accelerator-based Heterogeneous Datacenters**.
EU, Horizon 2020 (#H2020-687628). Co-PI. €6,283,895 (£4,467,972) (PI share: €663,625 (£471,850)).

[38] **[10/2015–09/2020] Principles and Practice of Near Data Computing**.
Royal Society Wolfson Research Merit Award (#WM150009). PI. £50,000.

[37] **[09/2015–09/2018] Meeting the Future Challenges of Heterogeneous and Extreme Scale Parallel Computing**.
SFI-DEL, Investigator Awards (#14/IA/2474). PI. £521,947.

[36] **[10/2015–10/2018] ECOSCALE: Energy Efficient Heterogeneous Computing at Exascale**.
EU, Horizon 2020 (#H2020-671632). Co-PI. €4,237,398 (£2,922,346) (PI share: €696,750 (£480,518)).

[35] **[10/2015–10/2018] ALLScale: An Exascale Programming, Multi-objective Optimization and Resilience Management Environment Based on Nested Recursive Parallelism**.
EU, Horizon 2020 (#H2020-671603). Co-PI. €3,366,196 (£2,463,217) (PI share: €438,578

(£320,930)).

[34] **[08/2015–09/2017] Crevinn Teoranta OpenCL Knowledge Transfer**.
Knowledge Transfer Partnership / Intertrade Ireland Fusion Project. Co-PI. £39,000.

[33] **[03/2015–03/2018] SERT: Scale-Free, Energy-Efficient and Resilient CSE Software for Mega-Core Systems**.
EPSRC (Software for the Future II) (#EP/M01147X/1). PI. £963,929 (PI share: £694,909).

[32] **[01/2015–01/2018] RAPID: Heterogeneous Secure Multi-level Remote Acceleration Service for Low-Power Integrated Systems and Devices**.
EU, Horizon 2020 (#H2020-644312). Co-PI. €2,203,800 (£1,695,231) (PI share: €326,925 (£251,481)).

[31] **[01/2015–01/2018] DIVIDEND: Distributed Heterogeneous Vertically Integrated Energy Efficient Data Centers**.
EPSRC, CHIST-ERA (#EP/M015742/1). Co-PI. €1,346,885 (£1,077,508) (PI share: €279,646 (£223,717)).

[30] **[10/2014–10/2017] HPDCJ: Heterogeneous Parallel and Distributed Computing with Java**.
EPSRC, CHIST-ERA (#EP/M015750/1). Co-PI. €1,721,010 (£1,376,808) (PI share: €178,159 (£142,527)).

[29] **[03/2014–03/2017] ASAP: An Adaptive, Highly Scalable Analytics Platform**.
European Commission, FP7-ICT (#FP7-619706). Co-PI. €2,245,128 (£1,909,122) (PI share: €407,720 (£346,701)).

[28] **[01/2014–02/2014] US-Ireland R&D Partnership Planning Grant: Cloud-based Electronic Integration of Patient Records (CLEAR)**.
(#PG20). Co-PI. £1,356.

[27] **[10/2013–10/2016] ENPOWER: Energy-Proportional Computing with Heterogeneous and Reconfigurable Processors**.
EPSRC (#EP/L004232/1). Co-PI. £741,043 (PI share: £348,325).

[26] **[09/2013–09/2016] ALEA: Abstraction-Level Energy Accounting and Optimization in Many-Core Programming Languages**.
EPSRC, System Approaches to Distributed and Embedded Architectures (#EP/L000555/1). PI. £669,561 (PI share: £394,025).

[25] **[09/2013–09/2016] NanoStreams: A Hardware and Software Stack for Real-Time Analytics on Fast Data Streams**.
European Commission, FP7-ICT, Objective 3 (#ICT-610509). PI. €3,300,000 (PI share: €723,565).

[24] **[10/2013–10/2016] CACTOS: Context-Aware Cloud Topology Optimization and Simulation**.
European Commission, FP7-ICT, Objective 1 (#ICT-610811). Co-PI. €3,215,751 (PI share: €583,330).

[23] **[08/2013–08/2019] SAP: PhD Project on High Availability for Petascale Systems**.
SAP AG (#UK-2013-009). PI. £12,436.

[22] **[06/2013–06/2016] SCORPIO: Significance-Based Computing for Reliability and Power Optimization**.
European Commission, FP7-FET-Open (#FP7-323872). Co-PI. €1,890,775 (PI share: €273,400).

[21] **[04/2013–04/2015] NovoSoft: Software Management of Non-Volatile Memory Hierarchies**.
European Commission, Marie Curie Intra-European Fellowship (#FP7-327744). Scientist in

Charge. €309,235.

[20] **[03/2013–03/2016] Characterizing and Optimizing In-Memory Database Systems for Emerging Memory Technologies**.
SAP UK Limited (#R502). Co-PI. £34,298 (PI share: £17,149).

[19] **[09/2012–09/2015] GEMSCLAIM: Greener Mobile Systems by Cross Layer Integrated Energy Management**.
EPSRC, CHIST-ERA (#EP/K017594/1). Co-PI. €1,776,688 (PI share: €436,884).

[18] **[11/2012–06/2013] Exascale Mesh Generation Runtime Systems**.
Royal Academy of Engineering, Distinguished Visiting Fellowships. Host PI. £4,100.

[17] **[01/2012–01/2016] HOLISTIC: Hardware and Software Techniques for Multicore Processor Architectures Reliability Enhancement**.
Greek Ministry of Education, Lifelong Learning and Religious Affairs, Thales Programme (#1103). Co-PI. €600,000 (PI share: €98,000).

[16] **[03/2010–03/2012] SCC–MR: Scalable and Energy-Efficient Runtime Support for the MapReduce Programming Model on the Intel SCC**.
Intel Corporation. PI. Equipment donation.

[15] **[06/2010–09/2012] TEXT: Towards Exascale Applications**.
European Commission, FP7-INFRASTRUCTURES Programme (#ICT-261580). Co-PI. €2,470,000 (PI share: €299,364).

[14] **[06/2010–06/2011] ReMap: Rearchitecting MapReduce for Multicore Systems with Explicit Communication**.
High Performance and Embedded Architectures and Compilers Network of Excellence, Cluster Collaboration Grant (#ICT-217068). PI. €3,000 share.

[13] **[03/2010–03/2013] ENCORE: Enabling Technologies for a Programmable Many-core**.
European Commission, FP7-ICT, Objective 3 (#ICT-248647). Co-PI. €2,533,000 (PI share: €533,000).

[12] **[08/2009–08/2013] Coupled Models of Diffusion and Individual Behavior over Extremely Large Scale Social Networks**.
National Science Foundation, PetaApps Program (#OCI-0904844). Co-PI. $1,182,798 (PI share: $200,000).

[11] **[01/2009–01/2013] I-Cores: Hypervisor-based Synthesis of Custom Execution Environments for Multi-core Systems**.
European Commission, FP7 Programme, Marie Curie International Reintegration Grants (#IRG-224759). PI. €100,000.

[10] **[01/2008–02/2008] HiPEAC Fellowship: Runtime Systems for Parallel Programming**.
European Commission, FP7 Programme, European Network of Excellence in High Performance and Embedded Architectures (#ICT-217068). PI. €8,600 share.

[9] **[07/2007–07/2010] VT-ASOS: Virtualization Technologies for Application-Specific Operating Systems on Many-Core HPC Systems**.
NSF Computer Systems Research Program (#CNS-0720673). PI. $300,000 (PI share: $150,000).

[8] **[07/2007–07/2010] Thermal Conductors: Runtime Software Support for Proactive Heat Management in Advanced Execution Systems**.
NSF Computer Systems Research Program (#CNS-0720750). Co-PI. $350,000 (PI share: $175,000).

[7] **[05/2007–05/2008] Models and Adaptive Runtime Systems for Accessible Parallel Programming on IBM Multi-Core Systems**.
IBM Faculty Award Program (#VTF-874197). PI. $15,000.

[6] **[07/2007–07/2008] MISER: A High-Performance, Power-Aware Cluster**.
NSF Computing Research Infrastructure Program (#CNS-0709025). Co-PI. $500,000 (PI share: $166,667).

[5] **[08/2006–08/2007] Faculty Startup Grant**.
Virginia Tech. PI. $100,000.

[4] **[08/2005–08/2008] MELISSES: Liquid Services for Scalable Multithreaded and Multicore Execution on Emerging Supercomputers**.
DOE Early Career Principal Investigator Award Program (#DE-FG02-06ER25751, DE-FG02-05ER25689). PI. $299,907.

[3] **[05/2005–05/2008] Acquisition of STEMS: A Laboratory for End-to-End Development of Software and Tools for Emerging Multigrain Supercomputers**.
NSF Major Research Instrumentation Program (#CNS-0521381). PI. $228,134 (PI share: $76,045).

[2] **[05/2005–08/2005] A Unified Framework for Multilevel Parallelization on Deep Computing Systems**.
NSF Research Experiences for Undergraduates Program (#CCF-0531887). PI. $6,000.

[1] **[01/2004–01/2009] A Unified Framework for Multilevel Parallelization on Deep Computing Systems**.
NSF CAREER Award Program (#CCF-0346867, CCF-0715051). PI. $419,835 (PI share: $225,000).

# Education

## Courses Taught

Evaluation is the Teaching Effectiveness score, out of 6.00.

| | Term | Course | Evaluation | Enrolled |
|---|---|---|---|---|
| *Virginia Tech* | | | | |
| [43] | **Spring 2026** | CS2506: Computer Organization II | N/A | 200 |
| [42] | **Spring 2025** | CS5914: Special Topics in Computer Science: LLMs for High-Performance Code Generation | 5.70 | 20 |
| [41] | **Fall 2024** | CS2506: Computer Organization II | 4.90 | 195 |
| [40] | **Spring 2024** | CS2506: Computer Organization II | 5.10 | 150 |
| [39] | **Fall 2023** | CS5510/ECE5510: Multiprocessor Programming | N/A | 45 |
| [38] | **Spring 2023** | CS2506: Computer Organization II | 4.87 | 192 |
| [37] | **Fall 2022** | CS5914: Warehouse Scale Computing | 5.79 | 17 |
| [36] | **Spring 2022** | CS2506: Computer Organization II | 5.16 | 216 |
| [35] | **Fall 2021** | CS5510/ECE5510: Multiprocessor Programming | 5.78 | 35 |
| [34] | **Spring 2021** | CS2506: Computer Organization II | 5.15 | 148 |
| [33] | **Fall 2020** | CS4284: Systems & Networking Capstone | 5.85 | 23 |
| [32] | **Spring 2020** | CS2506: Computer Organization II | 5.16 | 82 |
| *Queen's University Belfast* | | | | |
| [31] | **Spring 2015** | ECS1002: Design Projects | | |
| [30] | **Fall 2014** | ECS2001: Second Stage Design Projects | | |
| [29] | **Spring 2014** | ECS1002: Design Projects | | |
| [28] | **Fall 2013** | ECS2001: Second Stage Design Projects | | |
| [27] | **Spring 2013** | ECS1002: Design Projects | | |
| [26] | **Fall 2012** | ECS2001: Second Stage Design Projects | | |

| | Term | Course | Evaluation | Enrolled |
|---|---|---|---|---|
| *University of Crete* | | | | |
| [25] | **Fall 2011** | CS425: Computer Systems Architecture | | |
| [24] | **Fall 2011** | CS100: Introduction to Computer Science | | |
| [23] | **Spring 2011** | CS225: Computer Organization | | |
| [22] | **Spring 2011** | CS529: Multi-core Systems Programming | | |
| [21] | **Fall 2010** | CS425: Computer Systems Architecture | | |
| [20] | **Spring 2010** | CS225: Computer Organization | | |
| [19] | **Spring 2010** | CS529: Multi-core Systems Programming | | |
| [18] | **Fall 2009** | CS425: Computer Systems Architecture | | |
| [17] | **Spring 2009** | CS529: Multi-core Systems Programming | | |
| [16] | **Spring 2009** | CS225: Computer Organization | | |
| [15] | **Fall 2008** | CS425: Computer Systems Architecture | | |
| [14] | **Spring 2008** | CS425: Computer Systems Architecture | | |
| *Virginia Tech* | | | | |
| [13] | **Fall 2007** | CS5234: Advanced Parallel Computation | | |
| [12] | **Fall 2007** | CS2504: Introduction to Computer Organization | | |
| [11] | **Spring 2007** | CS2504: Introduction to Computer Organization | | |
| [10] | **Fall 2006** | CS4234: Parallel Computation | | |
| *College of William and Mary* | | | | |
| [9] | **Spring 2006** | CSCI644: Advanced Computer Architecture | | |
| [8] | **Fall 2005** | CSCI444/544: Principles of Operating Systems | | |
| [7] | **Spring 2005** | CSCI644: Advanced Computer Architecture | | |
| [6] | **Fall 2004** | CSCI444/544: Principles of Operating Systems | | |
| [5] | **Spring 2004** | CSCI644: Advanced Computer Architecture | | |
| [4] | **Fall 2003** | CSCI444/544: Principles of Operating Systems | | |
| [3] | **Spring 2003** | CSCI644: Advanced Computer Architecture | | |
| [2] | **Fall 2002** | CSCI444/544: Principles of Operating Systems | | |
| *University of Illinois at Urbana–Champaign* | | | | |
| [1] | **Spring 2002** | ECE291: Computer Engineering II | | |

## Seminars Taught

- **[Spring 2012]** Implementation of Multi-core Programming Models, Universitat Politècnica de Catalunya
- **[Spring 2010]** Multi-core Systems Programming and Optimization, Universitat Politècnica de Catalunya
- **[Spring 2008]** Multi-core Systems Programming and Optimization, Universitat Politècnica de Catalunya
- **[Spring 2007]** Multi-core Systems Programming and Optimization, Universitat Politècnica de Catalunya
- **[Spring 2004]** Multithreaded Architectures and Software, College of William and Mary

## Course and Curriculum Development

- CS2506: Computer Organization II — redeveloped to introduce a new open-source processor and tools (Virginia Tech, 2022)
- ECS2001: Software and Electronic Systems Engineering Design Projects (2nd stage) — developed from scratch (Queen's University Belfast, 2016)

- ECS1002: Software and Electronic Systems Engineering Design Projects (1st stage) — developed from scratch (Queen's University Belfast, 2016)
- CS529: Multi-core Processor Programming — developed from scratch (University of Crete, 2009)
- CS425: Computer Systems Architecture — major revision (University of Crete, 2009)
- CS225: Computer Organization — major revision (multi-core, cache coherence) (University of Crete, 2009)
- CS5234: Advanced Parallel Computation — developed from scratch (Virginia Tech, 2006)

## Teaching Grants

[4] **[05/2012–06/2012] Advanced Topics in Implementing Multicore Programming Models**. Universitat Politècnica de Catalunya. PI. €2,400.
[3] **[05/2010–06/2010] Multi-core Systems Programming**. Universitat Politècnica de Catalunya. PI. €2,400.
[2] **[05/2008–06/2008] Multi-core Systems Programming**. Universitat Politècnica de Catalunya. PI. €3,600.
[1] **[05/2007–06/2007] Multi-core Systems Programming and Optimization**. Universitat Politècnica de Catalunya. PI. €3,600.

## Supervision

### Ph.D. Thesis Students

#### Primary Advisor

[19] **Mona Moghadampanah** — Computer Science, Virginia Tech (in progress). Thesis area: *Scheduling of workloads in the continuum*.
[18] **Adib Rezaei** — Computer Science, Virginia Tech (in progress). Thesis area: *Accelerator virtualization and fault tolerance*.
[17] **Sina Heidari** — Computer Science, Virginia Tech (in progress). Thesis area: *Large-scale processing of graph neural networks*.
[16] **Jiakun Fan** — Computer Science, Virginia Tech (in progress). Thesis area: *Edge AI*.
[15] **Yunqi Shen** — Computer Science, Virginia Tech (in progress). Thesis area: *Tiered memory systems*.
[14] **Farhana Amin** — Computer Science, Virginia Tech (in progress). Thesis area: *Programming models for and by LLMs*.
[13] **Xiangchen Li** — Electrical and Computer Engineering, Virginia Tech (in progress). Thesis area: *Edge computing and networking for AI*.
[12] **Dr. Emadeldin Abdrabou** — Electrical and Computer Engineering, Virginia Tech (Dec. 2025). Thesis title: *On the optimization of edge server scaling and placement for Open Radio Access Network (O-RAN) slicing*. (Co-advised with Scott Midkiff.)
[11] **Dr. Esha Barlaskar** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Nov. 2020). Thesis title: *User-centric cloud application management*. (Co-supervised with Ivor Spence and Peter Kilpatrick.)
[10] **Dr. Nan Wang** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Sep. 2019). Thesis title: *Resource management for edge computing systems*.
[9] **Dr. Roxana Istrate** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2019). Thesis title: *Efficient neural network architecture search*.
[8] **Dr. Sakil Barbhuiya** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2018). Thesis title: *Anomaly detection in cloud and mobile devices*. (Co-supervised with Peter Kilpatrick.)

[7] **Dr. Charalambos Chalios** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2017). Thesis title: *Software-defined significance-based computing*. (Co-supervised with Hans Vandierendonck.)

[6] **Dr. Giorgis Georgakoudis** — Computer and Telecommunications Engineering, University of Thessaly (May 2016). Thesis title: *Scheduling and performance characterization on heterogeneous computing systems*. (Co-supervised with Spyros Lalis.)

[5] **Dr. Jae-seung Yeom** — Computer Science, Virginia Tech (May 2014). Thesis title: *Optimizing data accesses for scaling data-intensive scientific applications*. (Co-supervised with Madhav Marathe.)

[4] **Dr. Spyros Lyberis** — Computer Science, University of Crete (Jul. 2013). Thesis title: *Myrmics: A scalable runtime system for global address spaces*.

[3] **Dr. Scott Schneider** — Computer Science, Virginia Tech (Dec. 2010). Thesis title: *Shared memory abstractions for heterogeneous multicore processors*.

[2] **Dr. Filip Blagojevic** — Computer Science, Virginia Tech (May 2008). Thesis title: *Scheduling on asymmetric parallel architectures*.

[1] **Dr. Matthew Curtis-Maury** — Computer Science, Virginia Tech (Mar. 2008). Thesis title: *Improving the efficiency of parallel applications on multithreaded and multicore systems*. **Virginia Tech Outstanding Ph.D. Dissertation Award.**

### Co-Advisor

[17] **Kazi Hassan Ibn Arif** — Computer Science, Virginia Tech (in progress). Thesis area: *Sparsity in multimodal learning*.

[16] **Dr. Yuze Li** — Computer Science, Virginia Tech (May 2026). Thesis title: *Cross-stack improvement on memory efficiency*. (Co-supervised with Ali R. Butt.)

[15] **Shunyu Yao** — Computer Science, Virginia Tech (in progress). Thesis title (Dec. 2026): *State Elasticity for Distributed HPC Workloads*. (Co-advised with Ali R. Butt.)

[14] **Dr. Abdullahi Abubakar** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Dec. 2023). Thesis title: *Anomaly detection in longitudinal data with applications in cloud computing and healthcare*.

[13] **Dr. Ioannis Tsiokanos** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Dec. 2021). Thesis title: *Cross-layer instruction-aware timing error mitigation and evaluation for energy-efficient dependable architectures*. (Co-supervised with Georgios Karakonstantis.)

[12] **Dr. Konstantinos Tovletoglou** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jun. 2021). Thesis title: *Modeling and design of energy-efficient dependable memory subsystems*. (Co-supervised with Georgios Karakonstantis.)

[11] **Dr. Jiawen Sun** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2017). Thesis title: *The GraphGrind framework: fast graph analytics on large shared-memory systems*. (Co-supervised with Hans Vandierendonck.)

[10] **Dr. Stuart McCool** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2016). Thesis title: *Guidance environments for program parallelization and analysis*. (Co-supervised with Peter Kilpatrick.)

[9] **Dr. Ahmad Hassan** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Jul. 2016). Thesis title: *Software management of hybrid main memory systems*. (Co-supervised with Hans Vandierendonck.)

[8] **Dr. Eoghan O'Neill** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Oct. 2015). Thesis title: *A framework for managing shared accelerators in heterogeneous environments*. (Co-supervised with Peter Kilpatrick.)

[7] **Dr. Aleksandr Khasymski** — Computer Science, Virginia Tech (Feb. 2015). Thesis title: *Accelerated storage systems*. (Co-supervised with Ali R. Butt.)
[6] **Dr. Chun-Yi Su** — Computer Science, Virginia Tech (Dec. 2014). Thesis title: *Energy-aware thread and data management in heterogeneous multi-core and multi-memory systems*. (Co-supervised with Kirk W. Cameron.)
[5] **Dr. Vassilis Papaefstathiou** — Computer Science, University of Crete (Nov. 2013). Thesis title: *Architectural support for software-guided energy reduction of manycore communication*. (Co-supervised with Manolis Katevenis.)
[4] **Dr. Muhammad Mustafa Rafique** — Computer Science, Virginia Tech (Sep. 2011). Thesis title: *An adaptive framework for managing heterogeneous many-core clusters*. (Co-supervised with Ali R. Butt.)
[3] **Dr. Dong Li** — Computer Science, Virginia Tech (Jan. 2011). Thesis title: *Scalable and energy-efficient execution methods for multicore systems*. (Co-supervised with Kirk W. Cameron.)
[2] **Dr. John Christian Linford** — Computer Science, Virginia Tech (May 2010). Thesis title: *Accelerating atmospheric modeling through emerging multi-core technologies*. (Co-supervised with Adrian Sandu.)
[1] **Dr. Richard Tran Mills** — Computer Science, College of William & Mary (Nov. 2004). Thesis title: *Dynamic adaptation to CPU and memory load in scientific applications*. (Co-supervised with Andreas Stathopoulos.)

### Postdoctoral Research Fellows

[17] **Dr. Dimitris Spatharakis** — Visiting postdoctoral fellow, School of Electrical and Computer Engineering, National Technical University of Athens (04/2025–05/2025).
[16] **Dr. Sakil Barbhuiya** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (05/2018–08/2019). Research themes: Cloud-based distributed manufacturing; smart manufacturing as a service.
[15] **Dr. Jun-Kyu Lee** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (04/2017–12/2020). Research themes: Approximate computing.
[14] **Dr. Damon Fenacci** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2017–01/2018). Research themes: Memory management.
[13] **Dr. Giorgis Georgakoudis** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (05/2016–09/2018). Research themes: System software and hardware/software interface.
[12] **Dr. Blesson Varghese** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2016–06/2017). Research themes: Energy-efficient and resilient high-performance computing.
[11] **Dr. Kiril Dichev** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (10/2015–08/2019). Research themes: Exascale resilience.
[10] **Dr. Cheol-Ho Hong** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (06/2015–12/2017). Research themes: Accelerator virtualization.
[9] **Dr. Lev Mukhanov** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (04/2014–12/2020). Research themes: Abstraction-level energy accounting in many-core programming languages.
[8] **Dr. Zafeirios Papazachos** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2014–08/2019). Research themes: Data center performance and reliability monitoring and optimization.
[7] **Dr. Hemant Mehta** — Electronics, Electrical Engineering and Computer Science, Queen's

University Belfast (10/2015–04/2017). Research themes: Virtual machine energy accounting.

[6] **Dr. Yun Wu** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2014–01/2017). Research themes: Energy-proportional many-core computing systems.

[5] **Dr. Paul Harvey** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2016–09/2016). Research themes: Exascale programming models.

[4] **Dr. Ahmed Sayed** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (09/2014–02/2015). Research themes: System software for real-time in-memory analytics.

[3] **Dr. Konstantina Mitropoulou** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2014–03/2014). Research themes: System software for real-time in-memory analytics.

[2] **Dr. Hans Vandierendonck** — Institute of Computer Science, Foundation for Research and Technology (FORTH) (10/2010–10/2011). Research themes: Parallel programming; scheduling.

[1] **Dr. Christos D. Antonopoulos** — Computer Science, College of William & Mary (06/2004–06/2006). Research themes: Energy-efficient parallel computation; runtime systems; memory management.

### Pre-Doctoral Research Assistants

[4] **Kai Chen** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (in progress). Research area: *Multi-scale power and performance modeling*.

[3] **George Tzenakis** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (10/2012–02/2017). Research area: *Dynamic parallelism and elasticity*.

[2] **Chhaya Trehan** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (01/2015–05/2016). Research area: *Optimization algorithms for energy efficiency*.

[1] **Mahwish Arif** — Electronics, Electrical Engineering and Computer Science, Queen's University Belfast. Research area: *Performance portability*. (Co-supervised with Hans Vandierendonck.)

### Visiting Scholars

[2] **Prof. Oscar Garcia Lorenzo** — Department of Computer Architecture, University of Santiago de Compostela (Apr. 2016). Topic: *Hardware counter-based performance analysis, modeling, and improvement through thread migration in NUMA systems*.

[1] **Dr. Satoshi Imamura** — System LSI Laboratory, Kyushu University. Research area: *Energy-efficiency optimization on multicore NUMA servers*.

### M.Sc. Research Students

### Primary Advisor

[29] **Katelyn Crumpacker** — Computer Science, Virginia Tech (May 2026). Thesis title: *LLM-guided run-time parameter optimization for energy-efficient model inference*.

[28] **Timothy Coyne** — Computer Science, Virginia Tech (May 2025). Thesis title: *Performance portability of CUDA across NVIDIA GPU architectures*.

[27] **Max Fisher** — Computer Science, Virginia Tech (May 2025). Thesis title: *Analysis of memory access patterns and prefetching strategies for large language model inference*.

[26] **Moustafa Kahla** — Electrical and Computer Engineering, Virginia Tech (Dec. 2023). Thesis title: *Automatic source code transformation to pass compiler optimization*.

[25] **Manthan Shah** — Electrical and Computer Engineering, Virginia Tech (Jun. 2023). Thesis

title: *Analysis of YOLOv7 inference performance through a systems perspective and identification of optimization opportunities*.

[24] **Shreya Bhandare** — Computer Science, Virginia Tech (Jun. 2023). Thesis title: *Designing RDMA-based efficient communication for GPU remoting*.

[23] **Melissa Cameron** — Computer Science, Virginia Tech (Jun. 2023). Thesis title: *Parallel islands: a diversity-aware tool for parallel computing education*.

[22] **Ishaan Gulati** — Computer Science, Virginia Tech (Jun. 2023). Thesis title: *A scalable leader-based consensus algorithm*.

[21] **Matthew Jackson** — Computer Science, Virginia Tech (May 2023). Thesis title: *Computational offloading for real-time computer vision in unreliable multi-tenant edge systems*.

[20] **Daniel Moyer** — Computer Science, Virginia Tech (May 2021). Thesis title: *Punching holes in the cloud: direct communication between serverless functions using NAT traversal*.

[19] **Dimitris Chassapis** — Computer Science, University of Crete. Thesis title: *Static analysis for parallelism and correctness in task dataflow programming models*.

[18] **Ioannis Manousakis** — Computer Science, University of Crete (Jul. 2013). Thesis title: *TPROF: An energy profiler for task-parallel programs*.

[17] **Evangelos Kafentarakis** — Computer Science, University of Crete (Jul. 2013). Thesis title: *Lprof: A tool for profiling locality awareness in a task-based programming model*.

[16] **Christi Symeonidou** — Computer Science, University of Crete (Jul. 2013). Thesis title: *Distributed region-based allocation and synchronization*.

[15] **Kallia Chronaki** — Computer Science, University of Crete (Jun. 2013). Thesis title: *Exploiting pipelined parallelism with task dataflow programming models*.

[14] **Alexandros Labrineas** — Computer Science, University of Crete (Jun. 2013). Thesis title: *BDDT-SCC: A task-parallel runtime for the Single-Chip Cloud Computer*.

[13] **Anastasios Papagiannis** — Computer Science, University of Crete (Mar. 2013). Thesis title: *MapReduce on distributed-memory many-core architectures*.

[12] **Angelos Papatriantafyllou** — Computer Science, University of Crete (Mar. 2012). Thesis title: *Optimized block-based dependence analysis for task parallelism*.

[11] **Constantinos Koukos** — Computer Science, University of Crete (Aug. 2010). Thesis title: *Locality management in task-based parallel programming models*. (Co-supervised with Angelos Bilas.)

[10] **Pranav Tendulkar** — ALaRi Institute, Advanced Studies in Embedded Systems Design (Jun. 2010). Thesis title: *Runtime OpenMP support using hardware primitives on explicitly memory-managed multiprocessors*.

[9] **Michail Zampetakis** — Computer Science, University of Crete (Apr. 2010). Thesis title: *Runtime support for programming explicit-communication chip multiprocessors*.

[8] **Maria Katsamani** — Computer Science, University of Crete (Mar. 2010). Thesis title: *Software implementation of MPI primitives on multicore FPGA*. (Co-supervised with Manolis Katevenis.)

[7] **Benjamin Rose** — Computer Science, Virginia Tech (May 2009). Thesis title: *Intra- and inter-chip communication support for asymmetric multicore processors with explicitly managed memory hierarchies*.

[6] **Beran Nova Bryant** — Computer Science, Virginia Tech (May 2008). Thesis title: *Temperature-aware scheduling of parallel applications on shared-memory multiprocessors*.

[5] **Harshil Shah** — Computer Science, Virginia Tech (May 2008). Thesis title: *Application parallelization on the Cell/BE*.

[4] **Jyotirmaya Tripathi** — Computer Science, Virginia Tech (May 2008). Thesis title: *Scheduling parallel applications on paravirtualized shared-memory multiprocessors*.

[3] **Ankur Shah** — Computer Science, Virginia Tech (Apr. 2008). Thesis title: *Prediction models for multi-dimensional power-performance optimization on many cores*.
[2] **Scott Schneider** — Computer Science, College of William & Mary (Jun. 2005). Thesis title: *Factory: An object-oriented parallel programming substrate for deep multiprocessors*.
[1] **Robert McGregor** — Computer Science, College of William & Mary (May 2005). Thesis title: *Scheduling with bus bandwidth considerations on shared-memory multiprocessors*.

### Co-Advisor

[4] **Foivos Zakkak** — Computer Science, University of Crete (Mar. 2012). Thesis title: *SCOOP: Language extensions and compiler optimizations for task-based programming models*. (Co-supervised with Angelos Bilas.)
[3] **Ioannis Kesapides** — Computer Science, University of Crete (Mar. 2011). Thesis title: *Dynamic dependence analysis on multi-core processors*. (Co-supervised with Angelos Bilas.)
[2] **Michail Alvanos** — Computer Science, University of Crete (Jun. 2010). Thesis title: *Design and evaluation of a task-based parallel H.264 video encoder for the Cell processor*. (Co-supervised with Angelos Bilas.)
[1] **George Tzenakis** — Computer Science, University of Crete (Oct. 2009). Thesis title: *Tagged procedure calls (TPC): efficient runtime support for task-based parallelism on the Cell processor*. (Co-supervised with Angelos Bilas.)

### Undergraduate (BEng/BSc) Research Students

Supervised *43 undergraduate research students* at Virginia Tech, the University of Crete, the National Technical University of Athens, and the College of William & Mary.

## Service

### Professional Activities

#### Membership in Professional Societies

- **IEEE Computer Society:** Fellow (2024–present); Distinguished Visitor (2024–present); Distinguished Contributor (2022–present); Senior Member (2010–2024); Member (1995–2010).
- **Association for Computing Machinery (ACM):** Distinguished Member (2018–present); Senior Member (2011–2018); Member (1995–2011).
- **Asia-Pacific Artificial Intelligence Association (AAIA):** Fellow (2024–present).
- **International Artificial Intelligence Industry Alliance (AIIA):** Fellow (2024–present).
- **British Computer Society (BCS):** Fellow (2014–present).
- **The Institution of Engineering and Technology (IET):** Fellow (2017–present).
- **United Kingdom Council of Professors and Heads of Computing (CPHC):** Member (2012–2019).
- **ACM SIGARCH:** Member (2000–present).
- **ACM SIGOPS:** Member (2000–present).
- **ACM SIGHPC:** Member (2000–present).
- **Chartered Engineer (CEng):** 2017–present.
- **Technical Chamber of Greece:** Member (1996–present).

#### Conference Committee Activities

**Steering Committee Chair:**
[2] ACM International Conference on Supercomputing (ICS) — 2024–present

[1] IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) — 2022–2023

**Steering Committee Associate Chair:**

[1] ACM International Conference on Supercomputing (ICS) — 2023–2024

**Program Chair/Co-Chair:**

[9] ACM International Conference on High Performance Parallel and Distributed Computing (HPDC) — 2026
[8] ACM International Conference on Supercomputing (ICS) — 2023
[7] ACM International Conference on Supercomputing (ICS) — 2022
[6] International Workshop on Deployment and Use of Accelerators (DUAC) — 2021–2025
[5] IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID) — 2014
[4] International Workshop on Power-Aware Algorithms, Systems and Architectures (PASA) — 2013
[3] EuroMPI MPI Conference — 2011
[2] IEEE International Conference on Scalable Computing and Communications (ScalCom) — 2011
[1] Workshop on Parallel Programming for Accelerators (PPAC) — 2009–2011

**Program Vice Chair/Area Chair:**

[7] IEEE International Parallel and Distributed Processing Symposium (IPDPS) — Area Chair (Programming Models and Runtime Systems), 2026
[6] International European Conference on Parallel and Distributed Computing (EUROPAR) — Global Area Chair (Scheduling and Load Management), 2022
[5] International Conference on Embedded Computer Systems: Architecture, Modeling and Simulation (SAMOS) — 2016
[4] IEEE/ACM International Conference on High-Performance Computing, Networking, Architecture and Storage (SC) — 2014
[3] European Conference on Parallel Processing (EuroPar) — 2012
[2] IEEE International Parallel and Distributed Processing Symposium (IPDPS) — 2011
[1] International Conference on Parallel Processing (ICPP) — 2007

**General Chair:**

[10] IEEE International Conference on Cluster Computing (CLUSTER) — 2010; 2018
[9] IEEE International Symposium on the Performance Analysis of Systems and Software (ISPASS) — 2018
[8] ISC Conference Workshop on Approximate and Transprecision Computing on Emerging Technologies — 2018
[7] HiPEAC Conference Workshop on Approximate Computing — 2018
[6] SC Conference Workshop on Energy Efficient Supercomputing — 2013–2017
[5] ParCo Conference MiniSymposium on Edge Computing — 2017
[4] ParCo Conference MiniSymposium on Parallel Programming for Reliability and Energy Efficiency — 2016
[3] ParCo Conference MiniSymposium on Energy and Resilience in Parallel Programming — 2015
[2] HiPEAC Conference Workshop on Energy Efficient Heterogeneous Computing — 2015
[1] ICPP Conference Workshop on Power Aware Systems and Architectures — 2013

**Program Committee:**

[56] Annual Meeting of the Association for Computational Linguistics (ACL) — 2027 (External Review Committee)
[55] IEEE/ACM International Symposium on High-Performance Computer Architecture (HPCA) — 2027

[54] IEEE/ACM International Conference on High-Performance Computing, Networking, Storage and Analysis (SC) — 2012–2018; 2026
[53] ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) — 2021; 2026
[52] Annual AAAI Conference on Artificial Intelligence (AAAI) — Senior Program Committee Member, 2026
[51] ACM Symposium on Cloud Computing (SoCC) — 2026 (External Review Committee)
[50] ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP) — 2013; 2015; 2020–2021; 2025–2026
[49] International Conference on Parallel Processing (ICPP) — 2003–2004; 2008; 2014; 2016–2021; 2024–2025
[48] IEEE/ACM International Symposium on Microarchitecture (MICRO) — 2024–2025 (External Review Committee)
[47] IEEE International Conference on Parallel Architectures and Compilation Techniques (PACT) — 2009; 2022–2023; 2025
[46] International Symposium on Computer Architecture (ISCA) — 2025
[45] IEEE/ACM International Symposium on High Performance Distributed Computing (HPDC) — 2019–2021; 2025
[44] ACM International Conference on Supercomputing (ICS) — 2007; 2009; 2011–2012; 2014; 2017; 2020–2021; 2025
[43] IEEE/SBC International Symposium on Computer Architecture and High-Performance Computing (SBAC-PAD) — 2025
[42] IEEE International Parallel and Distributed Processing Symposium (IPDPS) — 2013–2014; 2017–2021; 2023–2024
[41] European Conference on Parallel Processing (EuroPar) — 2013–2014; 2016; 2020; 2023–2024
[40] IEEE International Conference on Big Data (BigData) — 2013–2018; 2023–2024
[39] IEEE International Conference on Edge Computing (EDGE) — 2022
[38] IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID) — 2013; 2015–2016; 2018–2022
[37] International Supercomputing Conference (ISC) — 2018–2020
[36] International Green and Sustainable Computing Conference (IGSC) — 2019–2020
[35] Workshop on Languages and Compilers for Parallel Computing (LCPC) — 2020
[34] IEEE International Conference on High-Performance Computing (HiPC) — 2014–2016; 2019
[33] Parallel Computing Conference (ParCo) — 2015; 2017; 2019
[32] IEEE Graph Computing Conference (GC) — 2019
[31] ACM International Conference on Distributed Event-Based Systems (DEBS) — 2018
[30] International Conference on Embedded Computer Systems: Architecture, Modeling, and Simulation (SAMOS) — 2014–2015; 2017–2018
[29] ISC Conference Workshop on Communication Architectures for HPC, Big Data, Deep Learning, and Clouds at Extreme Scale (ExaComm) — 2017–2018
[28] ACM Conference on Computing Frontiers (CF) — 2011; 2014–2015; 2017
[27] IEEE Conference on Cluster Computing (CLUSTER) — 2011–2013; 2015–2016
[26] ISC Workshop on Energy-Aware High-Performance Computing (EnaHPC) — 2014–2017
[25] IEEE International Conference on Ubiquitous Computing and Communications (IUCC) — 2012; 2016
[24] HiPEAC International Conference on High Performance and Embedded Architectures and Compilation — 2012–2015

[23] ACM PPoPP Conference Workshop on General Purpose Processing using GPU (GPGPU) — 2014–2015
[22] Feedback Computing Conference — 2010; 2015
[21] IEEE International Conference on Green Computing and Communications (GreenCom) — 2010–2011; 2013
[20] IEEE International Conference on Parallel and Distributed Systems (ICPADS) — 2004; 2006; 2010; 2012
[19] IEEE High-Performance Computing and Communications Conference (HPCC) — 2009; 2012
[18] Symposium on Application Accelerators for High-Performance Computing (SAAHPC) — 2009–2012
[17] IFIP Network and Parallel Computing Conference (NPC) — 2010–2012
[16] International Conference on Architecture of Computer Systems (ARCS) — 2010–2012
[15] IEEE Network and Storage Systems Conference (NAS) — 2009–2011
[14] IEEE International Conference on e-Business Engineering (ICEBE) — 2010–2011
[13] IEEE Green Computing Conference — 2010
[12] International Conference on Algorithms and Architectures for Parallel Processing (ICA3PP) — 2010
[11] IEEE International Conference on Scalable Computing and Communications (ScalCom) — 2009–2010
[10] IEEE Cloud Computing Conference (CloudCom) — 2009–2010
[9] IEEE International Conference on Computational Science and Engineering (CSE) — 2009–2010
[8] International Conference on Future Computational Technologies and Applications (Future Computing) — 2010
[7] International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT) — 2010
[6] IEEE Autonomic and Trusted Computing Conference (ATC) — 2006–2008
[5] Balkan Conference on Informatics (BCI) — 2007
[4] ACM SIGMETRICS International Conference on Measurement and Modeling of Computer and Communication Systems — 2006
[3] IEEE International Conference on Pervasive Systems (ICPS) — 2005
[2] International Symposium on High Performance Computing (ISHPC) — 2003
[1] International Conference on Computational Science (ICCS) — 2001

**Steering Committee Member:**

[3] ACM International Conference on Supercomputing (ICS) — 2022–present
[2] IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) — 2018–present
[1] IEEE International Conference on Cluster Computing (CLUSTER) — 2009–2011; 2017–2019

**Other Committee Service:**

[9] IEEE/ACM International Symposium on Microarchitecture (MICRO) — External Review Committee, 2024
[8] IEEE/ACM International Conference on High-Performance Computing, Networking, Storage and Analysis (SC) — Reproducibility Challenge Committee, 2021
[7] IEEE/ACM International Conference on High-Performance Computing, Networking, Storage and Analysis (SC) — HPC Impact Showcase Chair, 2017; Tutorials Committee, 2013–2014
[6] ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP) — External Review Committee, 2012; 2014
[5] International Supercomputing Conference (ISC) — Tutorials Committee, 2012–2014

[4] HiPEAC Conference — Workshop and Tutorials Chair, 2011
[3] EuroMPI Conference — Finance Chair, 2011
[2] ACM International Conference on Supercomputing (ICS) — Finance Chair, 2009; Workshops and Tutorials Chair, 2007
[1] SIAM Conference on Parallel Processing for Scientific Computing — Mini-Workshop Organizer, 2004; 2006

## Editorial Boards

[17] *Computer Physics Communications* — **HPC Specialist Editor**, 2017–2026
[16] *Journal of Computational Science* — **Editorial Board Member**, 2014–2026
[15] *International Journal of High Performance Computing Applications (IJHPCA)* — **Associate Editor**, 2012–2026
[14] *Sustainable Computing: Informatics and Systems (SUSCOM)* — **Editorial Board Member**, 2010–2026
[13] *Future Internet* — **Editorial Board Member**, 2020–2026
[12] *Frontiers in High-Performance Computing* — **Editor**, 2023–2024
[11] *IEEE Transactions on Parallel and Distributed Systems* — **Associate Editor**, 2018–2021
[10] *IEEE Transactions on Parallel and Distributed Systems* — **Best Paper Committee**, 2021
[9] *IEEE Transactions on Parallel and Distributed Systems* — **Artifact Evaluation Committee**, 2020–2021
[8] *International Journal of Parallel, Emergent and Distributed Systems (IJPEDS)* — **Associate Editor**, 2010–2019
[7] *International Journal of Information Technology, Communications and Convergence* — **Editorial Board Member**, 2009
[6] *Scientific Programming* — **Editorial Board Member**, 2015–2016
[5] *Concurrency and Computation: Practice and Experience (CCPE)* — **Editorial Review Board**, 2015
[4] *Parallel Computing (PARCO)* — **Guest Editor**, 2015
[3] *IET Computers and Digital Techniques* — **Guest Editor**, 2014
[2] *Sustainable Computing: Informatics and Systems (SUSCOM)* — **Guest Editor**, 2014
[1] *Journal of Autonomic and Trusted Computing* — **Editorial Board Member**, 2006–2007

## Significant University and Departmental Leadership and Service

(Selected leadership roles; full committee service available in the internal CV.)

[9] College of Engineering Promotion and Tenure Committee, Virginia Tech (Member 2025–present)
[8] Personnel Committee, Computer Science, Virginia Tech (Chair 2025–present)
[7] Director of the Queen's Global Research Institute on Electronics, Communications and Information Technology (ECIT) (2018–2019)
[6] Head of School of EEECS, Queen's University Belfast (2016–2018)
[5] Member of the University Academic Council, Queen's University Belfast (2016–2018)
[4] Vice Chancellor Selection Panel, Queen's University Belfast (2017)
[3] Chair of the University High-Performance Computing Advisory Group, Queen's University Belfast (2012–2019)
[2] Engineering (Unit of Assessment 12) REF Champion, Queen's University Belfast (2018–2019)
[1] Computer Science (UoA11) REF Champion, Queen's University Belfast (2015–2019)

## Significant External Leadership and Service

[45] Computing Research Association (CRA), Leadership Academy Committee (2024–present)
[44] IEEE Computer Society Senior Membership Reviewer (2025–2026)
[43] American Association for the Advancement of Science (AAAS), Session Reviewer (2024)
[42] External Accreditation and Evaluation Panel — Graduate Program (M.S. in Data Science and Machine Learning; M.S. in Information Systems), Hellenic Open University (2023)
[41] External Accreditation and Evaluation Panel — Undergraduate Program, Department of Informatics and Communications, University of Ioannina (2022)
[40] Scientific Advisory Board, Kelvin Tier-2 HPC Center, Queen's University Belfast (2021–present)
[39] Faculty Appointment Committee, Norwegian University of Science and Technology (2022)
[38] External Evaluation Panel, Institute of Informatics and Telecommunications, National Center for Scientific Research "Demokritos" (2022)
[37] Evaluation Panel, Institute of Computer Science, Foundation for Research and Technology — Hellas (Seed Grants) (2021–present)
[36] Scientific Advisory Board, Marie Curie Individual Fellowship (2022)
[35] Faculty Appointment and Promotion Committee, University of Ioannina (2022)
[34] Faculty Appointment Committee, Heidelberg University (2021)
[33] External Accreditation and Evaluation Panel — Undergraduate Program, Department of Information Systems, Ionian University (2021)
[32] External Accreditation and Evaluation Panel — Undergraduate Program, School of Electrical and Computer Engineering, National Technical University of Athens (2021)
[31] Scientific Advisory Board, Institute of Computer Science, Foundation for Research and Technology Hellas (FORTH) (2020–present)
[30] Faculty Appointment and Promotion Committee, National Technical University of Athens (2020)
[29] Evaluation Panel, Foundation for Research and Technology — Hellas (Synergy Grants) (2019–present)
[28] Faculty Appointment and Promotion Committee, Technical University of Crete (2019)
[27] Faculty Appointment and Promotion Committee, University of Thessaly (2016; 2018–2019)
[26] Engineering and Physical Sciences Research Council (EPSRC) Strategic Advisory Team (SAT) Member — e-Infrastructure (2018–present)
[25] European Commission Framework Programme Consultant on Cloud Computing, European Commission DG Unit (2018)
[24] Faculty Appointment and Promotion Committee, Université Paris-Sorbonne (2018)
[23] Knowledge Transfer Partnership in Blockchain Technology, Vox Financial Partnerships (2018)
[22] Knowledge Transfer Partnership in OpenCL Technology, Crevinn (2017)
[21] Foreign Direct Investment Consultant, Invest Northern Ireland (2017)
[20] Faculty Appointment and Promotion Committee, Aristotle University of Thessaloniki (2017)
[19] Faculty Appointment and Promotion Committee, Chalmers University of Technology (2017)
[18] Faculty Appointment and Promotion Committee, University of Crete (2016–2018; 2020)
[17] Scientific Advisory Board, European Commission Horizon 2020 INTERTWinE Project (2015–2018)
[16] Scientific Advisory Member, European Commission Horizon 2020 Programme (2016)
[15] Grant Proposal Evaluator, King Abdullah University of Science and Technology (2016)
[14] Faculty Appointment and Promotion Committee, National Technical University of Athens (2014; 2016)
[13] Faculty Appointment and Promotion Committee, Ionian University (2016)

[12] Faculty Appointment and Promotion Committee, University of Athens (2015)
[11] Faculty Appointment and Promotion Committee, Technological Educational Institute of Athens (2015)
[10] Faculty Appointment and Promotion Committee, Technological Educational Institute of Piraeus (2014–2015)
[9] Faculty Appointment and Promotion Committee, Technological Educational Institute of Western Greece (2014–2016)
[8] Industry-Academic Partnership Training, Westminster Higher Education Forum (2013)
[7] External examiner, School of Computing, University of Leeds (2012–present)
[6] Faculty Appointment and Promotion Committee, University of Crete (2013)
[5] Multiple industrial consultancies, Queen's University Belfast (2012–present)
[4] Faculty Appointment and Promotion Committee, University of Ioannina (2011)
[3] Faculty Appointment and Promotion Committee, Aristotle University of Thessaloniki (2011)
[2] Faculty Hiring Committee, Technical University of Denmark (2010)
[1] Faculty Hiring Committee, National Technical University of Athens (2010)

### Government Research Funding Panelist & Reviewer Service

[25] National Science Foundation (NSF) — Panelist, 2025 (two programs)
[24] Swiss National Science Foundation — Grant proposal reviewer, 2016; 2020; 2025
[23] French National Research Agency (ANR) — Panelist, 2025
[22] Hellenic Foundation for Research and Innovation (Greece) — Panelist, 2024–2025
[21] National Science Foundation (NSF), CISE — Panelist, 2021; 2023–2024
[20] Coastal Virginia Center for Cyber Innovation (COVA CCI, U.S.) — Grant proposal reviewer, 2021
[19] Knowledge Foundation (Sweden) — Grant proposal reviewer, 2021
[18] Italian National Research Council (CINECA) — Grant proposal reviewer, 2021
[17] European Research Council (ERC) — Grant proposal reviewer, 2020
[16] U.S. Department of Energy — Grant proposal reviewer, 2020
[15] Royal Academy of Engineering (U.K.) — Grant proposal reviewer, 2015; 2017
[14] Austrian Academy of Sciences — Grant proposal evaluator, 2017
[13] Technology Foundation STW (The Netherlands) — Grant proposal reviewer, 2016
[12] National Science Center (Poland) — Grant proposal reviewer, 2016
[11] Natural Sciences and Engineering Research Council of Canada (NSERC) — Discovery Grants panelist, 2014–2016
[10] University of Cyprus Research Foundation — Grant proposal reviewer, 2016
[9] Engineering and Physical Sciences Research Council (EPSRC, U.K.) — Grant proposal reviewer, 2012–2014; 2017–2018; 2021
[8] EPSRC (U.K.) — Platform Grant panelist, 2015
[7] European Commission FP7 Framework Programme — Project reviewer, 2013–2016
[6] European Commission FP7 Framework Programme — Grant proposal reviewer, 2012
[5] Greek Secretariat for Research and Technology — Grant proposal reviewer, 2010
[4] U.S.–Israel Binational Science Foundation — Grant proposal reviewer, 2009
[3] National Science Foundation (NSF) — Panelist, 2002–2004; 2008
[2] Natural Sciences and Engineering Research Council of Canada — Grant proposal reviewer, 2007
[1] Maryland Industrial Partnerships Program — Grant proposal reviewer, 2007

### Conference Panels

[8] *Fifth International Workshop on Extreme Scale Programming Models and Middleware* (SC 2020) — Panelist: "When one size doesn't fit all: programming models in the post-Moore era" (Nov. 2020)
[7] *EPSRC Workshop on Manycore Computing: Hardware and Software* — Panelist (Southampton, U.K., Jan. 2018)
[6] *6th Workshop on Parallel Programming and Run-Time Management Techniques for Many-core Architectures (PARMA-DITAM 2015)* — Panelist (Jan. 2015)
[5] *IBM Research ExaChallenge Symposium* — Panelist (Dublin, Ireland, Oct. 2012)
[4] *39th International Conference on Parallel Processing (ICPP)* — Panelist: "Accelerators: fad, fashion, or future?" (Sep. 2010)
[3] *Fall Creek Falls Conference* — Panelist: "Key challenges presented by next generation hardware systems" (Sep. 2007)
[2] *Microsoft Faculty Research Summit* — Invitee (Redmond, WA, Sep. 2007)
[1] *15th ACM International Conference on Supercomputing (ICS)* — Panelist: NSF Next Generation Systems Software Program (Jun. 2001)

### Public Engagement

[12] More to Know Podcast (Christopher Conover) — "The Future of Datacenters: AI, Energy Demand, and the Next Infrastructure Boom" (Feb. 2026). link
[11] TEDxMidAtlantic — "Why More AI Data Centers Aren't the Answer" (Nov. 2025). TEDx Editor's Pick. video
[10] Northwest Indiana Times — "Data Centers in Northwest Indiana: Key Issues and Concerns" (Dec. 2025). link
[9] FuelCell Energy — "On the societal and economic implications of datacenters" (Dec. 2025). link
[8] MSN / WFXR — "On the societal and economic implications of datacenters" (Dec. 2025). link 1; link 2
[7] Cardinal News — "On the societal and economic implications of datacenters" (Aug. 2025). link
[6] Roanoke Times (VT Science Corner) — "A Way to Democratize AI" (Jan. 2024). link
[5] Newswise — "Queen's Researchers in Bid to Develop the Fastest Supercomputers" (Feb. 2015). link
[4] NNTV Behind the Science — "On the importance of supercomputers" (2015)
[3] Horizon 2020 focus workshops in Northern Ireland and the U.K. (Mar.–Sep. 2014)
[2] CNN — "Faster than 50 Million Laptops: The Race Goes to Exascale" (Mar. 2012). link
[1] IEEE Computer (LEAV/Com) — "Big Iron Move Toward Exascale" (Oct. 2012). link

## Invited Seminars and Talks

[55] *SWEET: A Hardware & Software Stack for Wrist-to-Cloud Intelligence* — NSF CSR Principal Investigators Meeting (Boca Raton, FL, Sep. 2025)
[54] *SWEET: System Architectures and Programming Abstractions for Resilient and Efficient Edge AI* — Discover-US Webinar, European Commission (virtual, Apr. 2025)
[53] *Sustainable AI from the Viewpoint of an AI Skeptic* — Distinguished Speaker Seminar, Department of Computer Engineering and Informatics (virtual, Dec. 2024)
[52] *Programming Models for High-Performance Computing: Key Drivers and Barriers to Adoption* — Invited talk, First NI-HPC Users Conference (virtual, Oct. 2021)

[51] *Why Should We Still be Concerned about Energy Management in System Software?* — Invited talk, 12th International Green and Sustainable Computing Conference (virtual, Oct. 2021)
[50] *From Approximate to Significance-Driven to Transprecision Computing: Challenges and Opportunities* — Distinguished Speaker Seminar, Department of Computational Science, ETH Zurich (Apr. 2018)
[49] *Virtualizing Any Accelerator Anywhere* — Distinguished Speaker Seminar, Department of Computer Science, College of William & Mary (Jan. 2018)
[48] *The Jevons Paradox in Computing Systems Research* — Distinguished Lecture Series, Department of Computer Science, Virginia Tech (Nov. 2016)
[47] *Computational Significance and its Implications for HPC* — 13th Workshop on Clusters, Clouds, and Data for Scientific Computing (CCDSC 2016) (Chemin de Chanzé, France, Oct. 2016)
[46] *Computational Significance and its Implications for Computing Systems* — School of Electrical and Electronic Engineering, Newcastle University (Oct. 2016)
[45] *Scaling Up, Out, or Down* — School of Informatics, University of Edinburgh (Mar. 2016)
[44] *Significance-Driven Runtime Systems* — RoMoL 2016 Workshop (Barcelona, Spain, Mar. 2016)
[43] *Advances in Energy-Efficient and Resilient HPC: Scaling Up, Out, or Back?* — Cardiff University (Mar. 2016)
[42] *Variability: Why should we care?* — Birds-of-a-Feather session on Variability in Large-Scale Computing Systems (SC 2015) (Austin, TX, Nov. 2015)
[41] *New Approaches to Energy-Efficient and Resilient HPC* — Department of Computer Science, Old Dominion University (Nov. 2015)
[40] *HPDC Research at Queen's: An Overview* — ARM High-Performance Computing Group (Manchester, U.K., Nov. 2015)
[39] *Server Resource Provisioning for Real-Time Analytics using Iso-Metrics* — Workshop on Performance Modeling: Methods and Applications (ISC 2015) (Frankfurt, Germany, Jul. 2015)
[38] *Evaluating Servers using Iso-Metrics: Power, Performance and Programmability Implications* — 8th Workshop on Programmability Issues for Heterogeneous Multicores (MULTIPROG 2015) (Amsterdam, The Netherlands, Jan. 2015)
[37] *The Challenges and Opportunities of Micro-Servers in the HPC Ecosystem* — 12th Workshop on Clusters, Clouds, and Data for Scientific Computing (CCDSC 2014) (Chemin de Chanzé, France, Oct. 2014)
[36] *NVRAM as a User-Level Object Store* — HiPEAC Autumn Computing Systems Week (Athens, Greece, Oct. 2014)
[35] *On the Viability of Microservers for Real-Time Data Analytics* — HiPEAC Autumn Computing Systems Week (Athens, Greece, Oct. 2014)
[34] *NanoStreams: A Hardware and Software Stack for Real-Time Analytics on Fast Data Streams* — Horizon 2020: the HPC Opportunity (London, U.K., Mar. 2014)
[33] *GEMSCLAIM: Greener Mobile Systems by Cross-Layer Energy Management* — CHIST-ERA Projects Seminar (Istanbul, Turkey, Mar. 2014)
[32] *Searching for Data: The Ever-Increasing Role of Memory Hierarchies on the Performance and Sustainability of Computing Systems* — Inaugural lecture, Queen's University Belfast (Mar. 2013)
[31] *Energy as a Resource in Parallel Programs* — Supercomputing 2012 Birds-of-a-Feather: *Cool Supercomputing* (Nov. 2012)
[30] *Block-Level Dynamic Dependence Analysis for Task-Based Parallelism* — Workshop on Perspectives on Parallel Numerical Linear Algebra (Manchester, U.K., Jul. 2012)
[29] *Software Techniques for Energy Conservation in High-End Computing Systems* — Invited seminar,

School of Computer Science, University of Manchester (U.K., Mar. 2012)

[28] *Energy Efficiency at Extreme Scale: Tools and Challenges* — Supercomputing 2011 Birds-of-a-Feather: *Energy-Efficiency* (Nov. 2011)

[27] *Rearchitecting MapReduce for Heterogeneous Multicore Processors with Explicitly Managed Memories* — School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (Oct. 2010)

[26] *Determinism in Parallel Software and Architectures* — HiPEAC Systems Week Cluster Meetings (Barcelona, Oct. 2009)

[25] *Parallelizing Non-trivial Applications with Multiple Programming Models* — HiPEAC Systems Week Cluster Meetings (Barcelona, Oct. 2009)

[24] *Uniform Evaluation of Programming Models* — HiPEAC Systems Week Cluster Meetings (Paris, Nov. 2008)

[23] *Unifying Layered Parallelism on the Cell BE* — Supercomputing 2007 Birds-of-a-Feather: *Unleashing the Power of the Cell Broadband Engine Processor for HPC* (Nov. 2007)

[22] *Unified Scheduling of Polymorphic Parallelism on Asymmetric Multi-core Systems* — Lawrence Livermore National Laboratory (Oct. 2007)

[21] *System Software for Scaling on Many Cores* — Oak Ridge National Laboratory (Sep. 2007)

[20] *Design and Implementation of Time- and Power-Efficient Software Stacks for Multicore Processors* — IBM Thomas J. Watson Research Center (Dec. 2006)

[19] *Design and Implementation of Time- and Power-Efficient Software Stacks for Multicore Processors* — Department of Computer Science, North Carolina State University (Sep. 2006)

[18] *Hardware Event-Driven Scalability Predictors: Improving Energy-Efficiency under Hard Performance Constraints on Multi-core and Multi-threaded Architectures* — Department of Electronic and Computer Engineering, Technical University of Crete (Jun. 2006)

[17] *Addressing the Challenges of Chip Multiprocessors using Autonomic Software* — Department of Computer Science, University of California, Riverside (Apr. 2006)

[16] *Addressing the Challenges of Chip Multiprocessors using Autonomic Software* — Department of Electrical and Computer Engineering, University of British Columbia (Mar. 2006)

[15] *High-Performance Power-Efficient Runtime Environments for Dense Computing Systems* — Department of Computer Science, Virginia Tech (Feb. 2006)

[14] *High-Performance Power-Efficient Runtime Environments for Dense Computing Systems* — Institute of Computer Science, Foundation for Research and Technology — Hellas (FORTH) (Jun. 2005)

[13] *High-Performance Power-Efficient Runtime Environments for Dense Computing Systems* — Department of Computer Engineering and Informatics, University of Patras (Jun. 2005)

[12] *A Unified Programming Framework for Multigrain Multithreaded Architectures* — Institute of Computer Science, Foundation for Research and Technology — Hellas (FORTH) (Jun. 2004)

[11] *A Unified Programming Framework for Multigrain Multithreading* — School of Electrical and Computer Engineering, National Technical University of Athens (Jun. 2004)

[10] *A Unified Programming Framework for Multigrain Multithreading* — Department of Computer Science, University of California, Riverside (Apr. 2004)

[9] *A Unified Programming Framework for Multigrain Parallel Architectures* — Department of Electrical and Computer Engineering, Northwestern University (Feb. 2004)

[8] *Program Transformations and Scheduling Algorithms for Managing Shared Caches on Multithreaded Processors* — Department of Informatics, Athens University of Economics and Business (Jun. 2003)

[7] *Program Transformations and Scheduling Algorithms for Managing Shared Caches on SMT Processors* — IBM Thomas J. Watson Research Center (Mar. 2003)

[6] *Building Adaptive Programs with Local Sensing of Execution Conditions* — Department of Computer Science, Texas A&M University (Mar. 2003)
[5] *Interoperable System Software* — Department of Information and Computer Sciences, University of California, Irvine (Apr. 2002)
[4] *Interoperable System Software* — Department of Computer Science, College of William & Mary (Mar. 2002)
[3] *Scaling Shared-Memory Programming Models beyond Shared-Memory Architectures* — Department of Computer Science, University of Houston (Nov. 2001)
[2] *Some Steps towards Simple, Scalable and Portable Parallel Programming Models* — Department of Computer Science, College of William & Mary (Oct. 2001)
[1] *A Case for User-Level Page Migration* — Coordinated Sciences Laboratory, University of Illinois at Urbana–Champaign (Jan. 2001)